

Aula 2

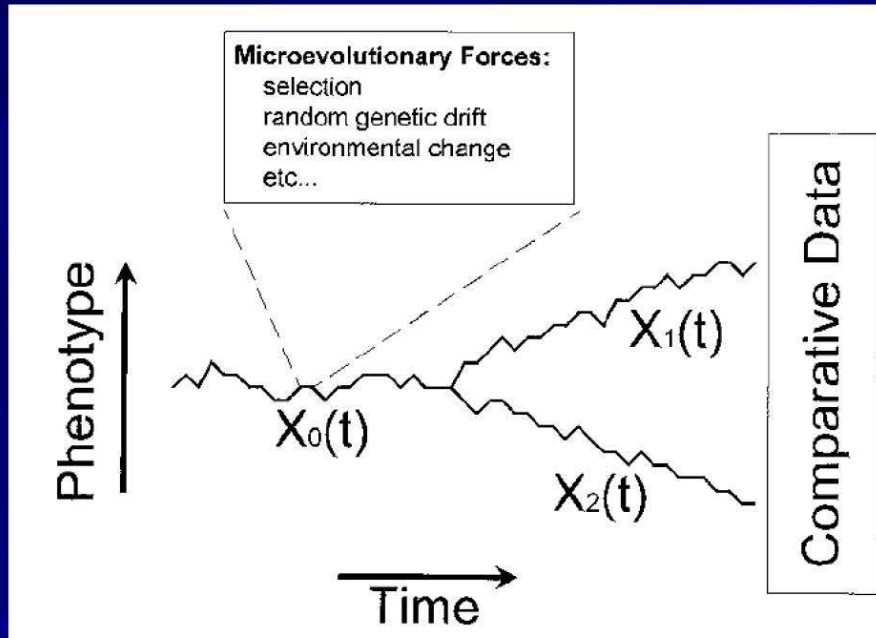
Modelos de evolução fenotípica de
caracteres contínuos e categóricos

PGLS

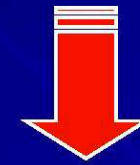
Prática

- Introdução ao R
 - Importação e manipulação de dados
 - Importação e manipulação de $\neq$ formatos de filogenias
- Timetree.org
- Phyndr
- Phylotargeting
- rotl
- Rphylopars

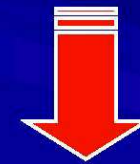
EVOLUTIONARY MODELS



Mechanisms (selection, drift, mutations...)



Evolutionary models



Interspecific data

The analytical core of comparative analysis

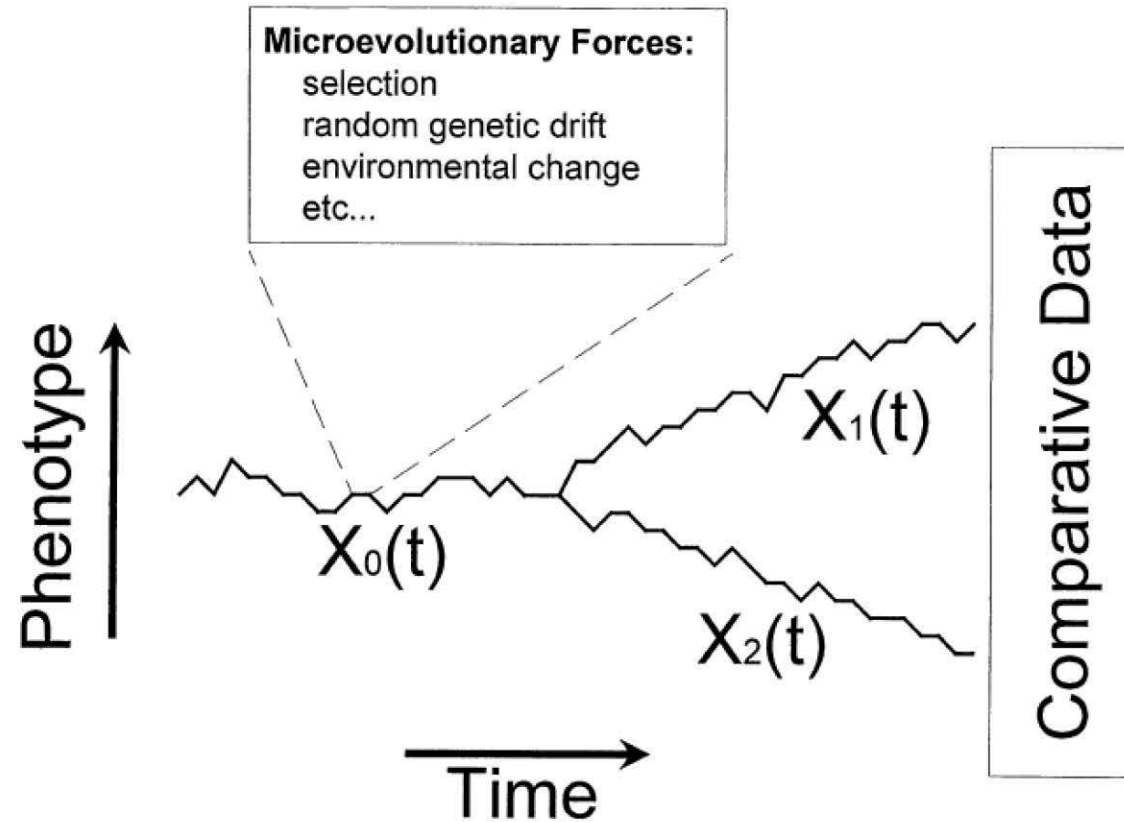
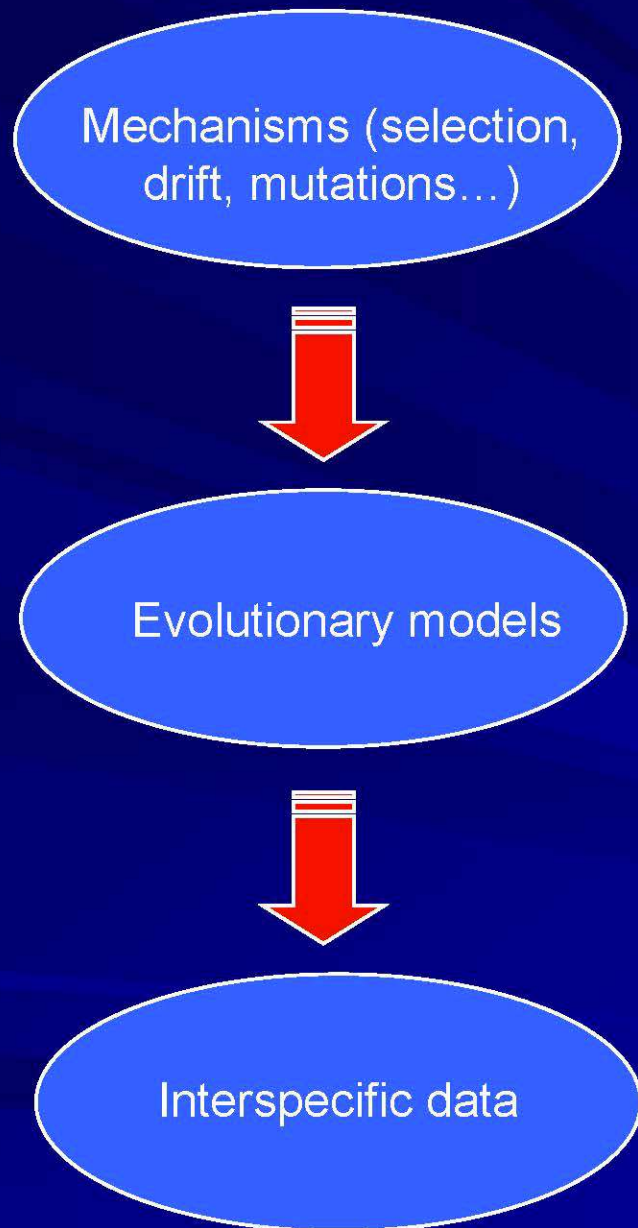


FIG. 1. An illustration of our basic model for translating between microevolutionary forces acting on a short time scale and the macroevolutionary pattern observed in a set of comparative data.



Phylogenetic Signal, Evolutionary Process, and Rate

LIAM J. REVELL,¹ LUKE J. HARMON,² AND DAVID C. COLLAR¹

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E-mail: lrevell@fas.harvard.edu (L.J.R.)

²Department of Biological Sciences, University of Idaho, Moscow, Idaho 83844, USA

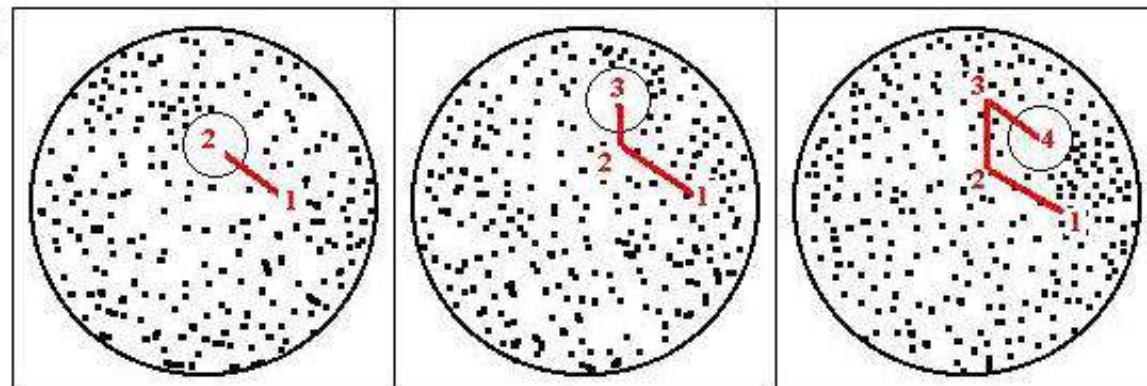
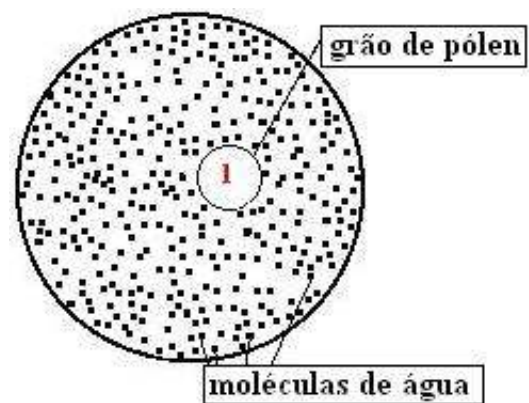
The path from evolutionary mechanisms (selection, drift, mutation and so on) to interspecific variation is a conceptual idea, but it may be hard (or even impossible) to reverse it and actually recover such processes from empirical data...

Modelos de evolução fenotípica

- Movimento Browniano (BM) => Muitos métodos usam este modelo. Tratabilidade analítica
- Early Burst (EB) => Incorpora a idéia de Radiação adaptativa
- Trend => Incorpora a idéia de tendência vista em fósseis
- Ornstein-Uhlenbeck (OU) => Incorpora a idéia de restrições em evolução

Movimento Browniano

- O que é o Movimento Browniano?
- Quando caracteres evoluem de forma parecida com o Movimento Browniano?
- Simulando BM em filogenias



Robert Brown, botânico

Movimento Browniano

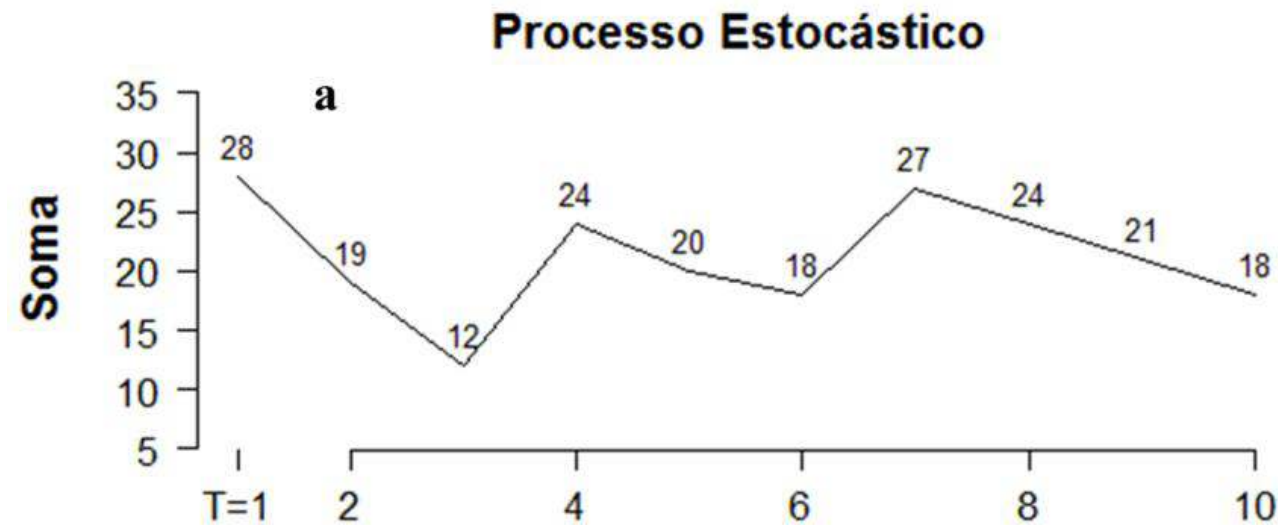
- Modelo para a evolução de caracteres contínuos
 - Seguindo um random walk
- Estados de caracter mudam continuamente ao longo do tempo
 - Processo estocástico de tempo contínuo
- Depois de um tempo e múltiplas simulações, o estado do caracter seguem uma distribuição normal

**Phylogenetic Analysis
Models and Estimation Procedures**

L. L. CAVALLI-SFORZA AND A. W. F. EDWARDS*

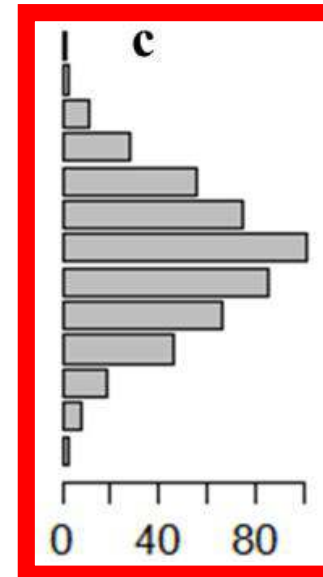
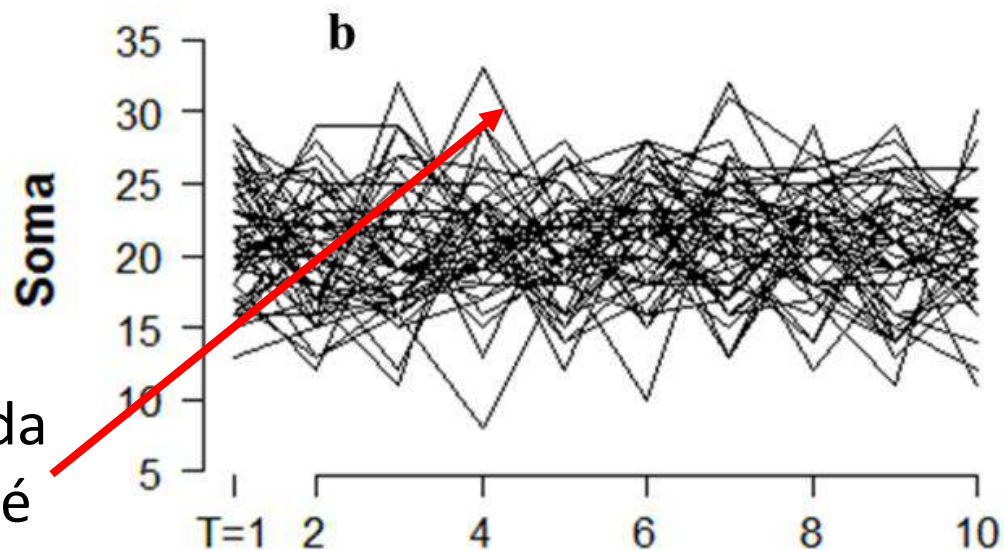
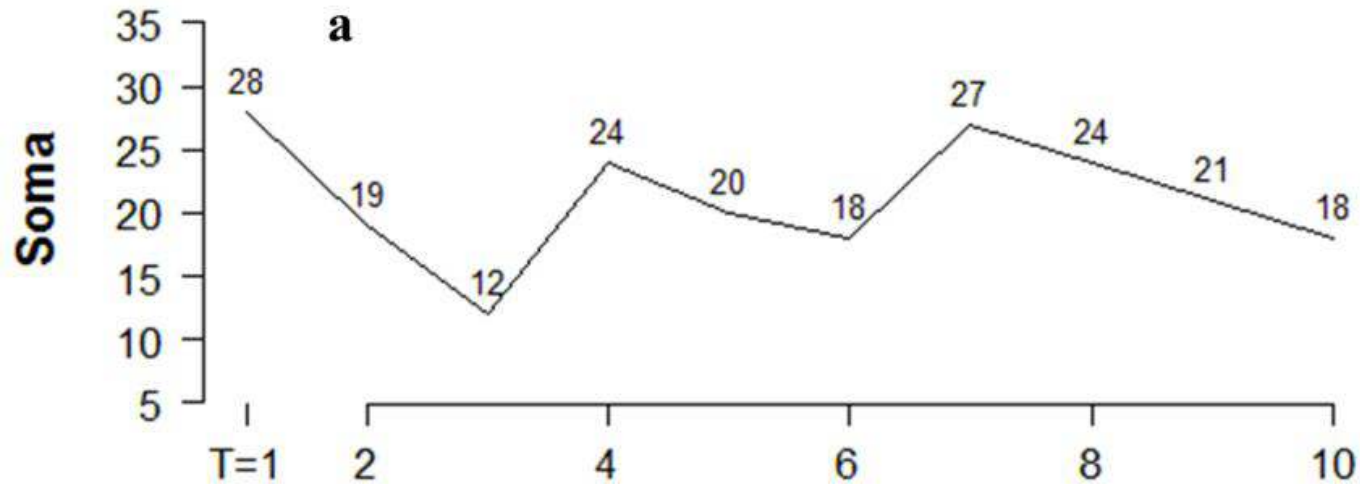
AMERICAN JOURNAL OF HUMAN GENETICS, VOL. 19, NO. 3, PART I (MAY), 1967

...mas whattahell é um processo estocástico de tempo contínuo?



Soma dos
resultados de
2 dados (6-36)

...mas whattahell é um processo estocástico de tempo contínuo?



Mas a distribuição de probabilidade de um conjunto de curvas é conhecida (aprox. distribuição normal) e permite predizer eventos futuros

Movimento Browniano

- Dois parâmetros => valor inicial (θ) e taxa de mudança (σ^2)
- Três fatos descrevem um BM (Processo de Wiener)
 - $W(t)$ é o valor do caracter no tempo t . Logo:
 - $E[W(t)] = W(0)$
 - Incrementos (passos sucessivos) são independentes ("pouca memória")
 - $W(t) \sim N(W(0), \sigma^2 t)$
- Traduzindo: o valor do caracter W no tempo t é aproximado pela distribuição normal com média 0 (*standard normal distribution*) e variância igual à taxa de mudança no tempo t

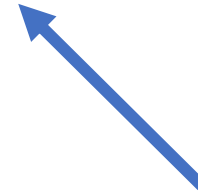
$$dX(t) = \sigma dB(t)$$

$$dX(t) = \sigma dB(t)$$



Quantidade de mudança no caracter X
entre o tempo do tempo t para $t + dt$

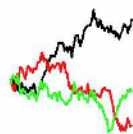
$$dX(t) = \sigma dB(t)$$

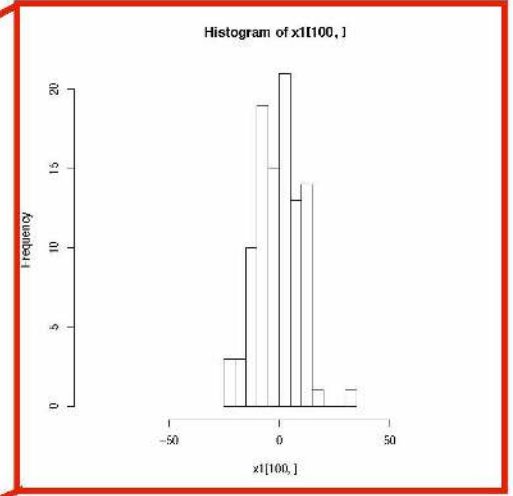
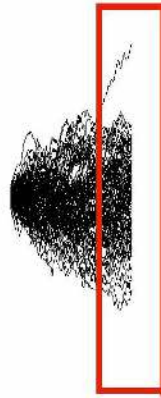


Descreve o "random walk" na variável,
que se aproxima de uma distribuição
normal com média 0 e variância dt



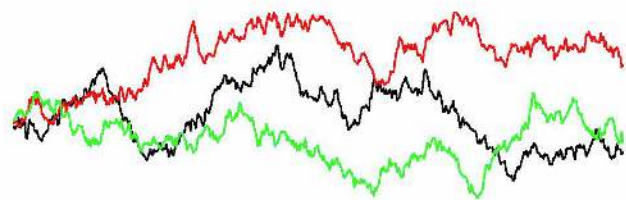


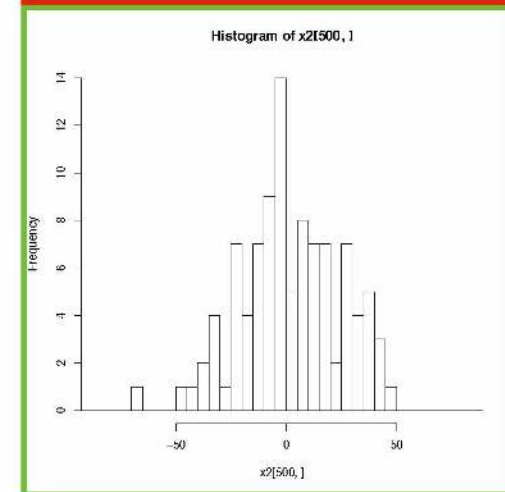
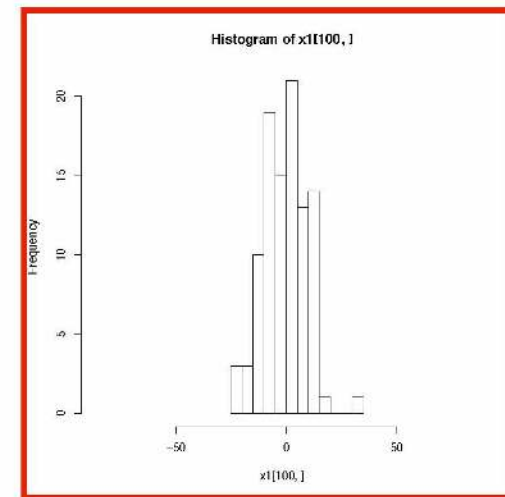
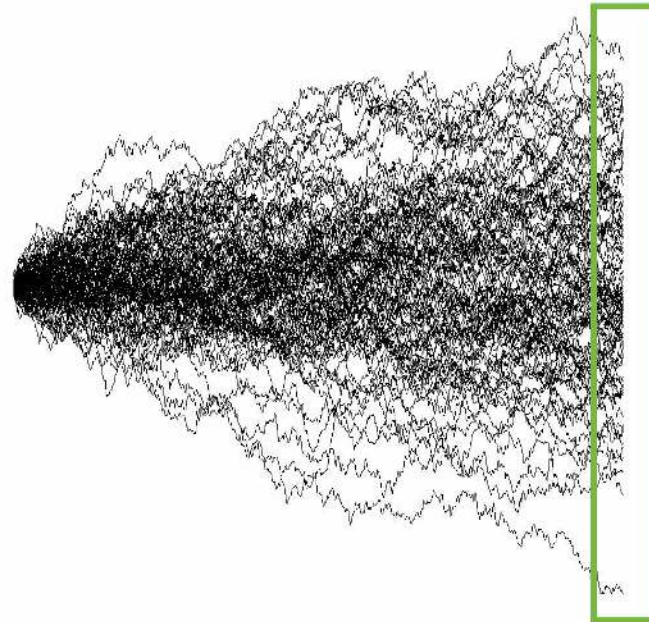


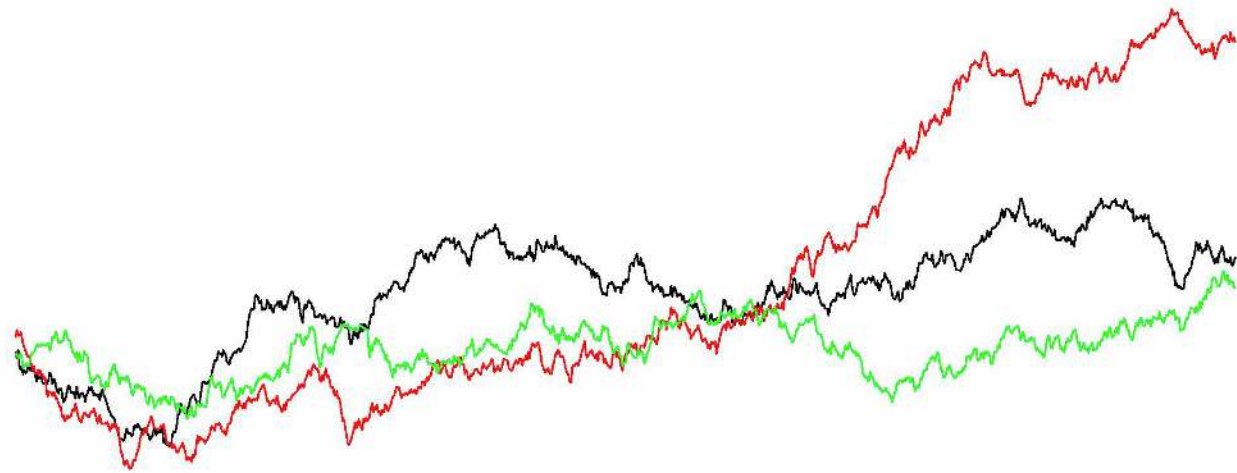


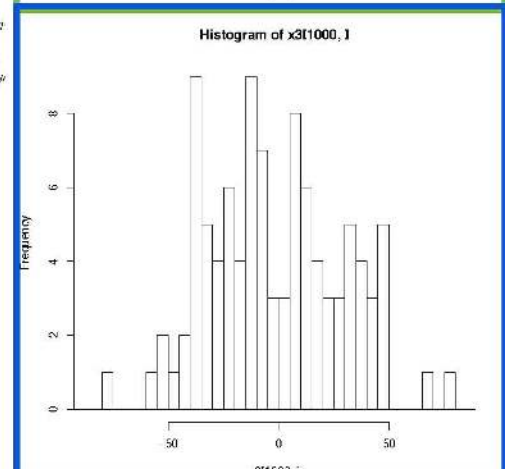
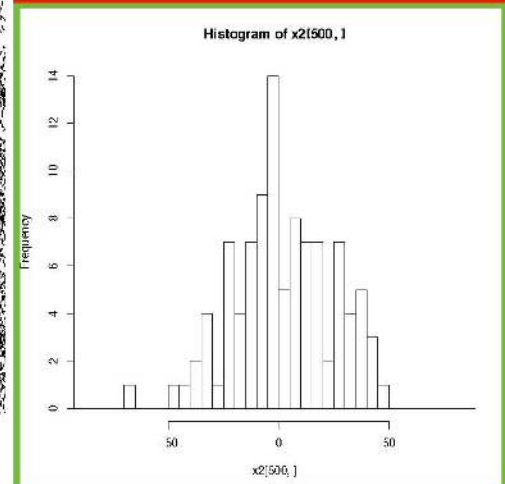
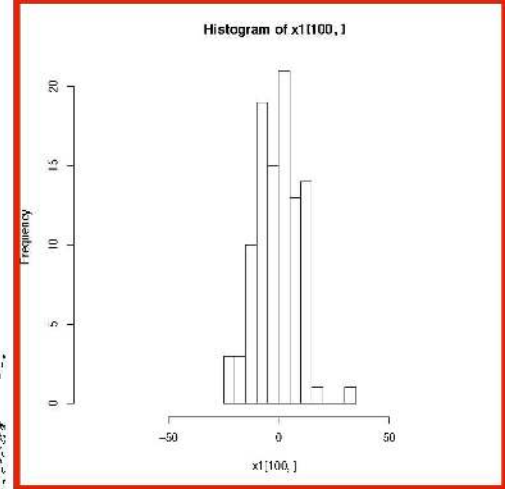
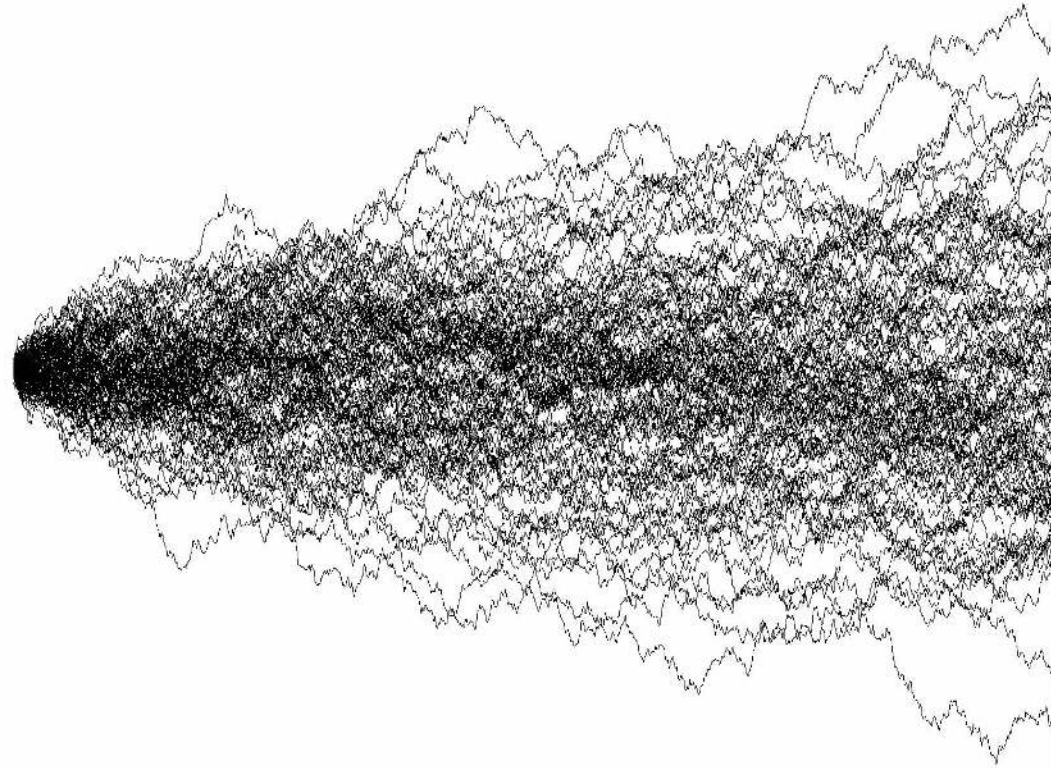


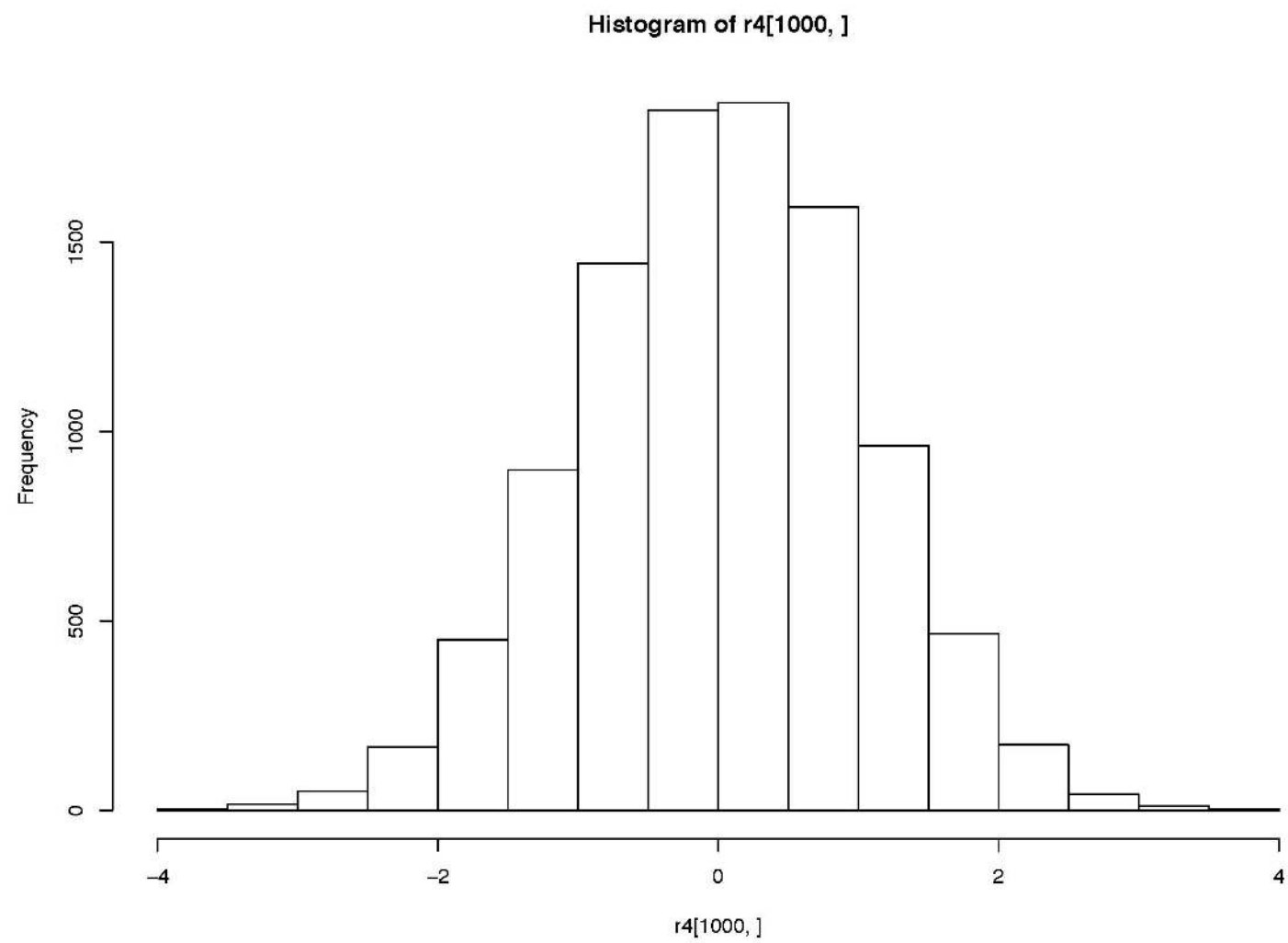


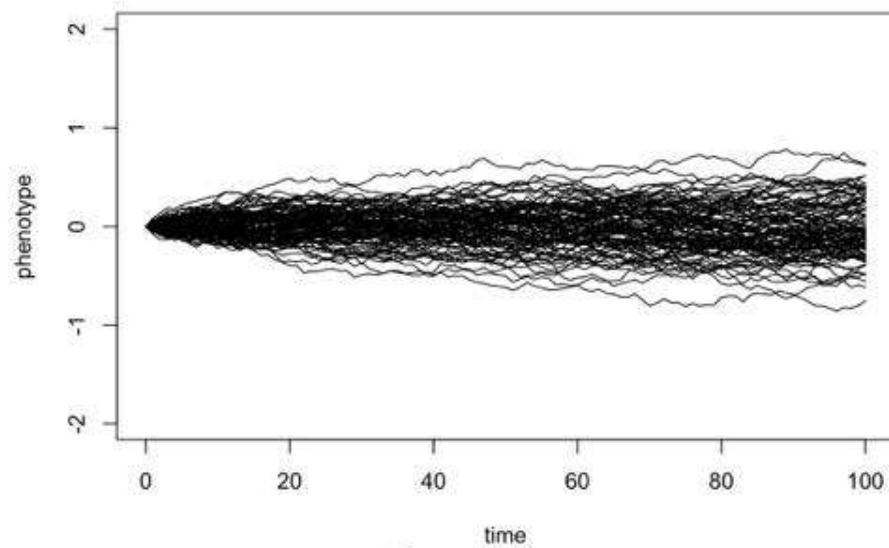




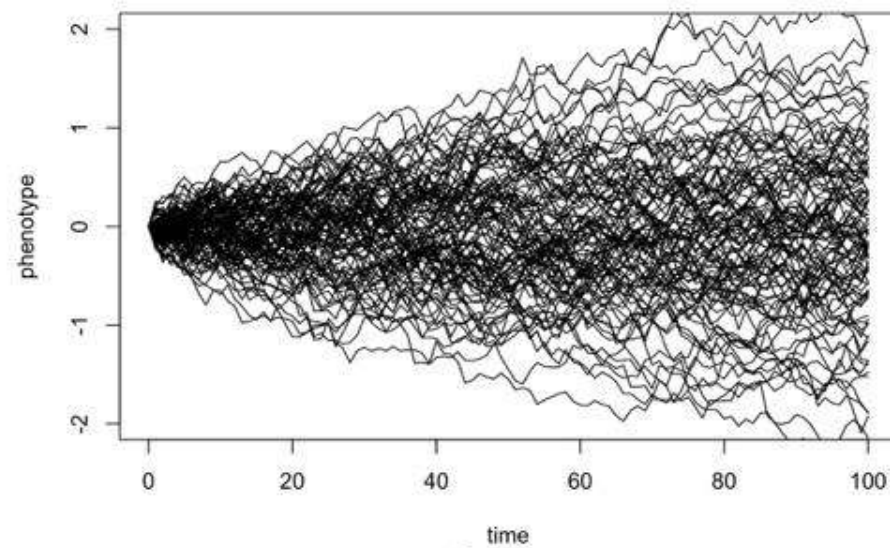




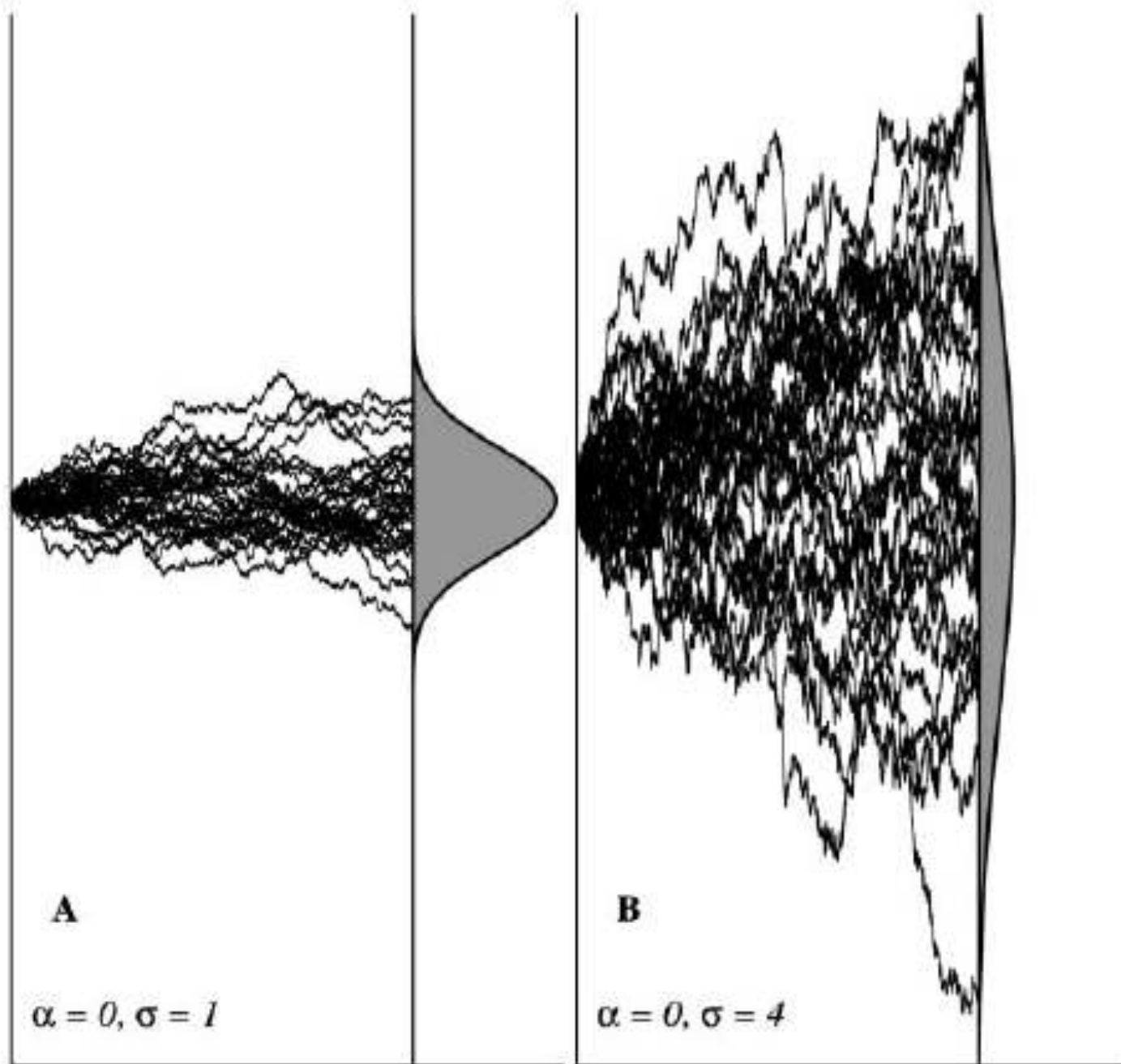




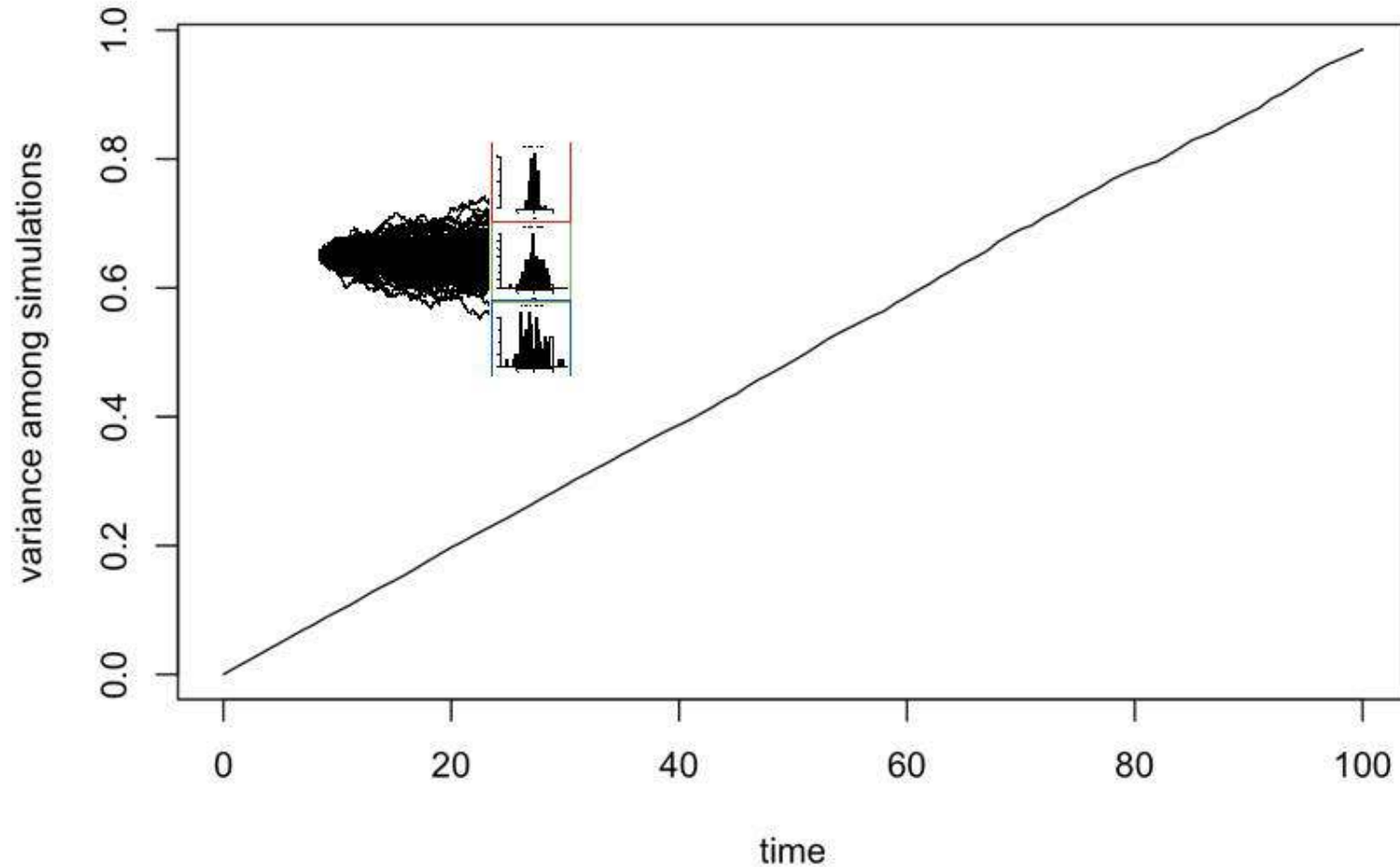
$$\sigma^2 = 0.01$$



$$\sigma^2 = 0.1$$



variance increases with time



$$\text{Var}(X) = \sigma^2 * \textit{time}$$

Quando a evolução é descrita por BM?

- Deriva genética
- Mudança pontuada aleatória
- Quando a seleção é fraca em relação à escala de tempo considerada
- Quando seleção muda de direção aleatoriamente ao longo do tempo

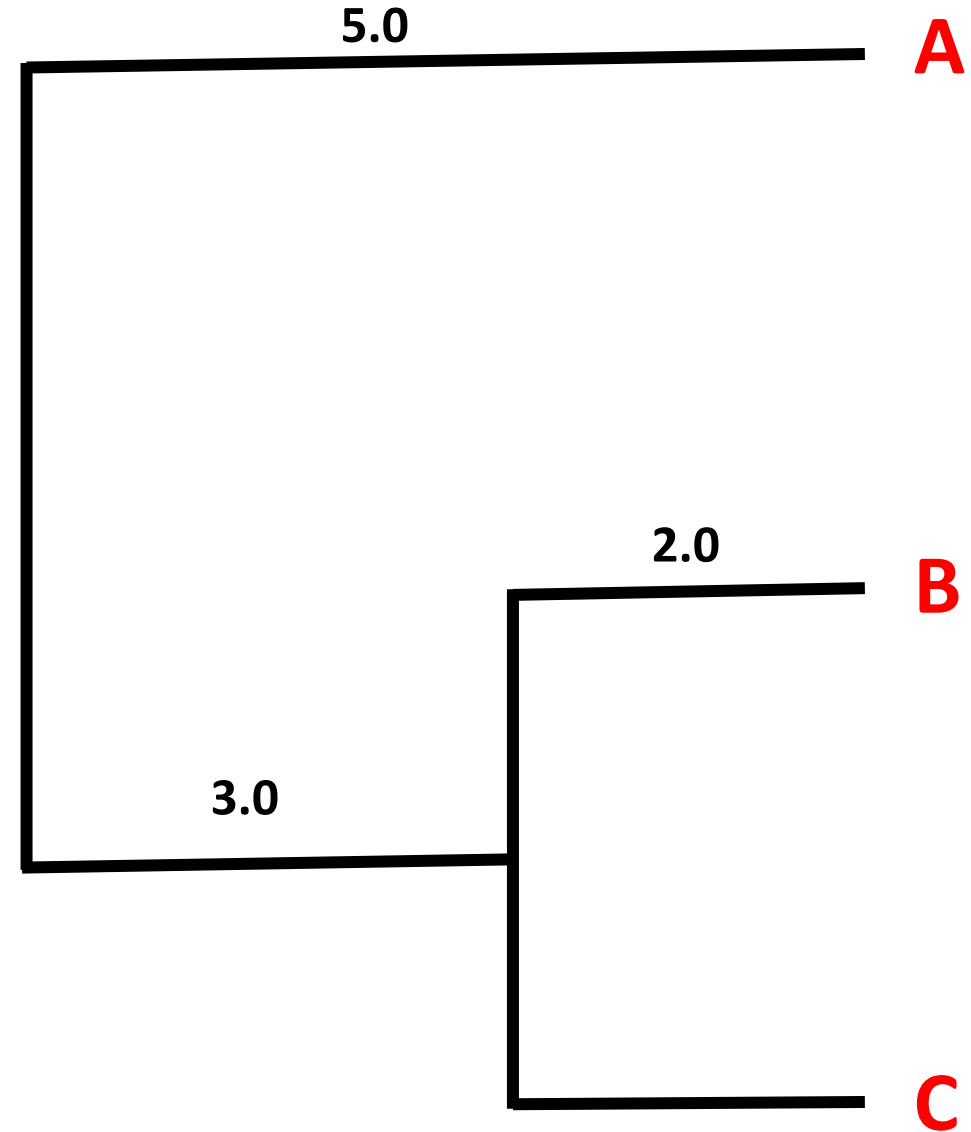
Como a filogenia influencia a evolução de fenótipos sob um processo BM?

O parentesco comum aumenta ou diminui a similaridade entre espécies em termos de fenótipo?

... Mas antes, vamos entender o que é uma matriz de distância filogenética (patrística) e de variância-covariância

Matriz de distância patrística

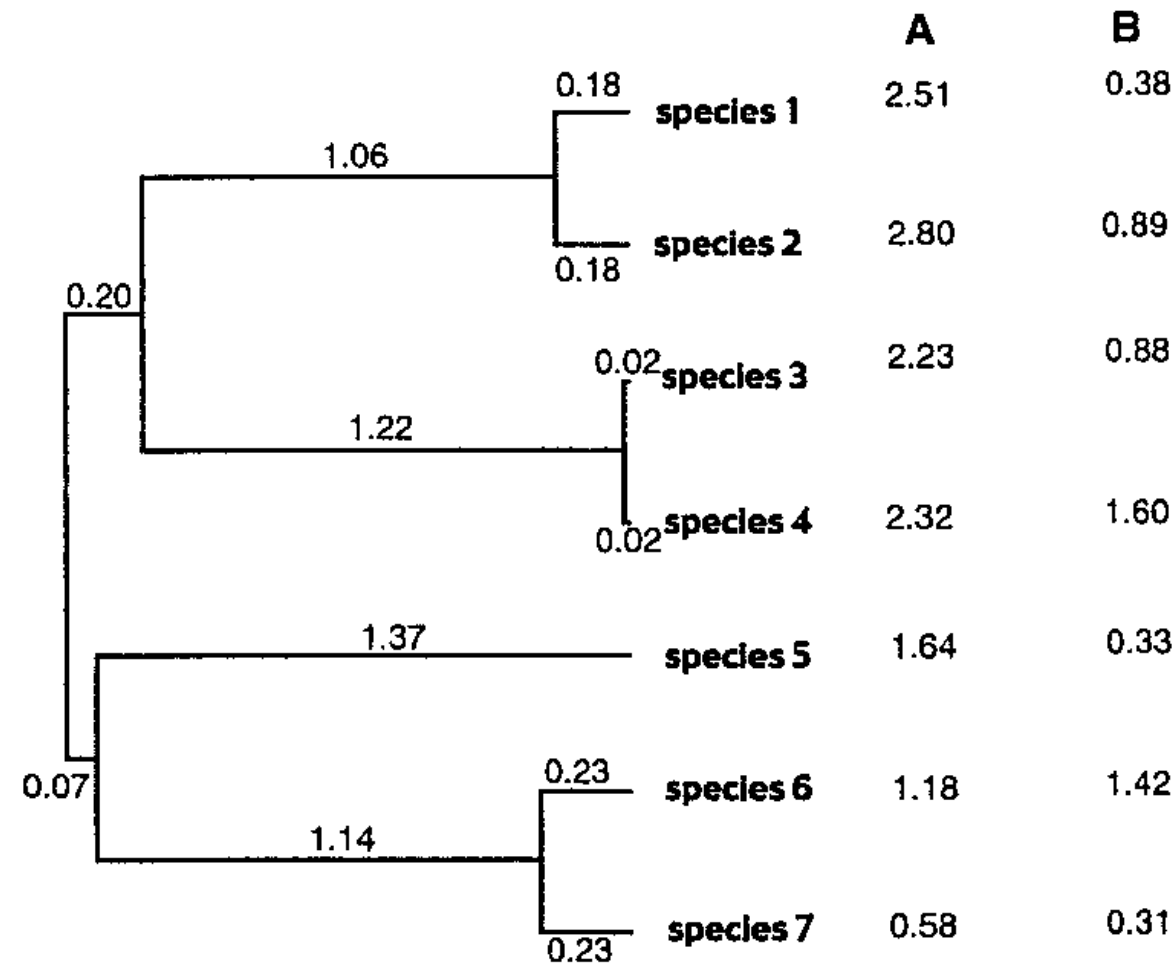
	A	B	C
A	0	10	10
B	10	0	4
C	10	4	0



Matriz de variância-covariância

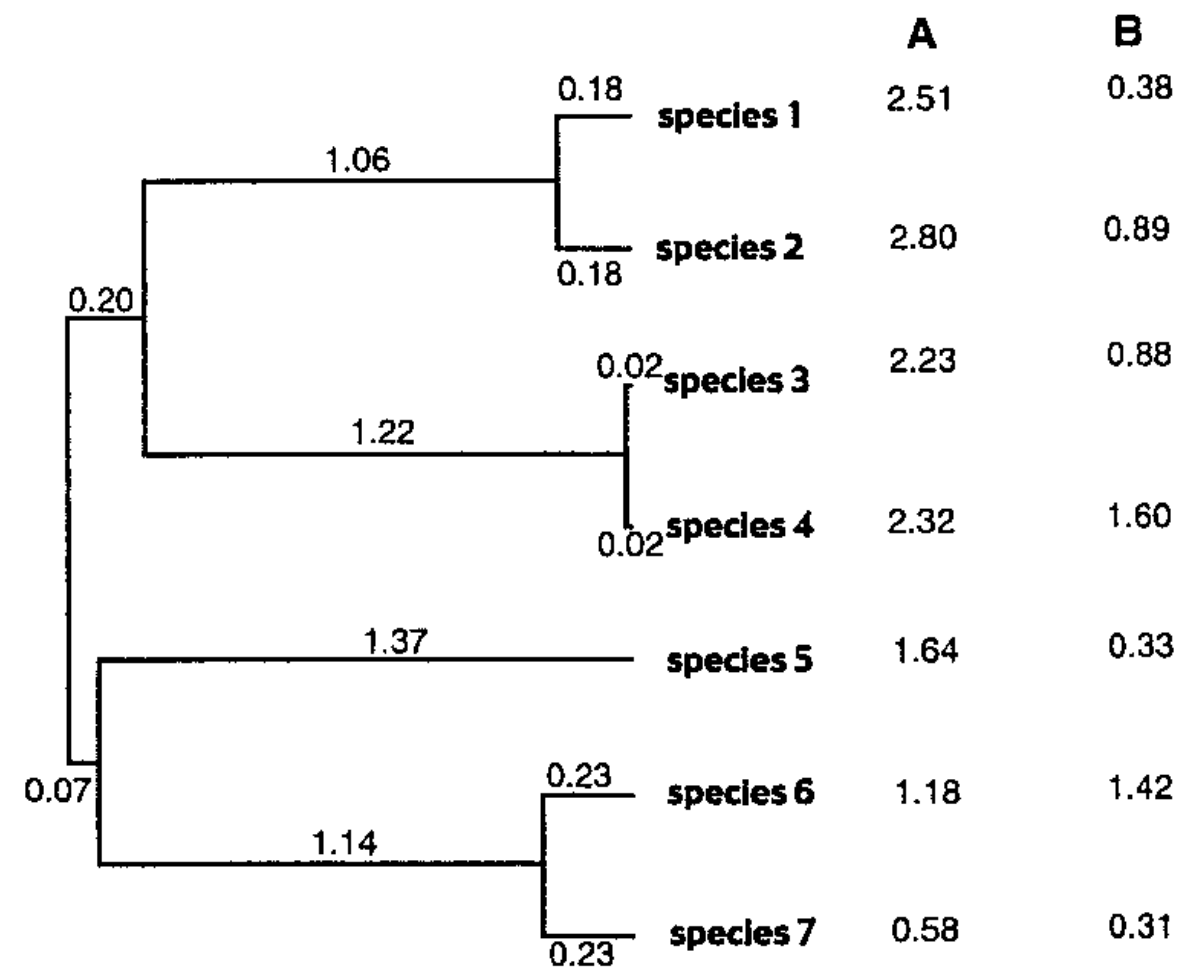
)

	species 1	species 2	species 3	species 4	species 5	species 6	species 7
species 1	1.44	1.26	0.20	0.20	0	0	0
species 2	1.26	1.44	0.20	0.20	0	0	0
species 3	0.20	0.20	1.44	1.42	0	0	0
species 4	0.20	0.20	1.44	1.44	0	0	0
species 5	0	0	0	0	1.44	0.07	0.07
species 6	0	0	0	0	0.07	1.44	1.21
species 7	0	0	0	0	0.07	1.21	1.44



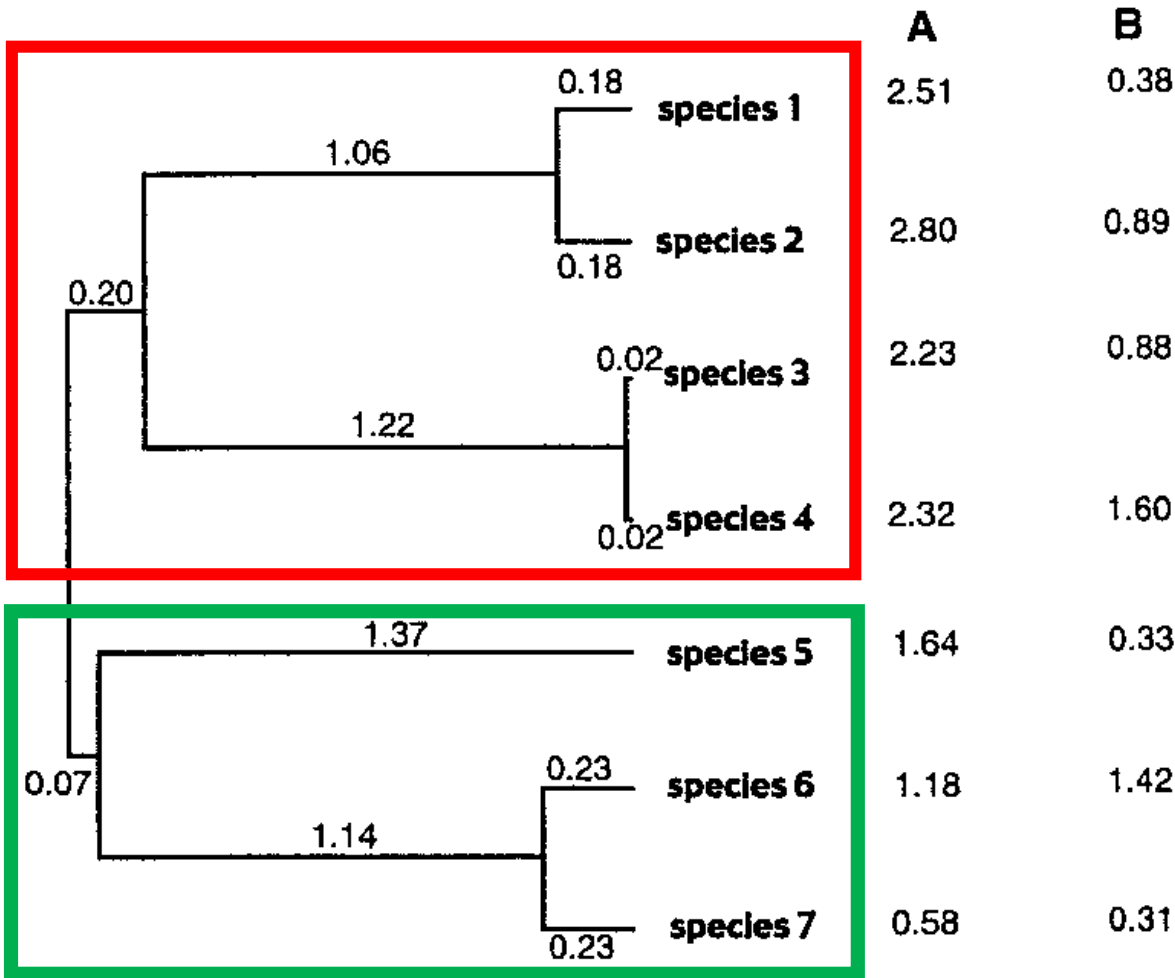
Matriz de variância-covariância

	species 1	species 2	species 3	species 4	species 5	species 6	species 7
species 1	var	COV	COV	COV	COV	COV	COV
species 2	COV	var	COV	COV	COV	COV	COV
species 3	COV	COV	var	COV	COV	COV	COV
species 4	COV	COV	COV	var	COV	COV	COV
species 5	COV	COV	COV	COV	var	COV	COV
species 6	COV	COV	COV	COV	COV	var	COV
species 7	COV	COV	COV	COV	COV	COV	var

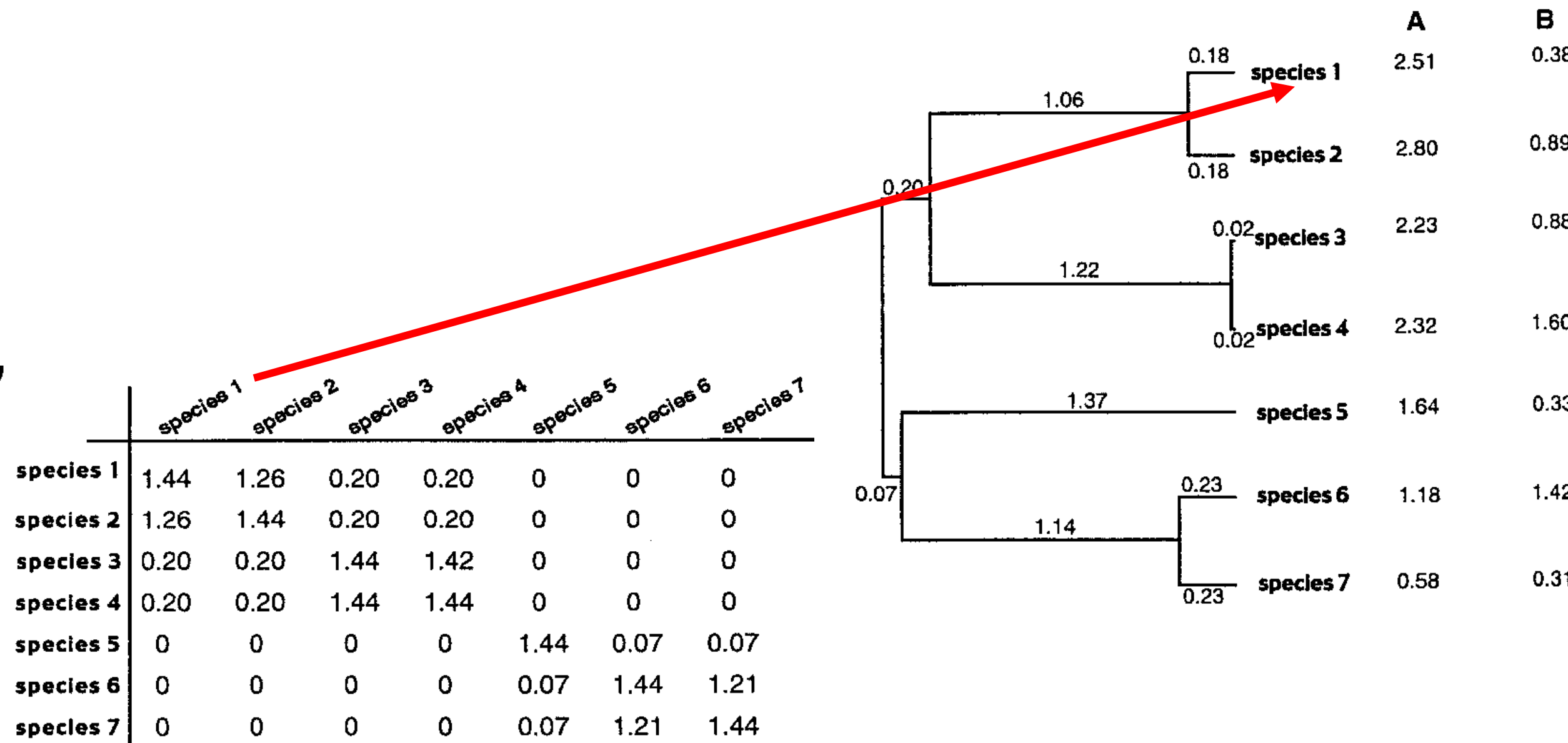


Matriz de variância-covariância

	species 1	species 2	species 3	species 4	species 5	species 6	species 7
species 1	1.44	1.26	0.20	0.20	0	0	0
species 2	1.26	1.44	0.20	0.20	0	0	0
species 3	0.20	0.20	1.44	1.42	0	0	0
species 4	0.20	0.20	1.44	1.44	0	0	0
species 5	0	0	0	0	1.44	0.07	0.07
species 6	0	0	0	0	0.07	1.44	1.21
species 7	0	0	0	0	0.07	1.21	1.44



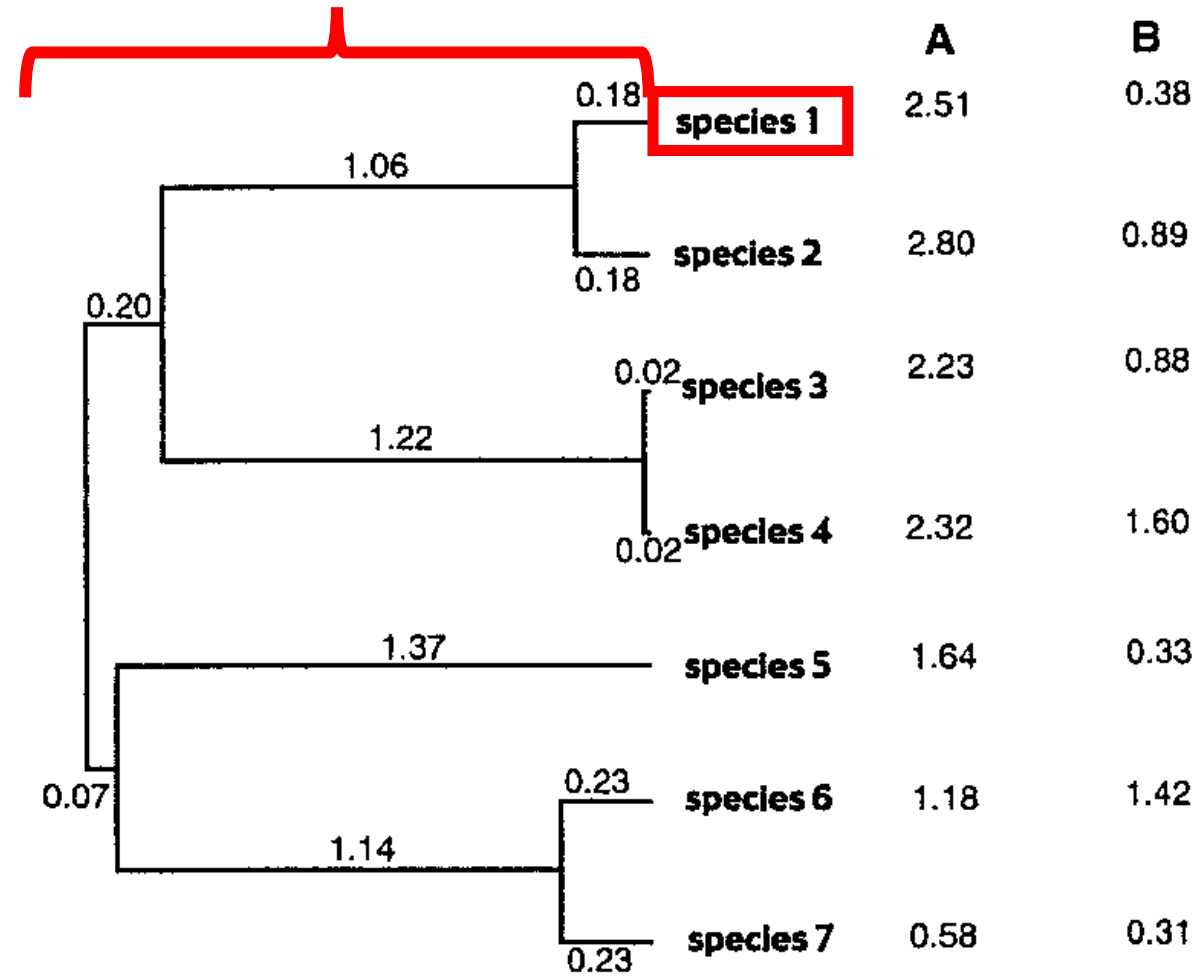
Matriz de variância-covariância



Matriz de variância-covariância

$$\text{var}(\text{species1}) = 0.18 + 1.06 + 0.20 = 1.44$$

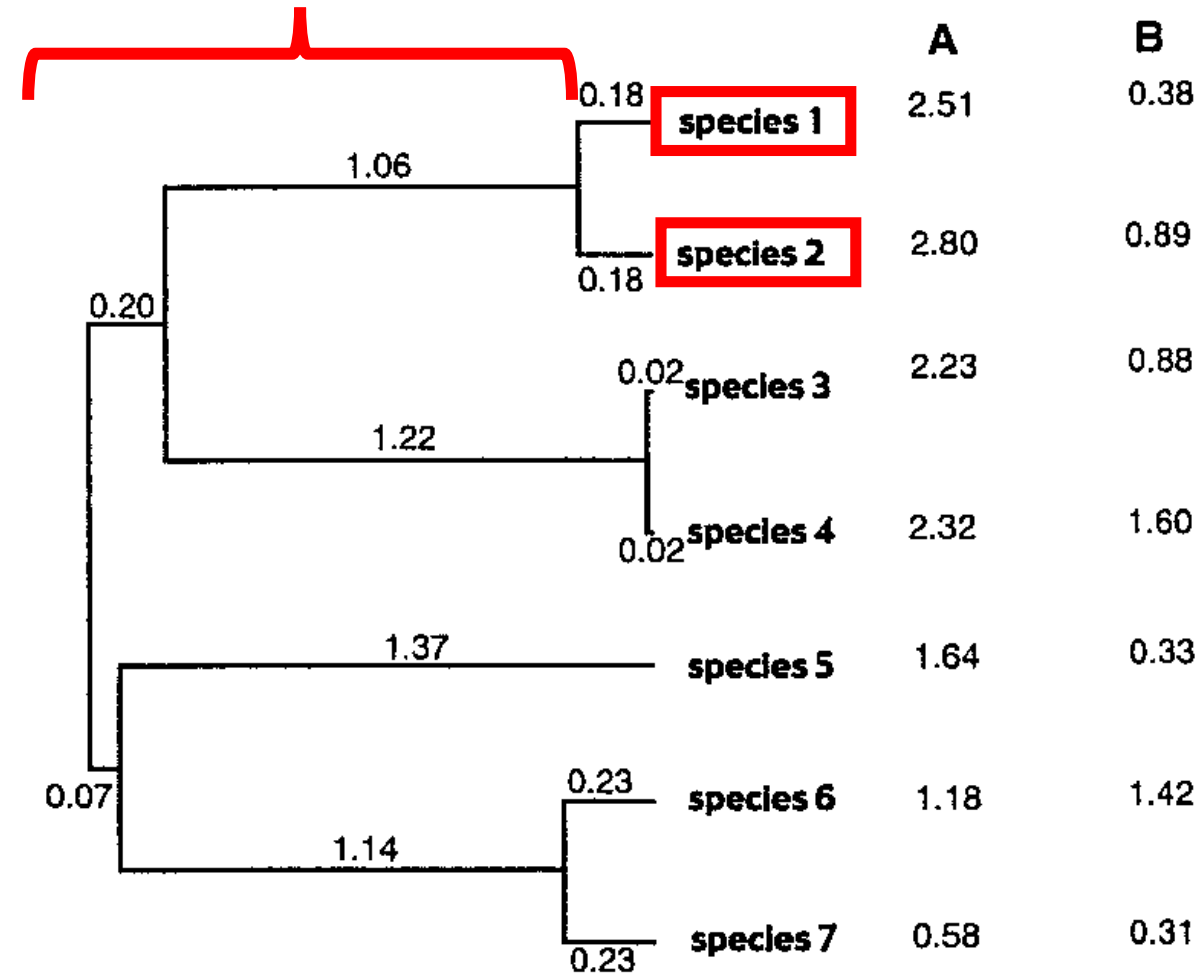
	species 1	species 2	species 3	species 4	species 5	species 6	species 7
species 1	1.44	1.26	0.20	0.20	0	0	0
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species 4	0.20	0.20	1.44	1.44	0	0	0
species 5	0	0	0	0	1.44	0.07	0.07
species 6	0	0	0	0	0.07	1.44	1.21
species 7	0	0	0	0	0.07	1.21	1.44



Matriz de variância-covariância

$$\text{cov}(1,2)=1.06+0.20=1.26$$

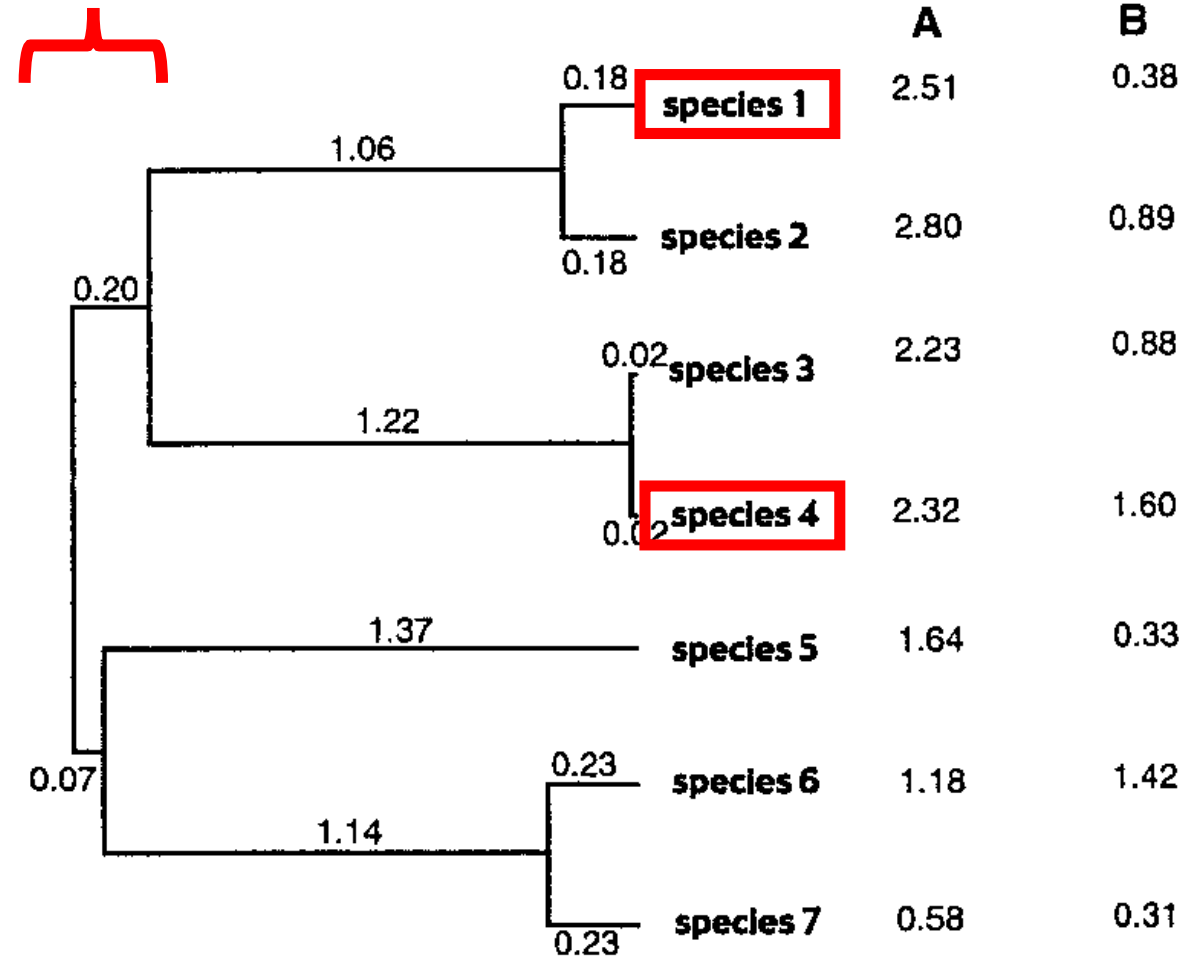
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species 5	0	0	0	0	1.44	0.07	0.07
species 6	0	0	0	0	0.07	1.44	1.21
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Matriz de variância-covariância

$cov(1,4)=0.20$

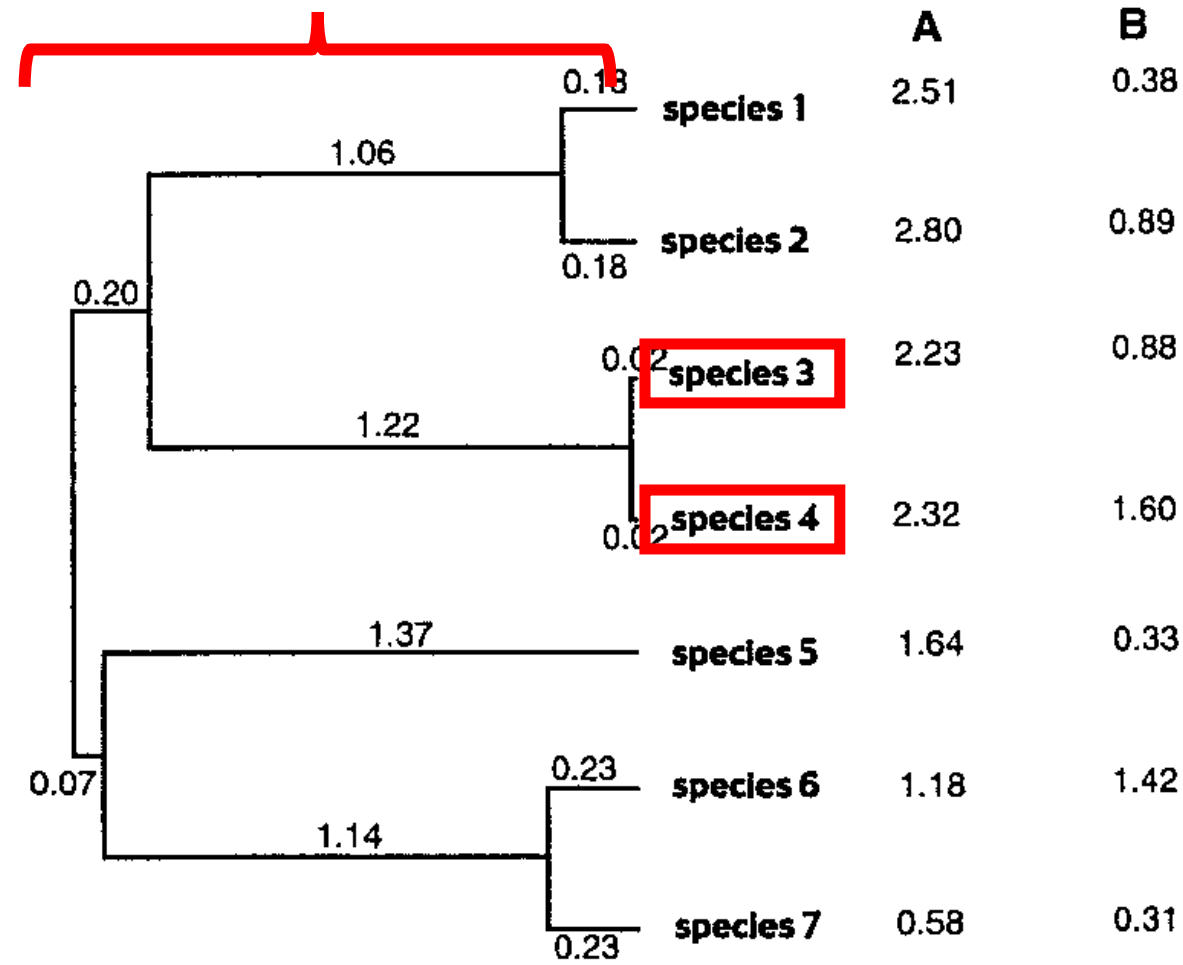
	species 1	species 2	species 3	species 4	species 5	species 6	species 7
species 1	1.44	1.26	0.20	0.20	0	0	0
species 2	1.26	1.44	0.20	0.20	0	0	0
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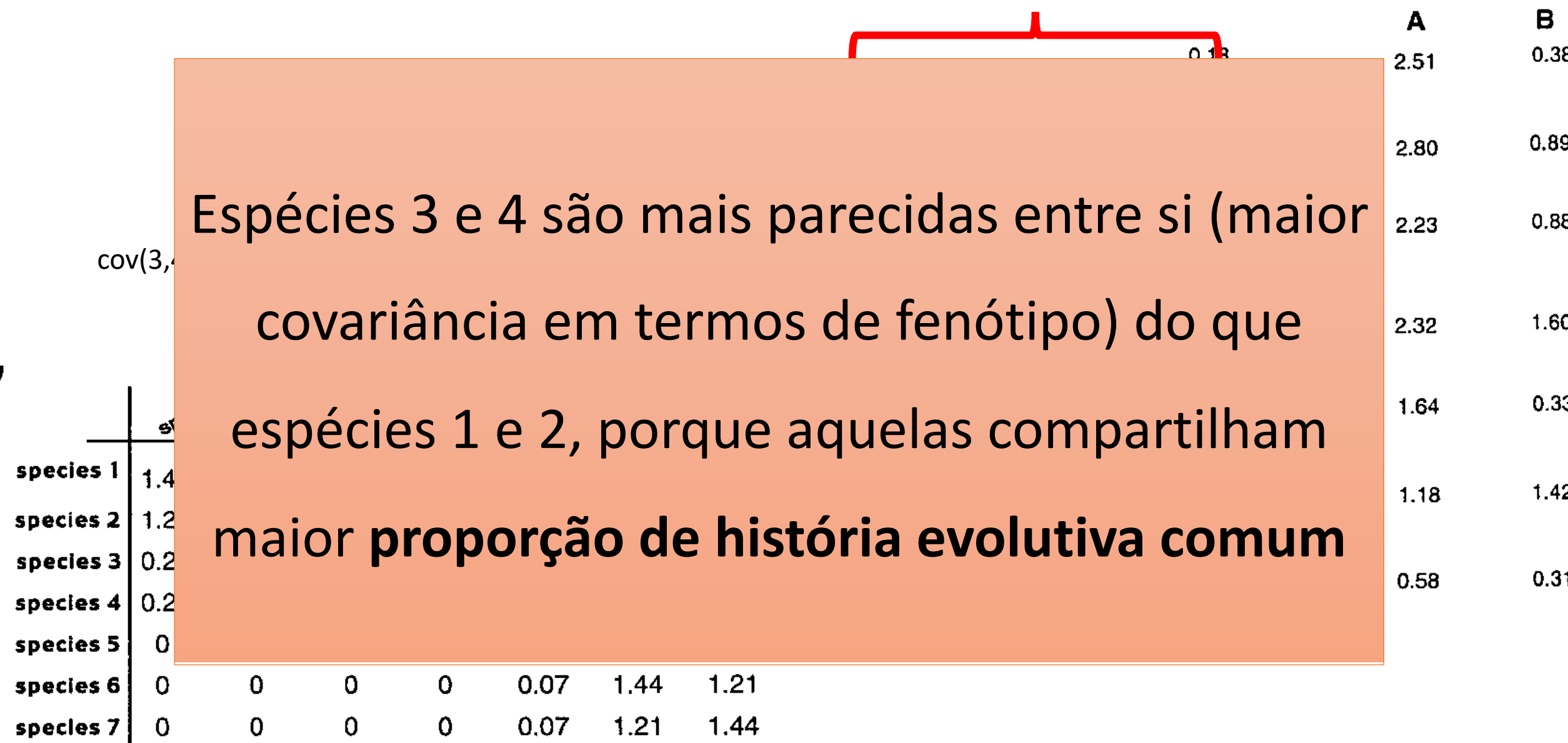
Matriz de variância-covariância

$$\text{cov}(3,4)=0.20+1.22+0.02=1.44$$

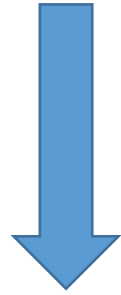
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species 4	0.20	0.20	1.44	1.44	0	0	0
species 5	0	0	0	0	1.44	0.07	0.07
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Matriz de variância-covariância



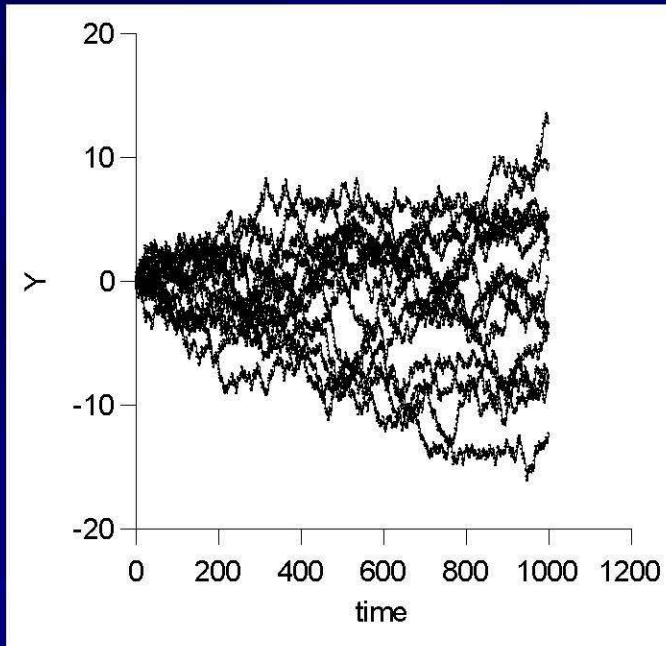
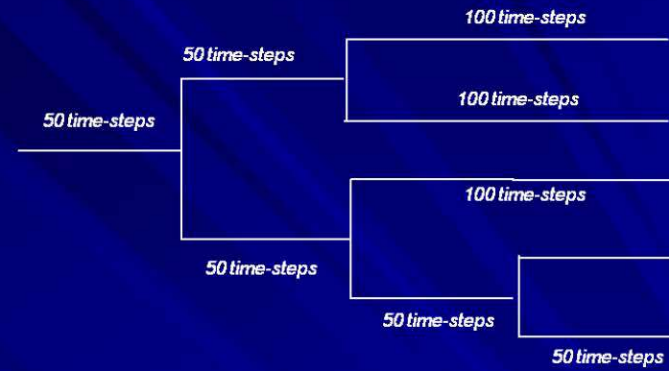
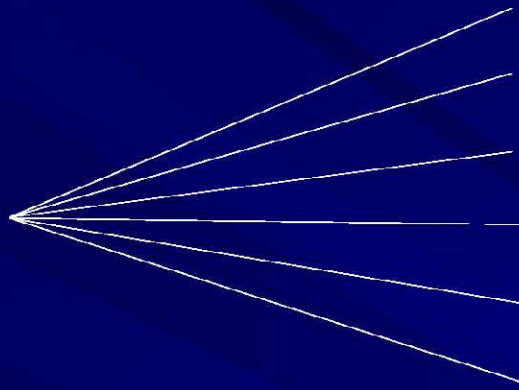
A mesma filogenia
pode gerar diferentes
matrizes VCV



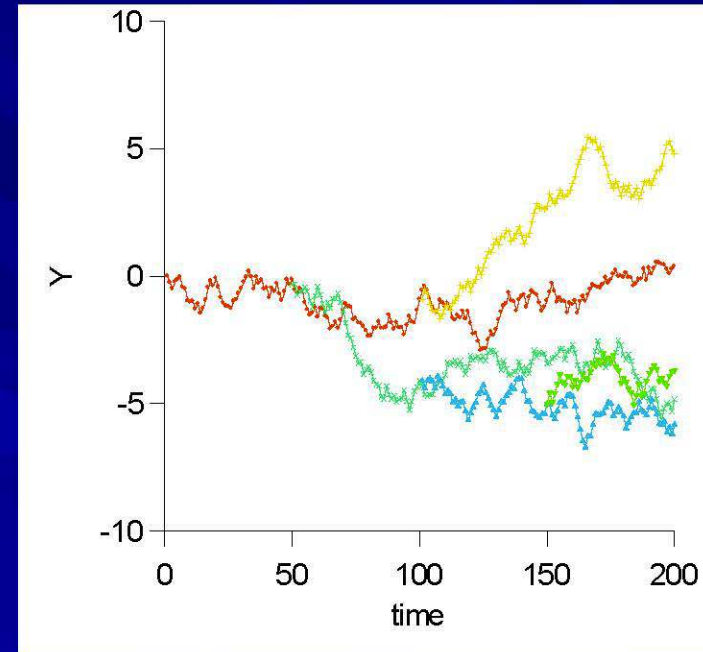
Modelos evolutivos

Como a filogenia influencia a evolução de fenótipos sob um processo BM?

O parentesco comum aumenta ou diminui a similaridade entre espécies em termos de fenótipo?



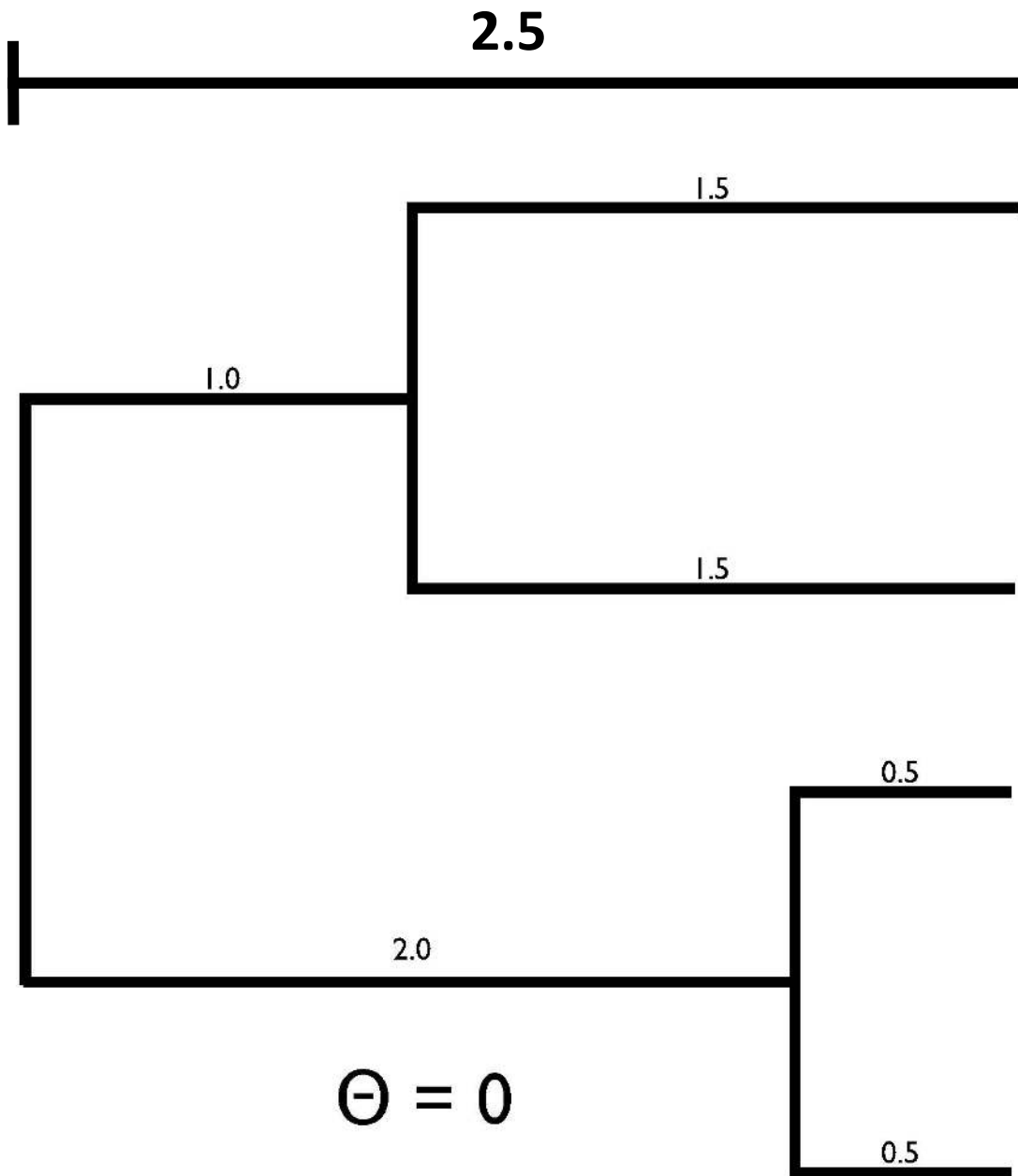
Here we assumed that species are *INDEPENDENT* (the started all at the root)



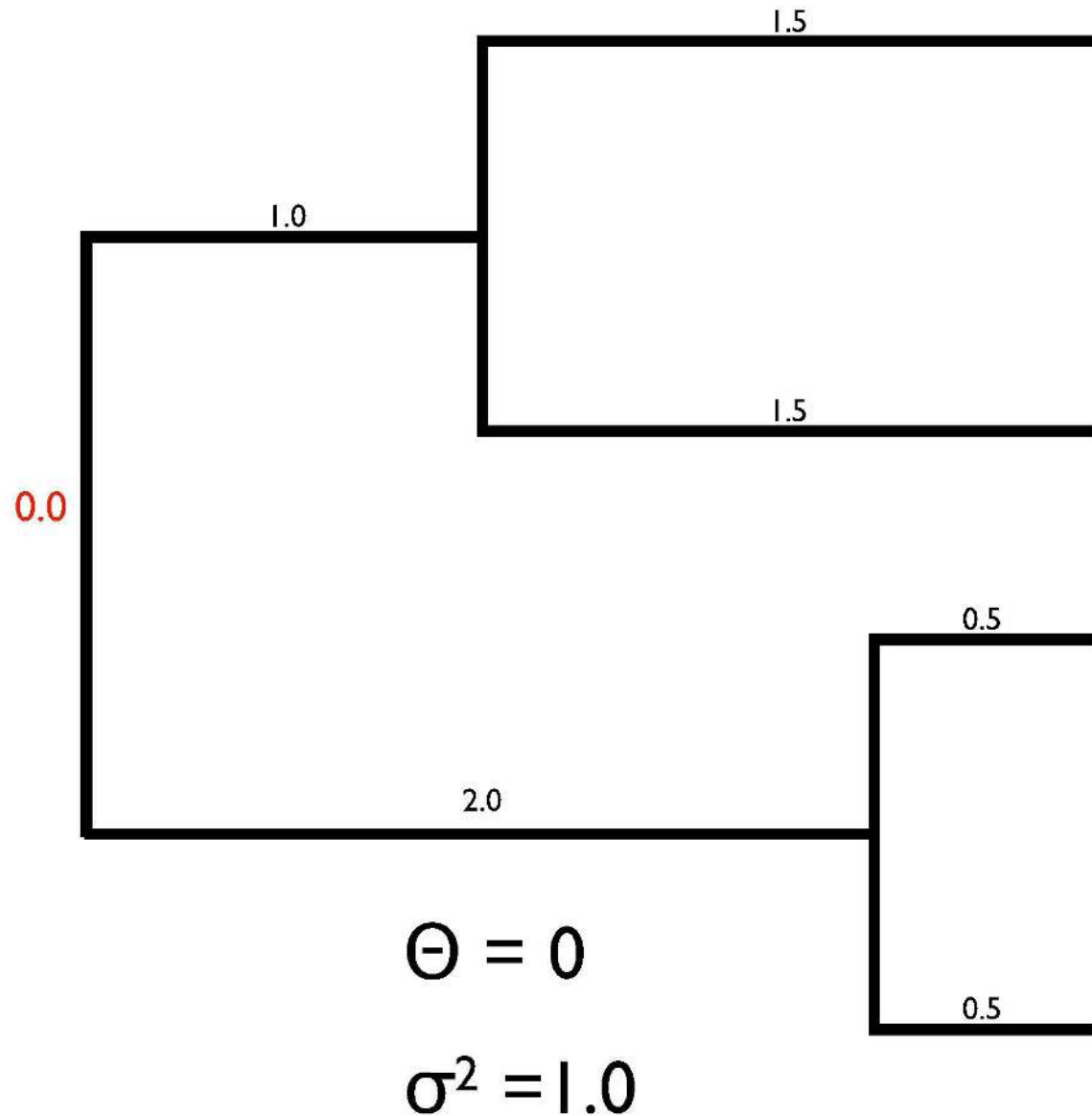
Here species are *PHYLOGENETICALLY STRUCTURED*

Simulando um processo BM

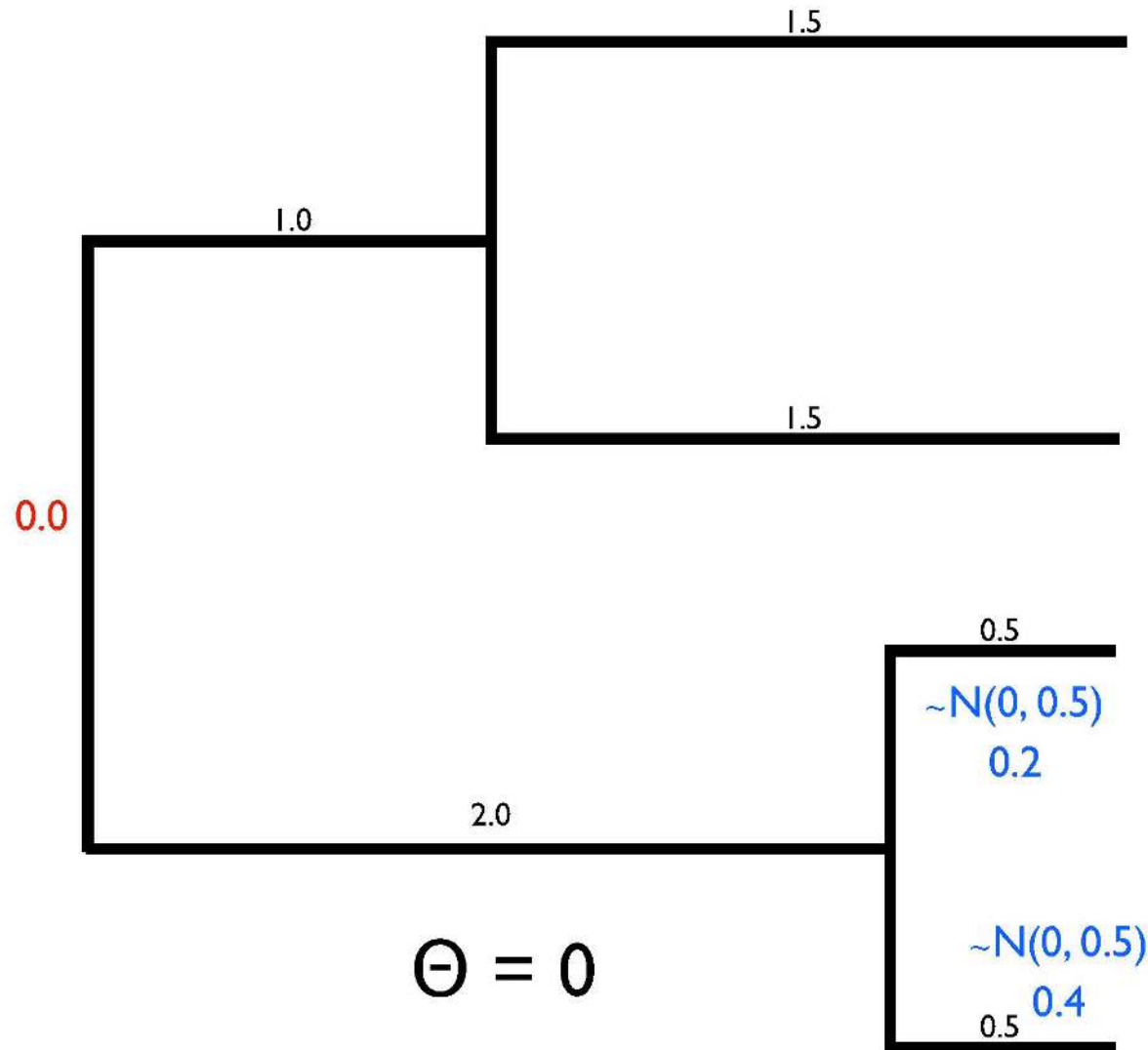
- Sortear valores de uma distribuição normal
- Variância depende do σ^2 e t
- Valores dos ramos adjacentes são adicionados da raíz para os tips da filogenia



1. Set root state
2. Draw random normal deviate for each branch
3. Add along path from root to each tip to get tip values



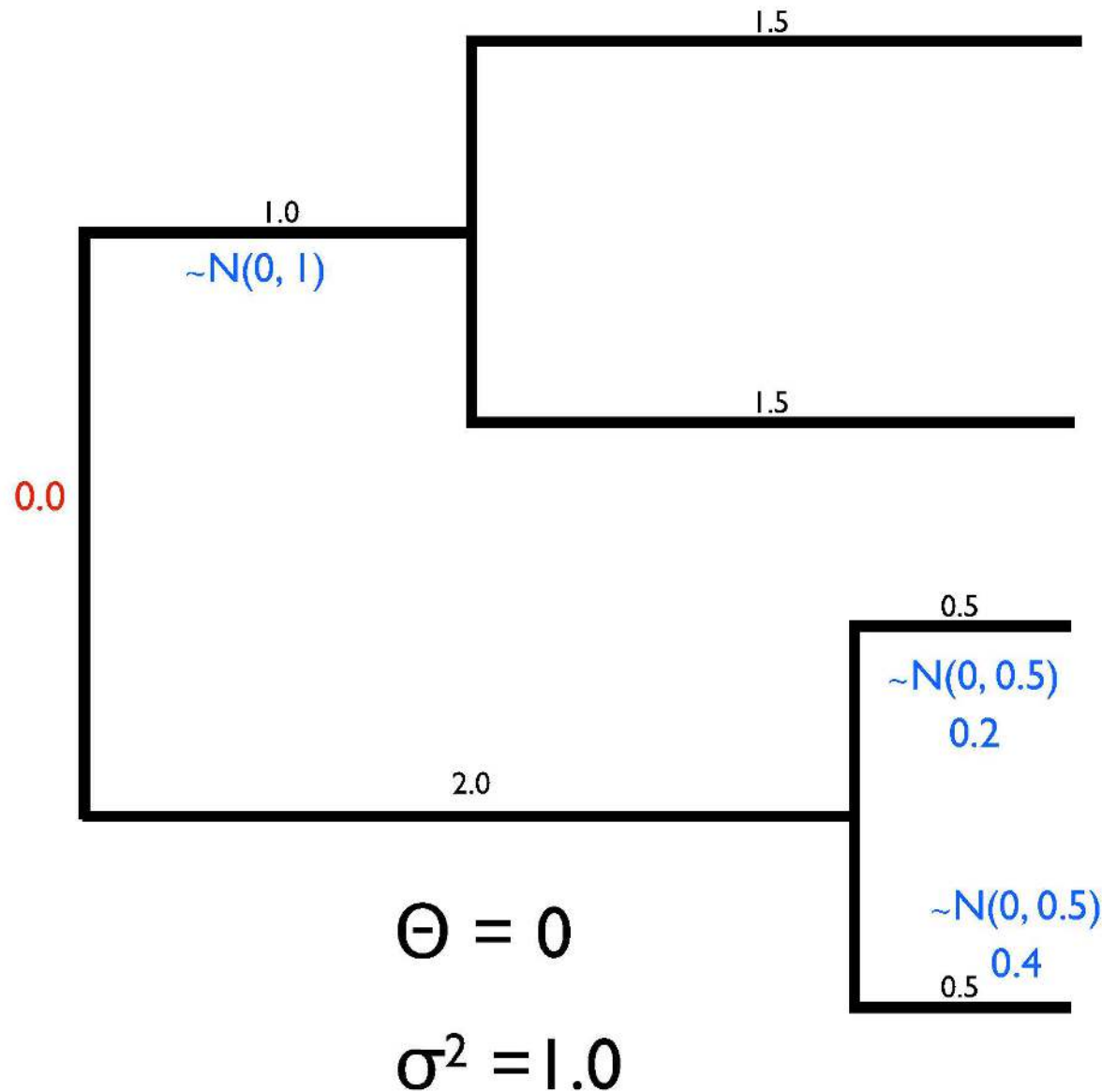
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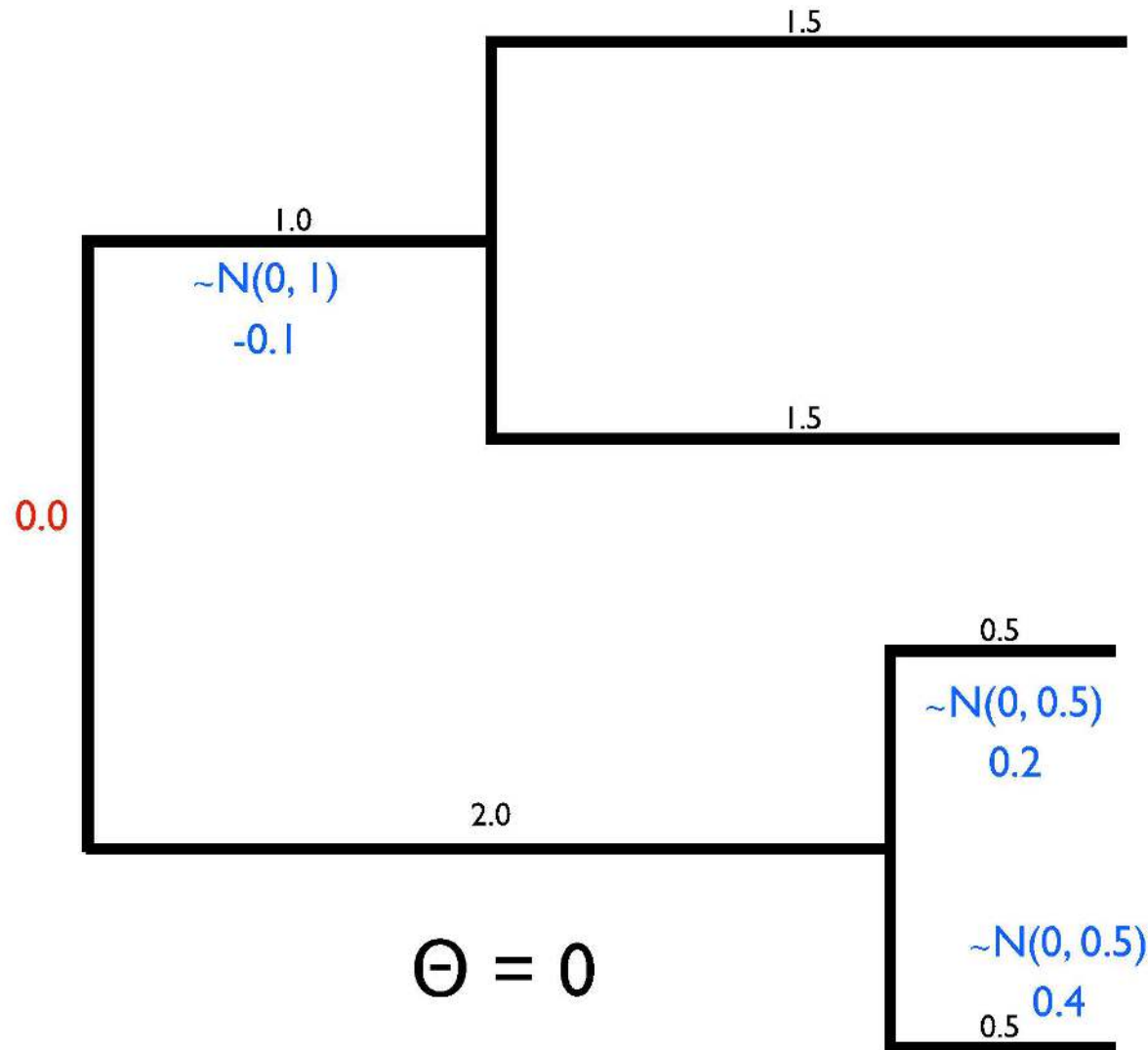
$$\Theta = 0$$

$$\sigma^2 = 1.0$$

1. Set root state
2. Draw random normal deviate for each branch
3. Add along path from root to each tip to get tip values



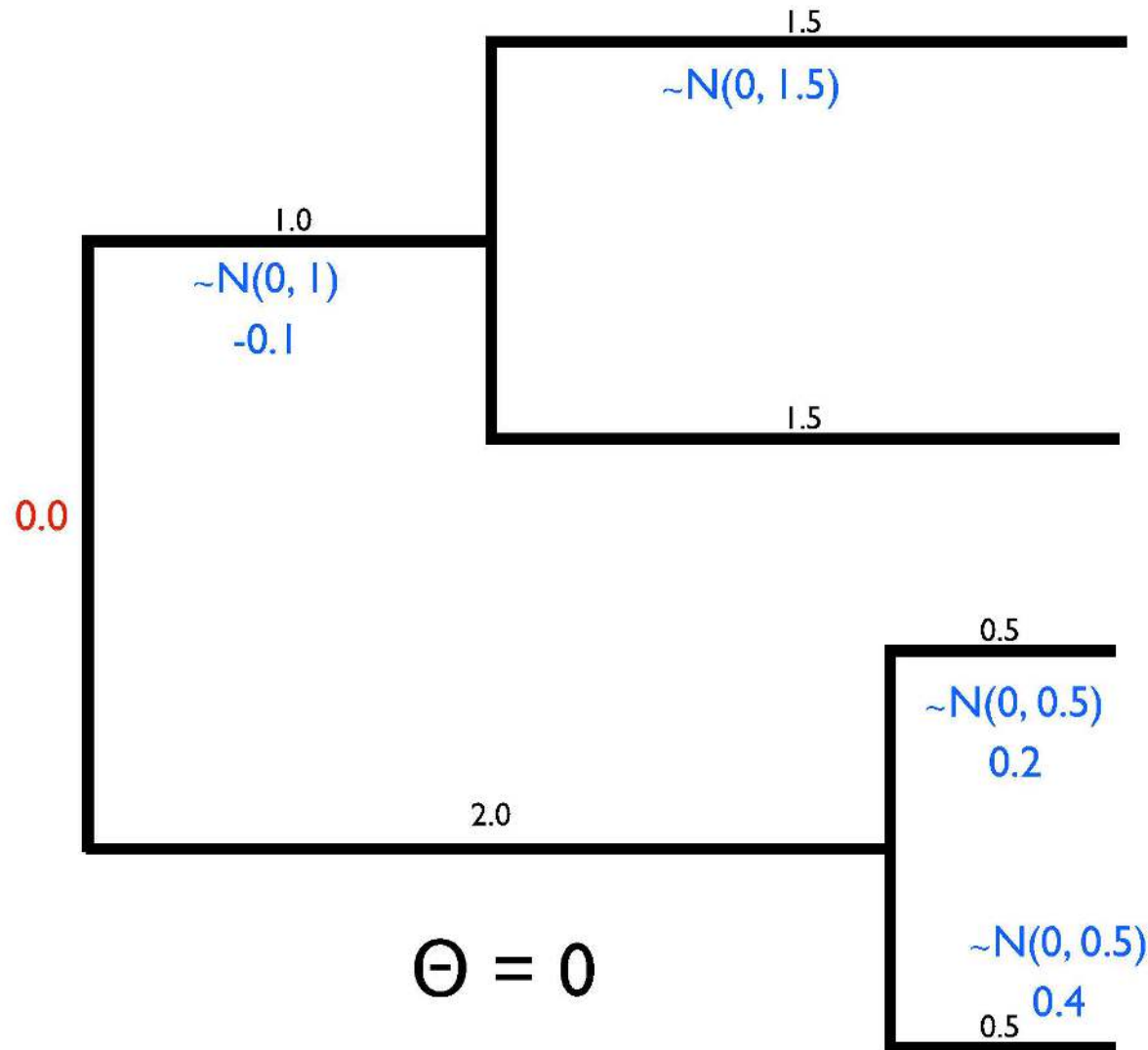
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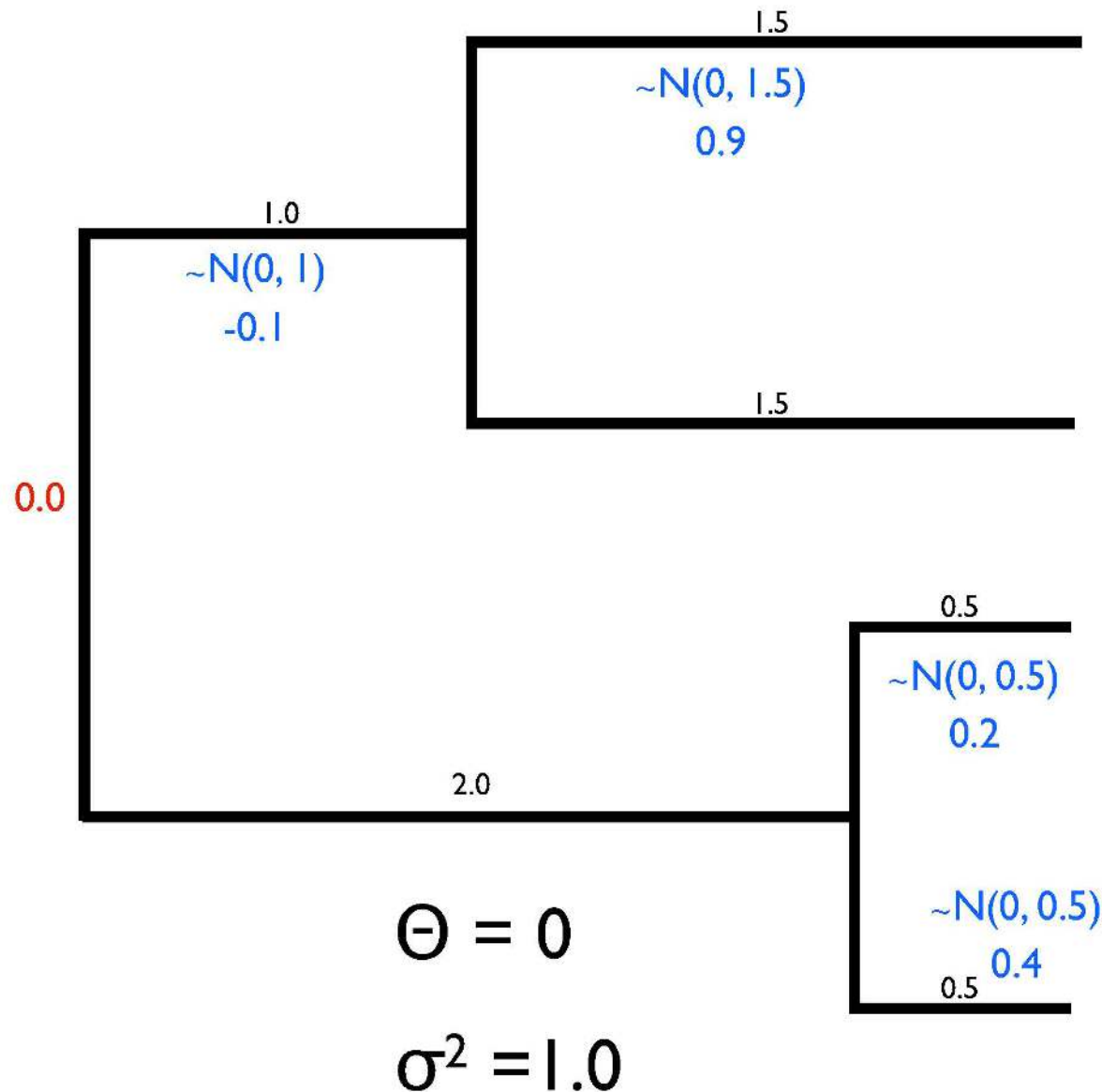
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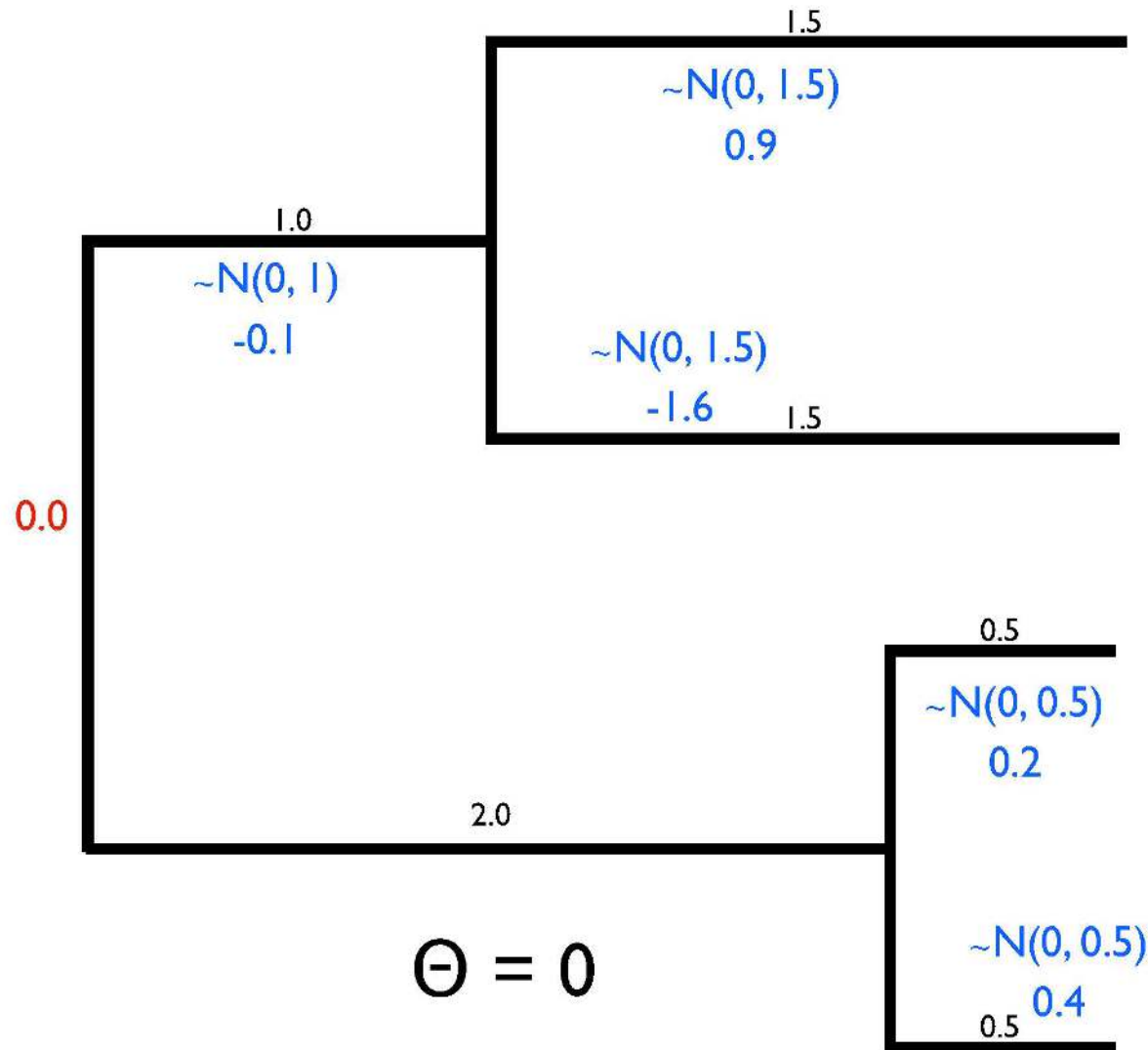
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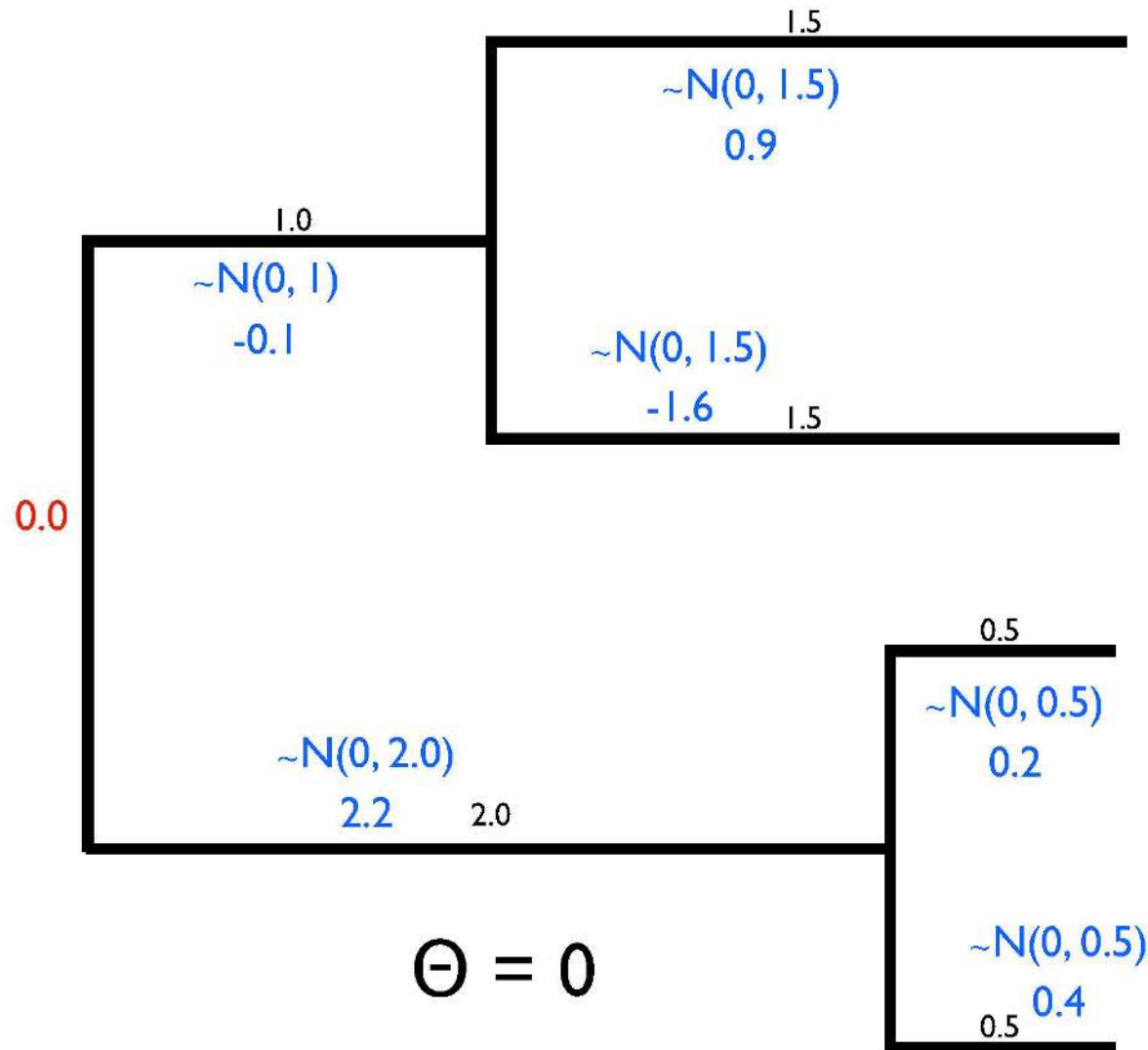
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$$\sigma^2 = 1.0$$

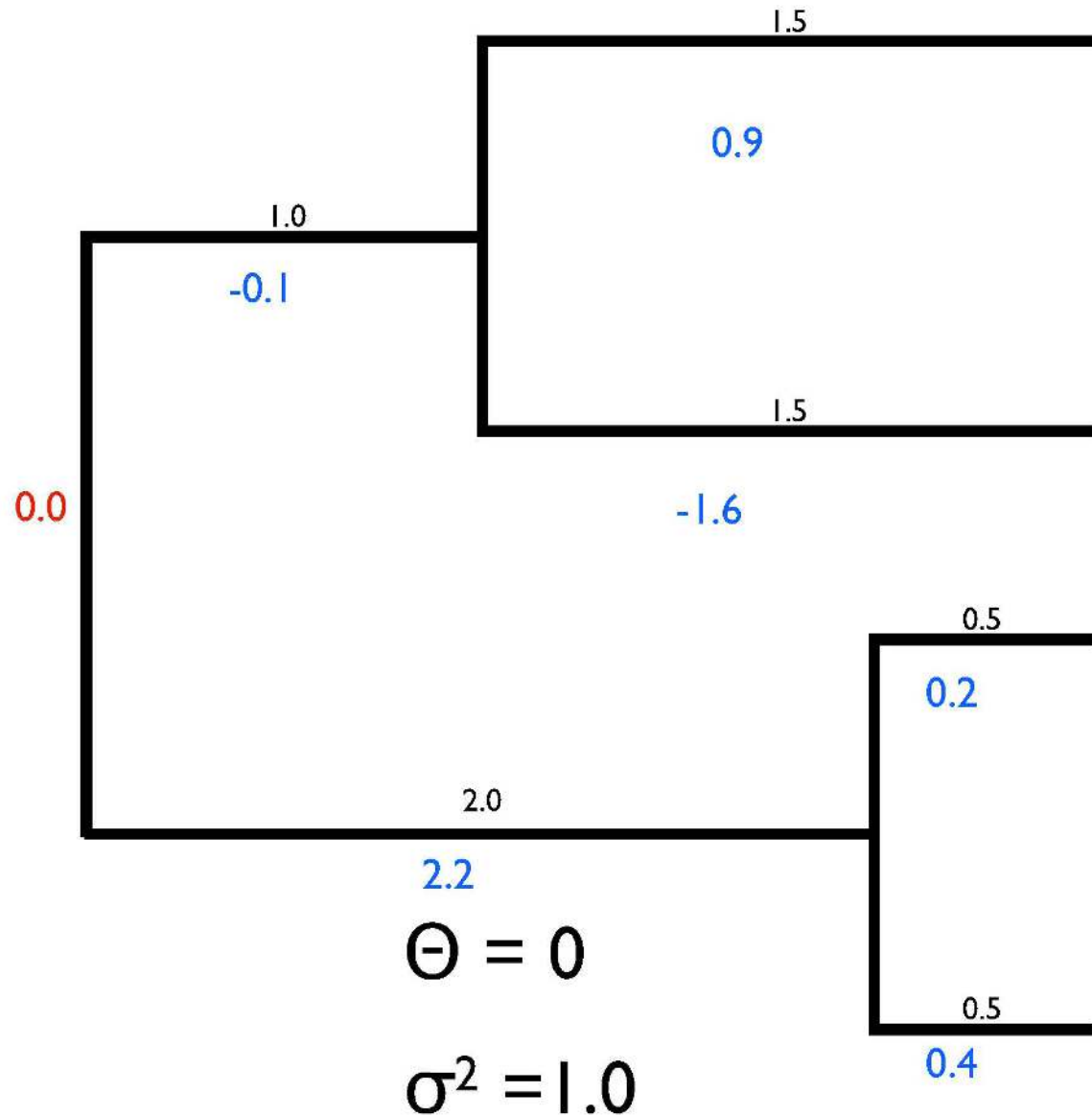
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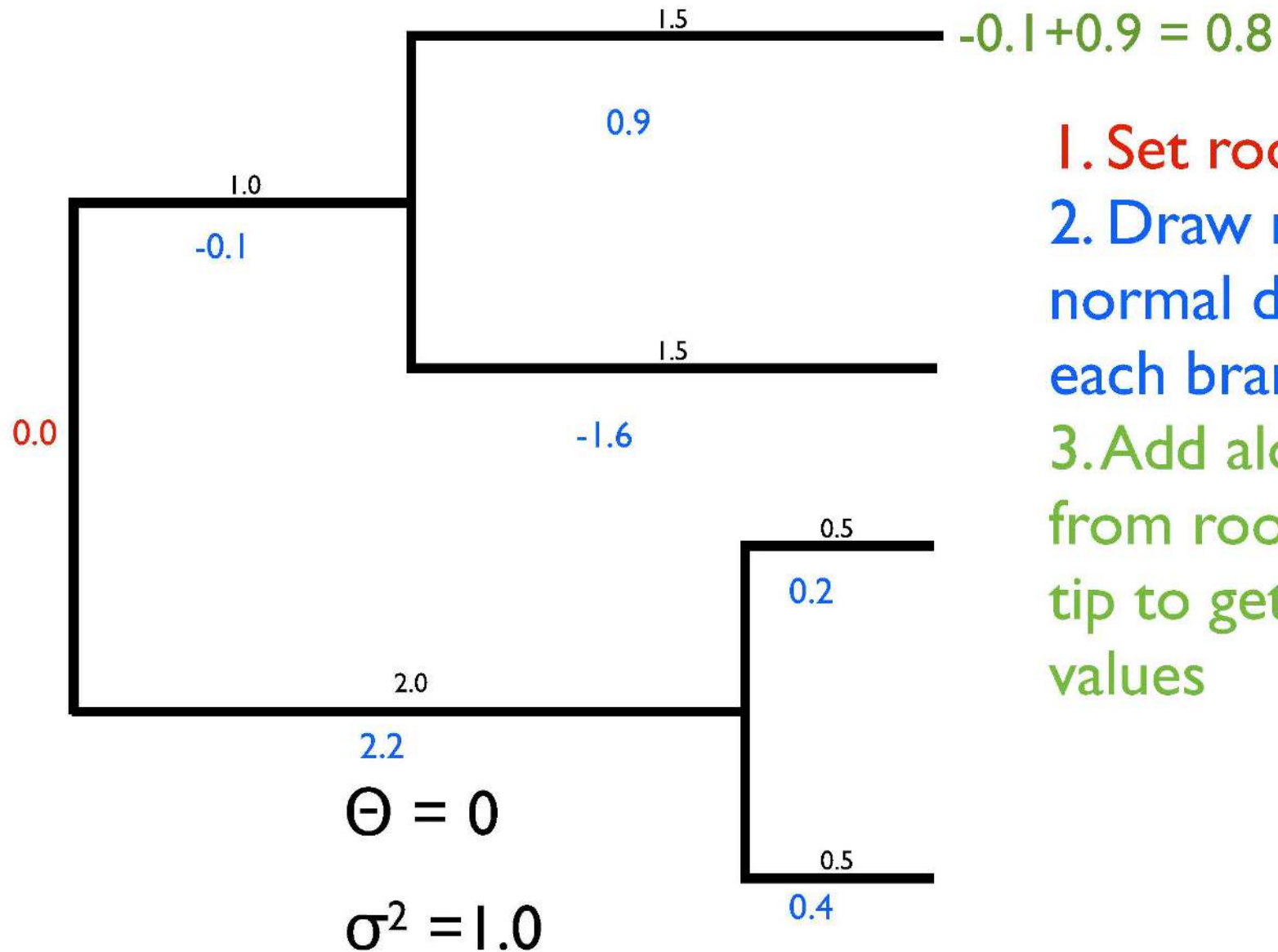
$$\Theta = 0$$

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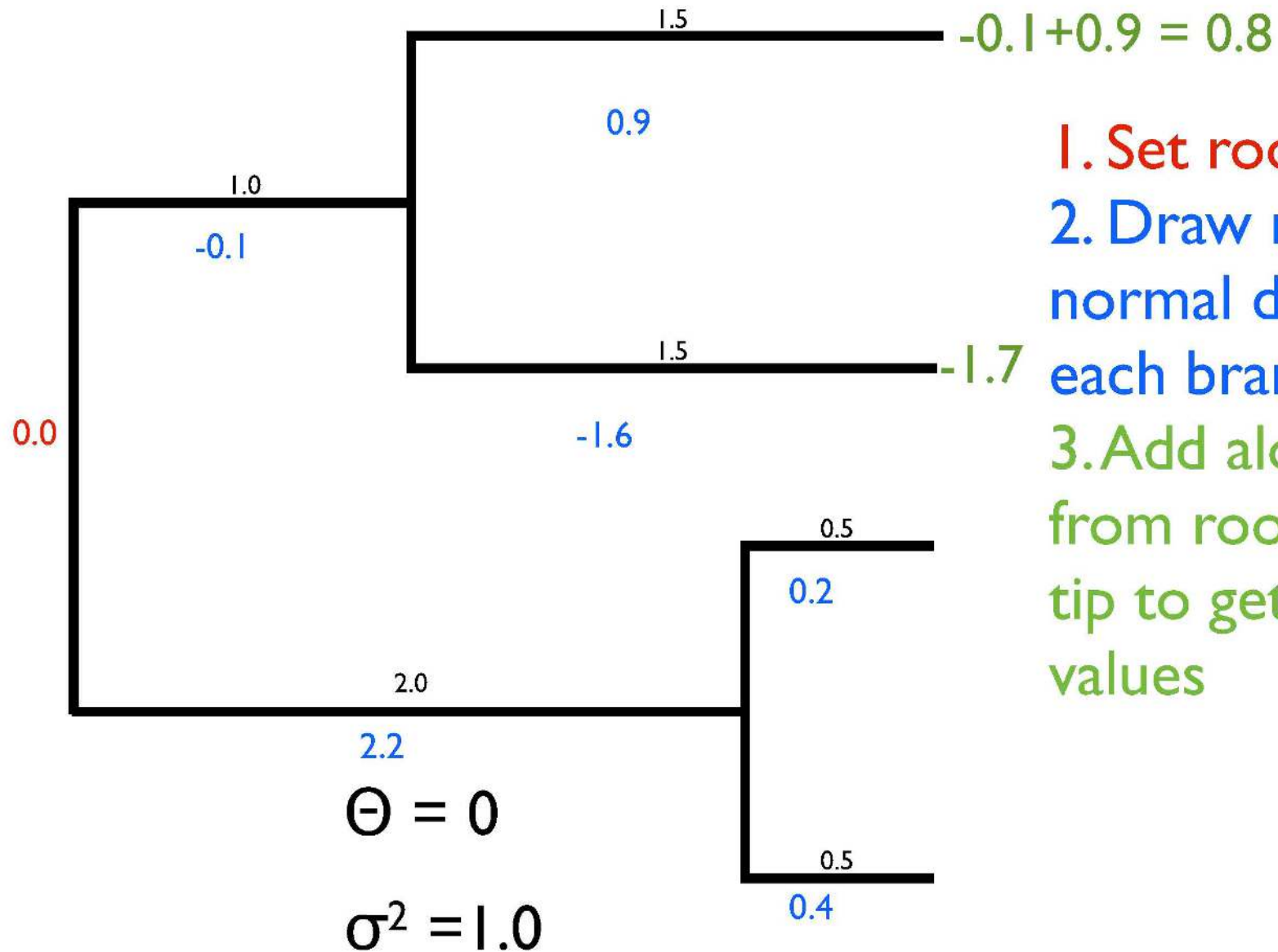
1. Set root state
2. Draw random normal deviate for each branch
3. Add along path from root to each tip to get tip values



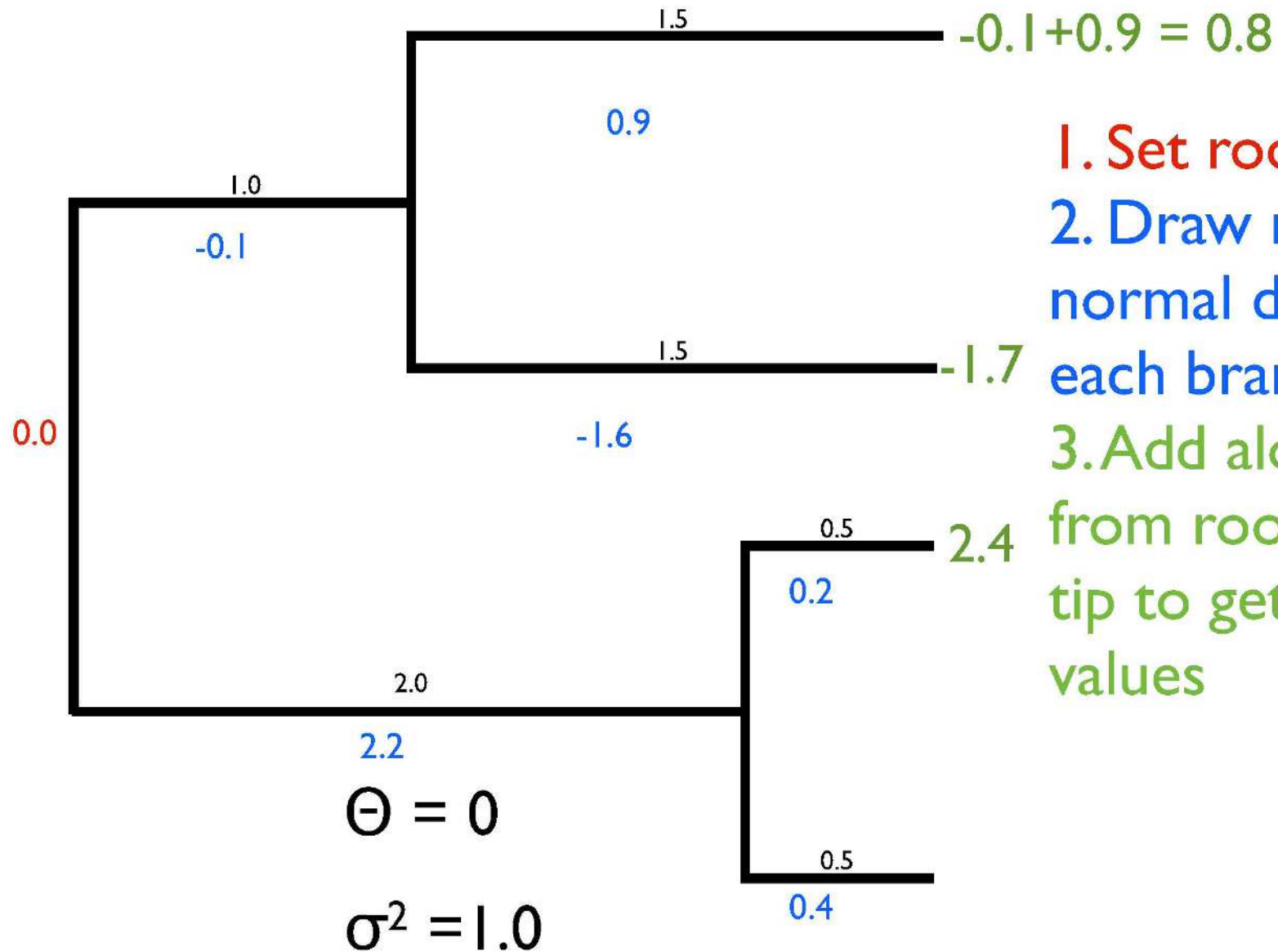
1. Set root state
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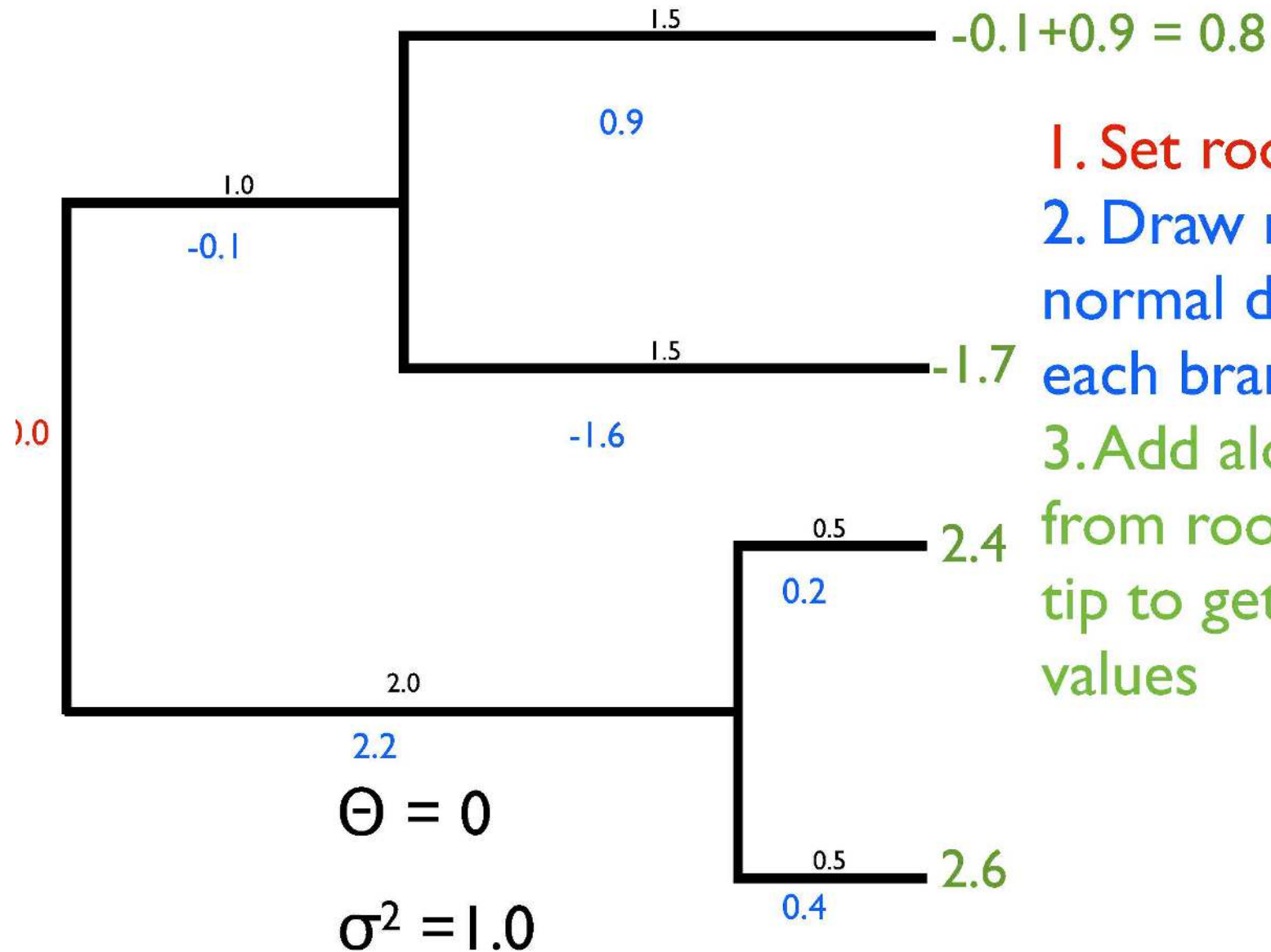
1. Set root state
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1. Set root state
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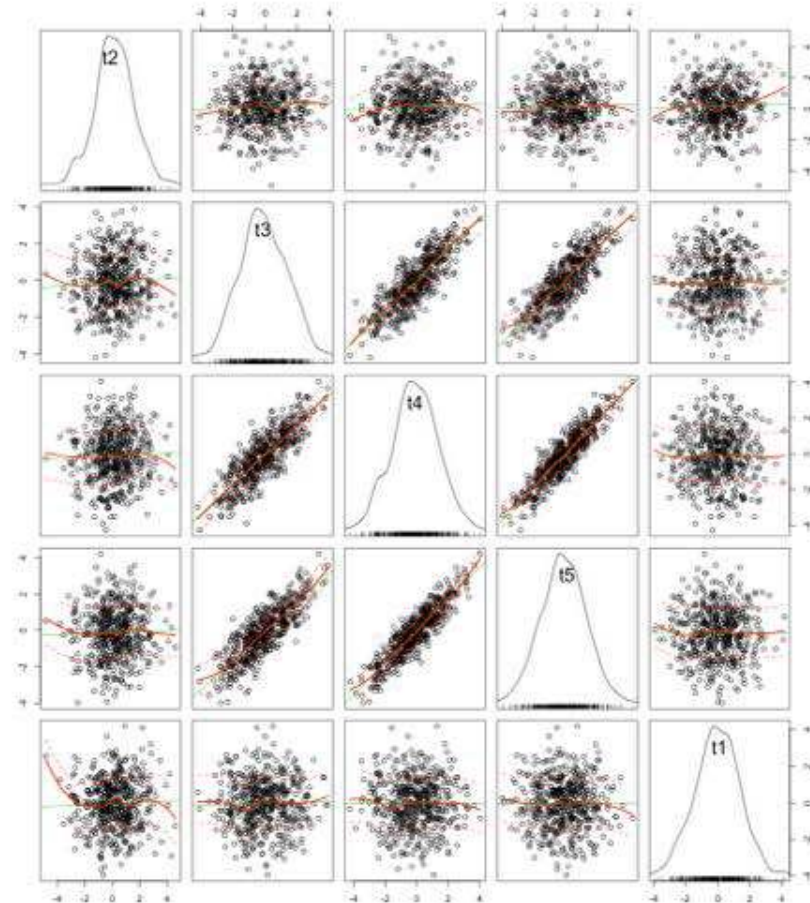
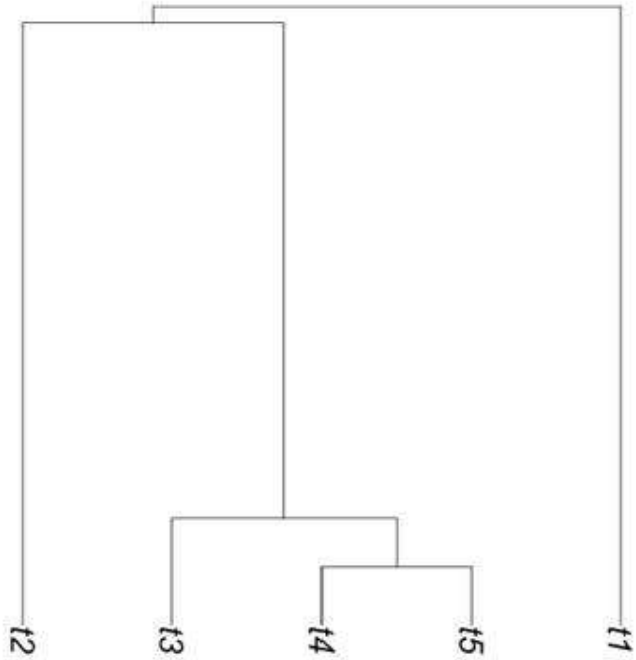


1. Set root state
2. Draw random normal deviate for each branch
3. Add along path from root to each tip to get tip values



1. Set root state
2. Draw random normal deviate for each branch
3. Add along path from root to each tip to get tip values

shared ancestry creates covariation across tips





YES!

**AGORA BORA BOTAR A MÃO NA
MASSA!**

Ornstein-Uhlenbeck (OU)

- Captura a *idéia* de evolução estabilizadora
 - Valor de um caracter contínuo tem uma tendência a se mover para um valor médio
- Movimento Browniano com um "salto"
- Tem 3 parâmetros
 - Valor inicial (θ)
 - Taxa (σ^2)
 - Parâmetro de restrição (α)



Leonard Ornstein

Ornstein-Uhlenbeck



George Uhlenbeck

- Physics-based model of Brownian motion under friction
- also referred to as mean reverting
- can be modeled with attraction to a different mean than the starting state

**species evolving towards adaptive peaks
that may impose limits on trait diversification**

Ornstein-Uhlenbeck (OU)

$$dX_{(t)} = \alpha[\Theta - X_{(t)}]dt + \sigma dB_{(t)}$$

Ornstein-Uhlenbeck (OU)

$$dX_{(t)} = \alpha[\Theta - X_{(t)}]dt + \sigma dB_{(t)}$$

brownian
motion

Ornstein-Uhlenbeck (OU)

$$dX_{(t)} = \alpha[\Theta - X_{(t)}]dt + \sigma dB_{(t)}$$

change
towards
optimum

brownian
motion

Ornstein-Uhlenbeck (OU)

$$dX_{(t)} = \alpha[\Theta - X_{(t)}]dt + \sigma dB_{(t)}$$

↑
optimal value

Ornstein-Uhlenbeck (OU)

$$dX_{(t)} = \alpha[\Theta - X_{(t)}]dt + \sigma dB_{(t)}$$

pull towards "optimum"



Ornstein-Uhlenbeck (OU)

$$dX_{(t)} = \alpha[\Theta - X_{(t)}]dt + \sigma dB_{(t)}$$

strength of selection is
proportional to
distance of trait from
optimal value

Quanto maior a distância entre
valor do trait e o ótimo



Maior o "puxão"
em direção ao ótimo

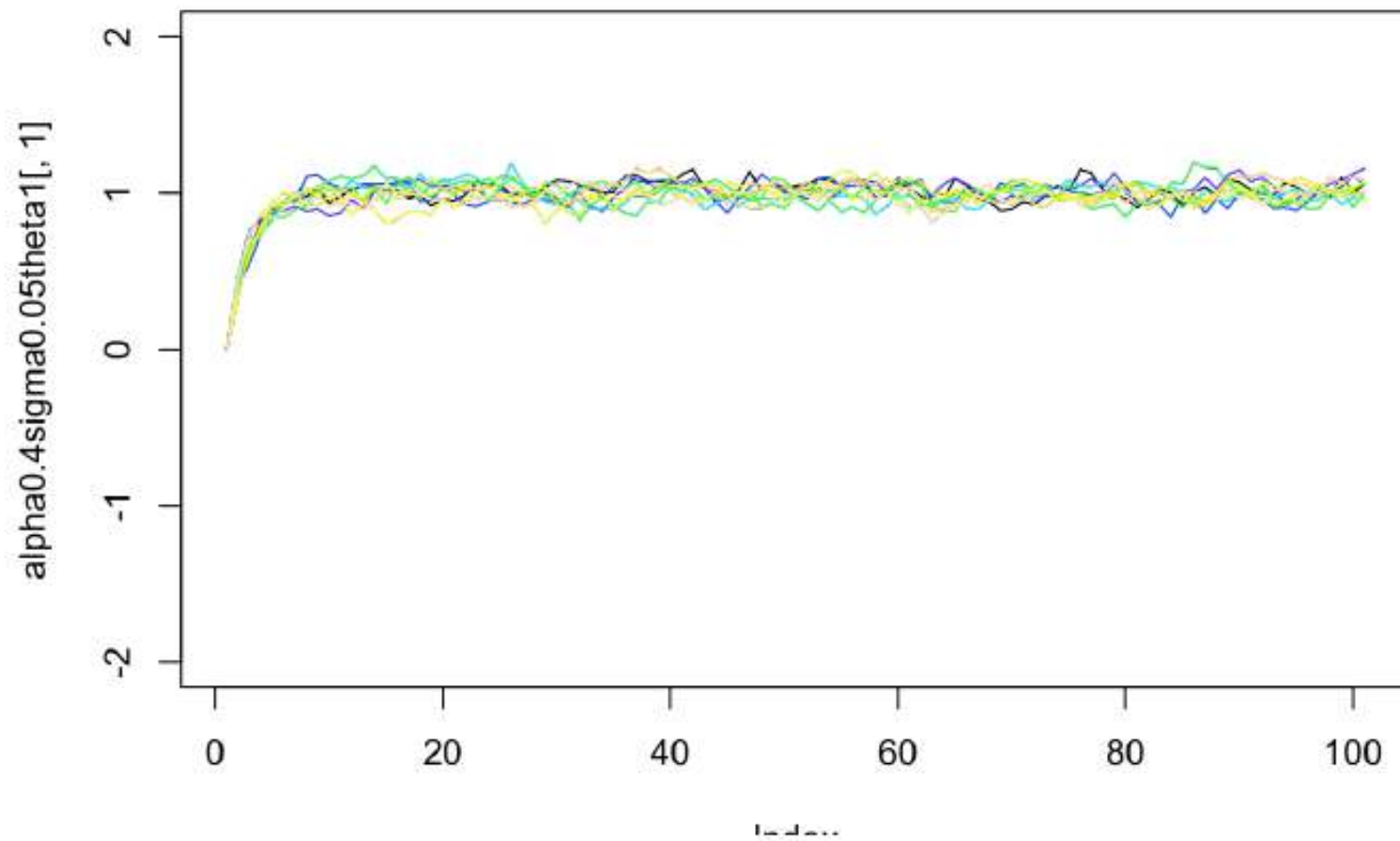
Ornstein-Uhlenbeck (OU)

$$dX_{(t)} = \alpha[\Theta - X_{(t)}]dt + \sigma dB_{(t)}$$

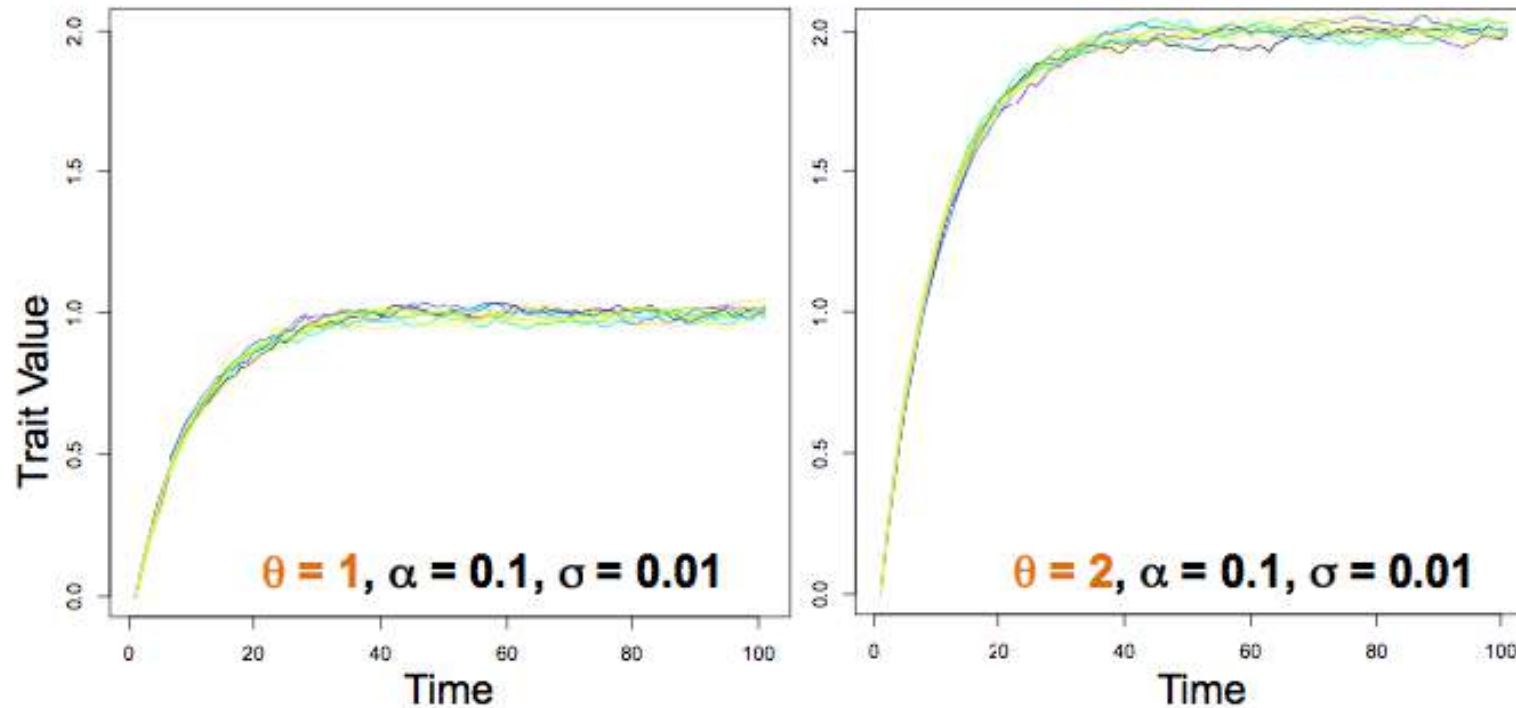
when alpha is 0, OU
becomes BM

OU evolution

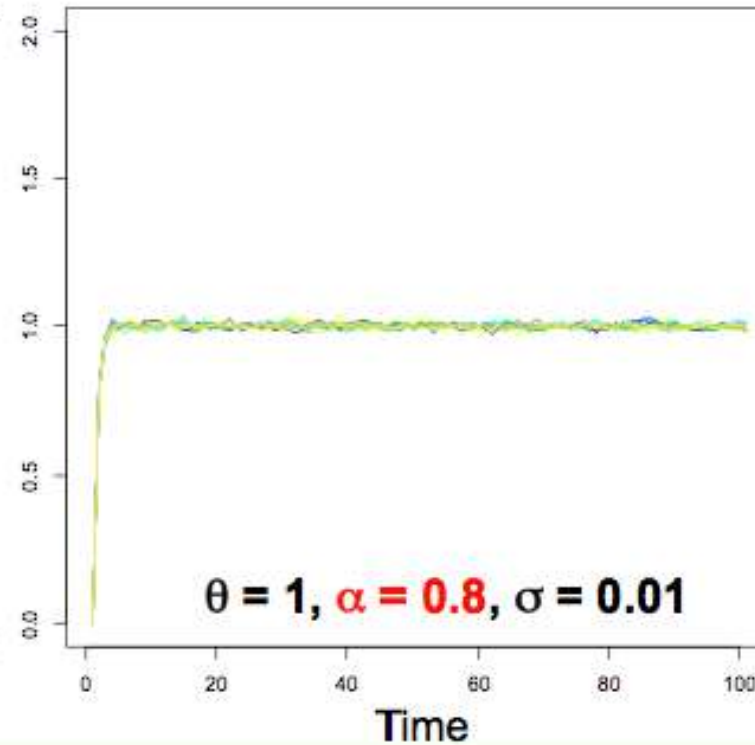
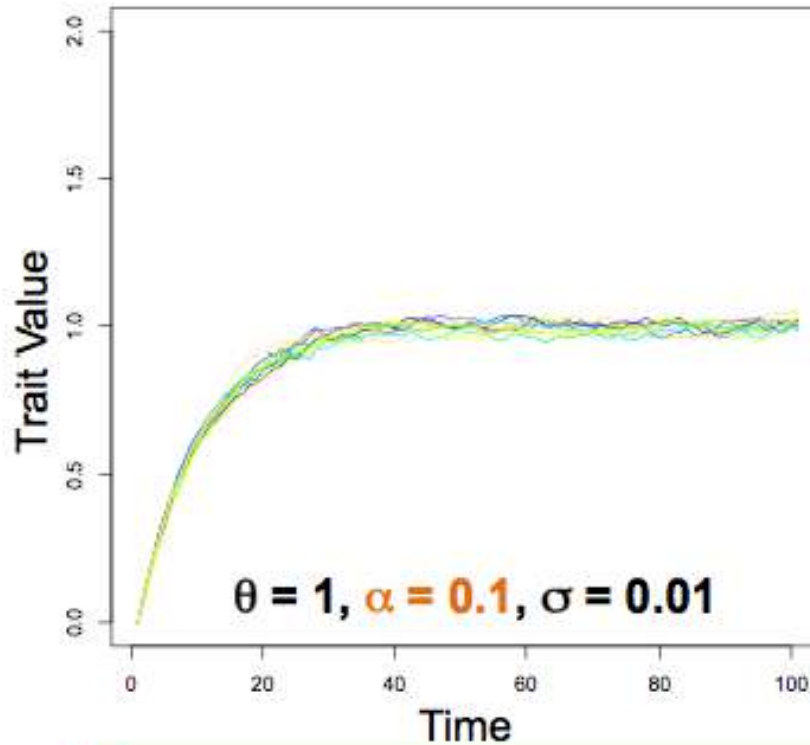
$\alpha 0.4$ $\sigma=0.05$



$$dX_{(t)} = \alpha[\theta - X_{(t)}]dt + \sigma dB_{(t)}$$

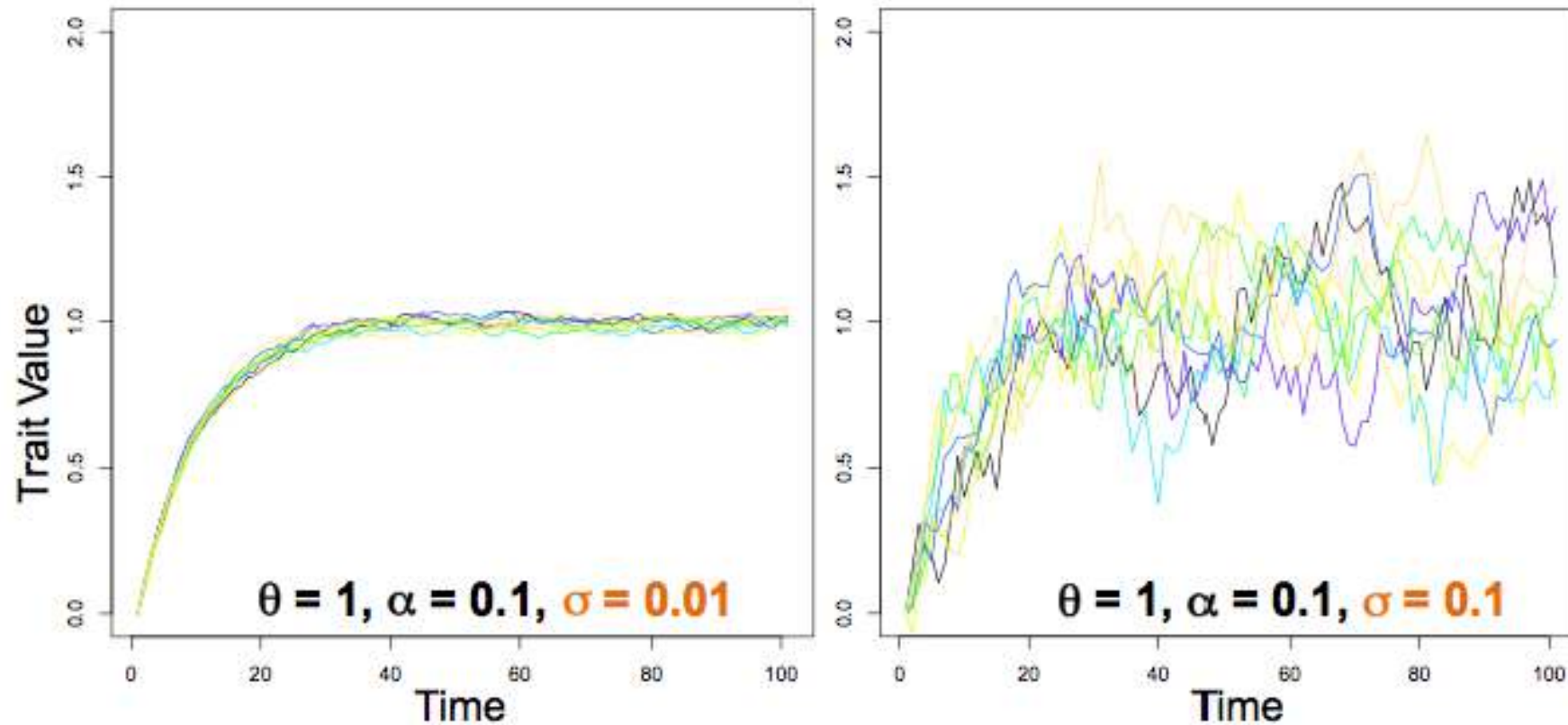


$$dX_{(t)} = \alpha[\theta - X_{(t)}]dt + \sigma dB_{(t)}$$

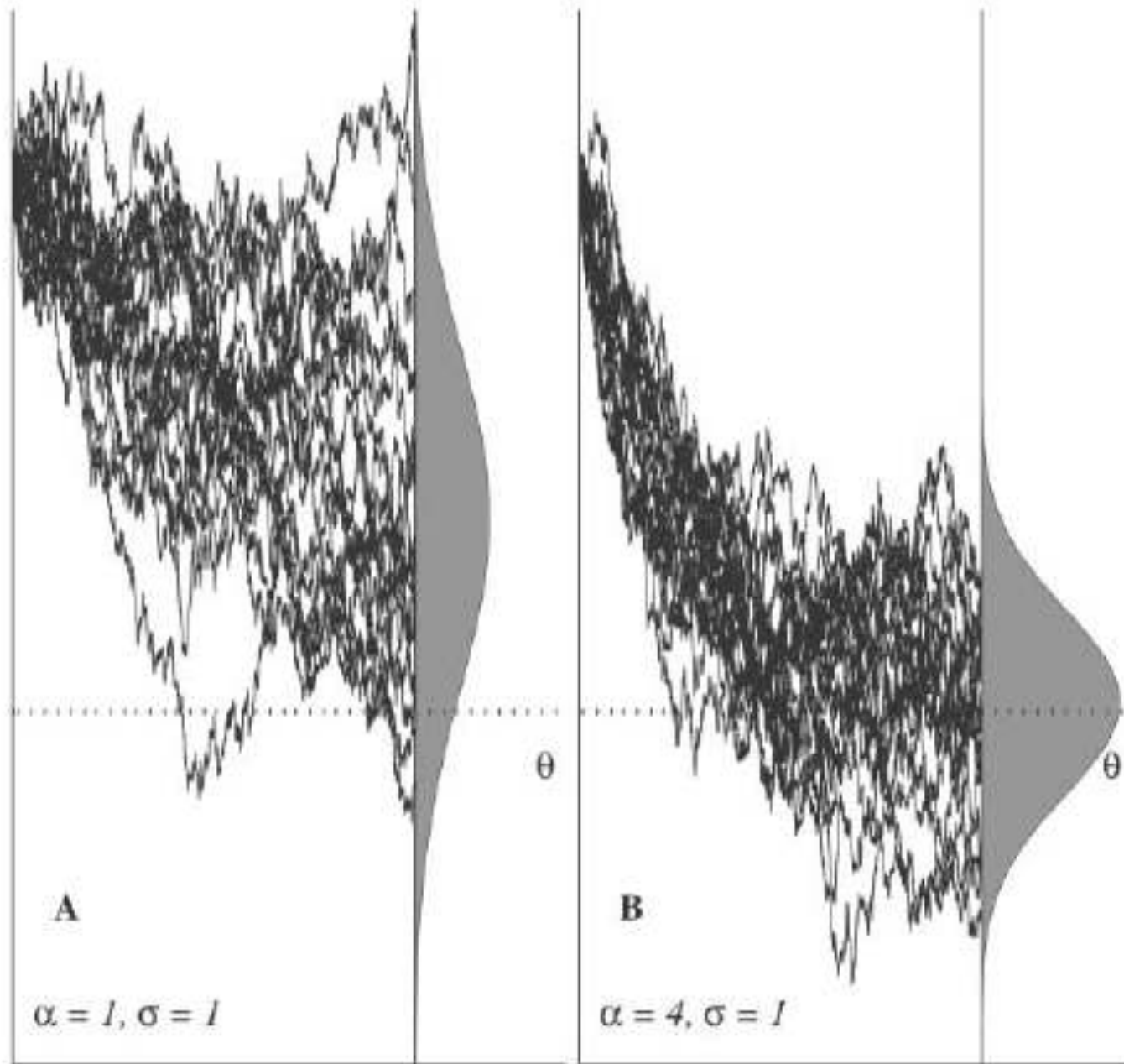


Stronger α the more quickly the optima is reached and the lower variance

$$dX_{(t)} = \alpha[\theta - X_{(t)}]dt + \sigma dB_{(t)}$$



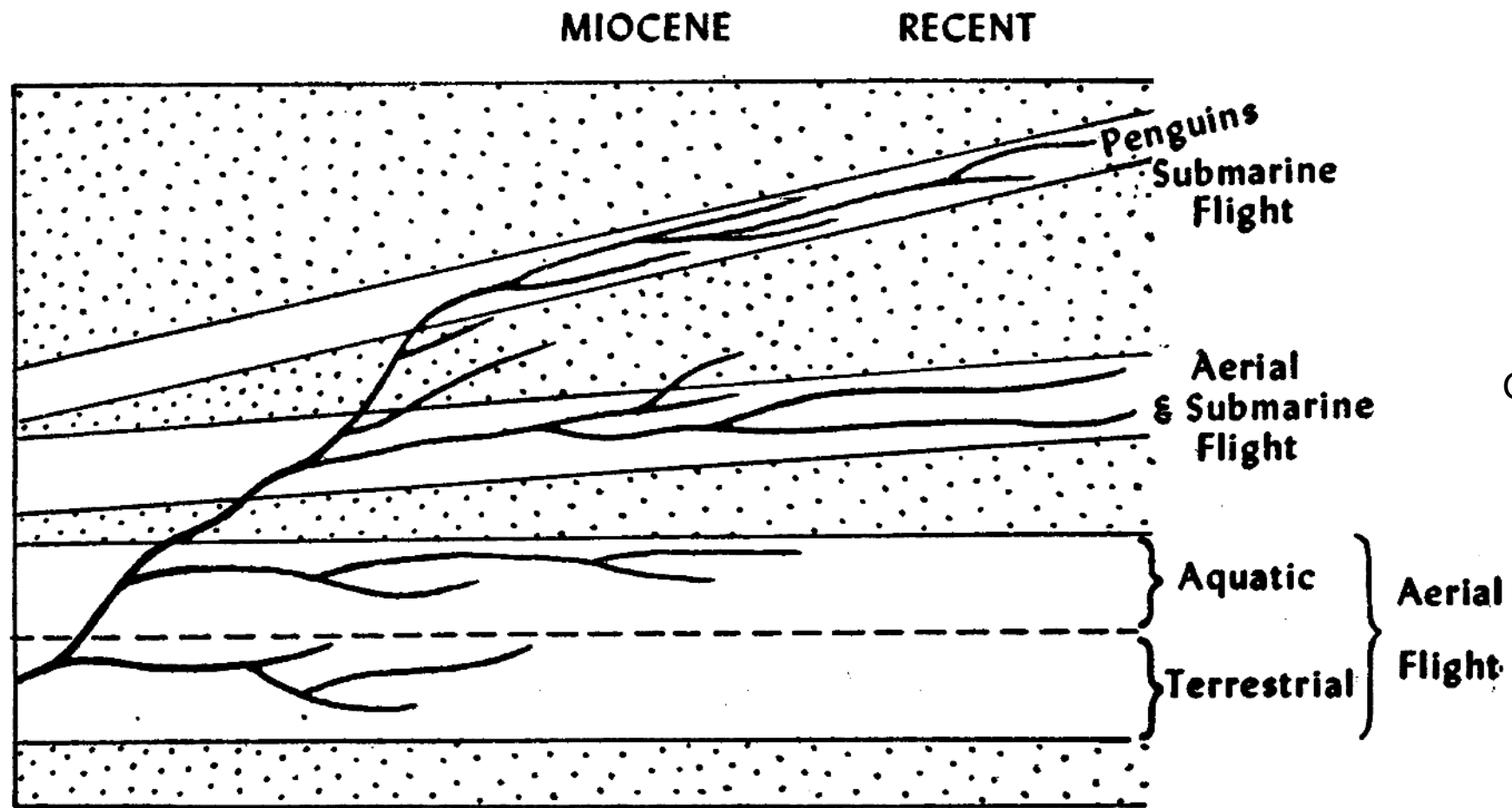
Higher σ the greater the variance



Early Burst (EB)

- Taxa de evolução desacelera com o tempo
- Maior taxa é na raiz da filogenia (valor inicial)
- Corresponde à idéia de **radiação adaptativa**
 - Espera-se que linhagens que entram em novas zonas adaptativas diversifiquem rápido
 - Taxas diminuem à medida que nichos são preenchidos
- Descrito por 3 parâmetros:
 - Valor inicial (θ)
 - Taxa inicial (σ_o^2)
 - Taxa de mudança (r)

$$V_{ij} = \int_0^{s_{ij}} \sigma_o^2 e^{rt} dt = \sigma_o^2 \left(\frac{e^{rs_{ij}} - 1}{r} \right).$$



George Gaylord Simpson

FIG. 31. Grid diagram of the theory of penguin evolution proposed in this paper.

Evolution, 57(4), 2003, pp. 717–745

TESTING FOR PHYLOGENETIC SIGNAL IN COMPARATIVE DATA: BEHAVIORAL TRAITS ARE MORE LABILE

SIMON P. BLOMBERG,¹ THEODORE GARLAND, JR.,^{1,2} AND ANTHONY R. IVES^{3,4}

EARLY BURSTS OF BODY SIZE AND SHAPE EVOLUTION ARE RARE IN COMPARATIVE DATA

Luke J. Harmon,^{1,2,3} Jonathan B. Losos,⁴ T. Jonathan Davies,⁵ Rosemary G. Gillespie,⁶ John L. Gittleman,⁷ W. Bryan Jennings,⁸ Kenneth H. Kozak,⁹ Mark A. McPeck,¹⁰ Franck Moreno-Roark,¹¹ Thomas J. Near,¹² Andy Purvis,¹³ Robert E. Ricklefs,¹⁴ Dolph Schluter,² James A. Schulte II,¹¹ Ole Seehausen,^{15,16} Brian L. Sidlauskas,^{17,18} Omar Torres-Carvajal,¹⁹ Jason T. Weir,² and Arne Ø. Mooers²⁰

2385

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Evolution 64-8: 2385–2396

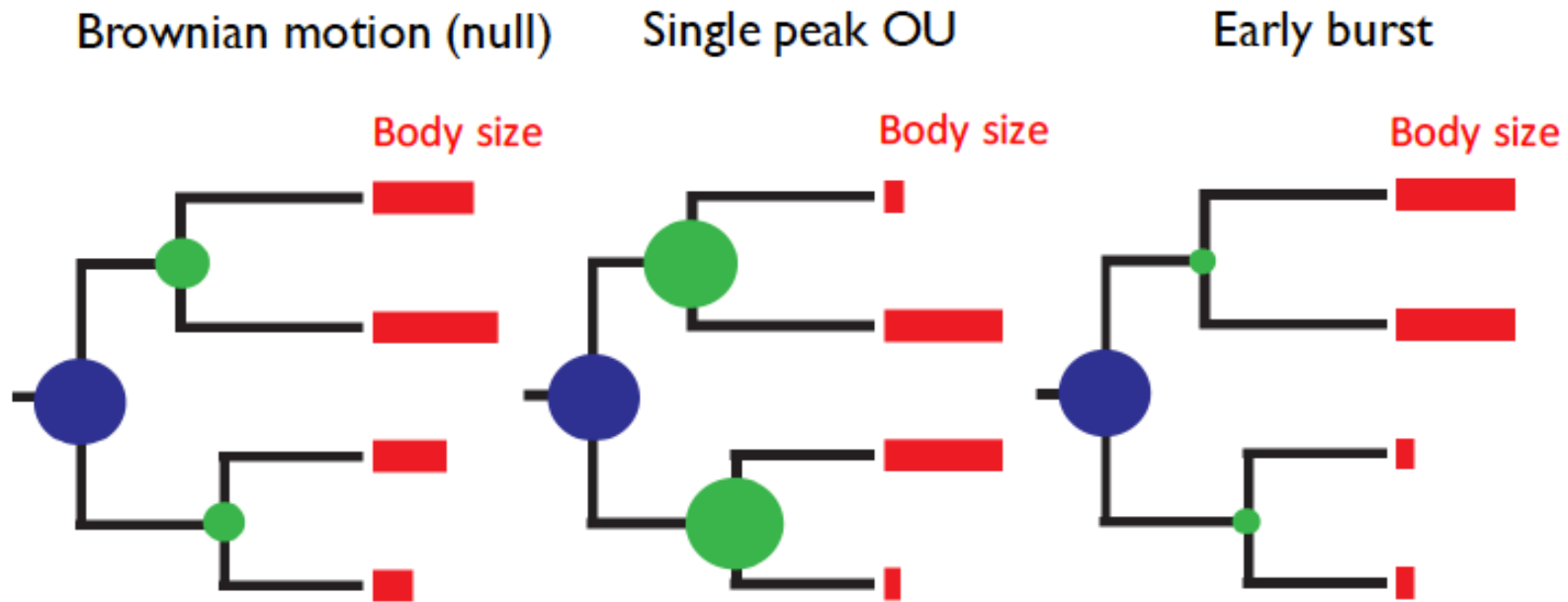
Early burst

Late burst

increases (accelerates, AC) or decreases (decelerates, DC) in rate over time. The covariance relationship among the characters can be represented by the formulae:

$$V\{X_i\} = \frac{1 - g^{-(\tau_i + \tau_{ij})}}{1 - g^{-1}} \sigma_\gamma^2 \quad \text{and} \quad (8a)$$

$$\text{cov}\{X_i, X_j\} = \frac{1 - g^{-\tau_{ij}}}{1 - g^{-1}} \sigma_\gamma^2, \quad (8b)$$

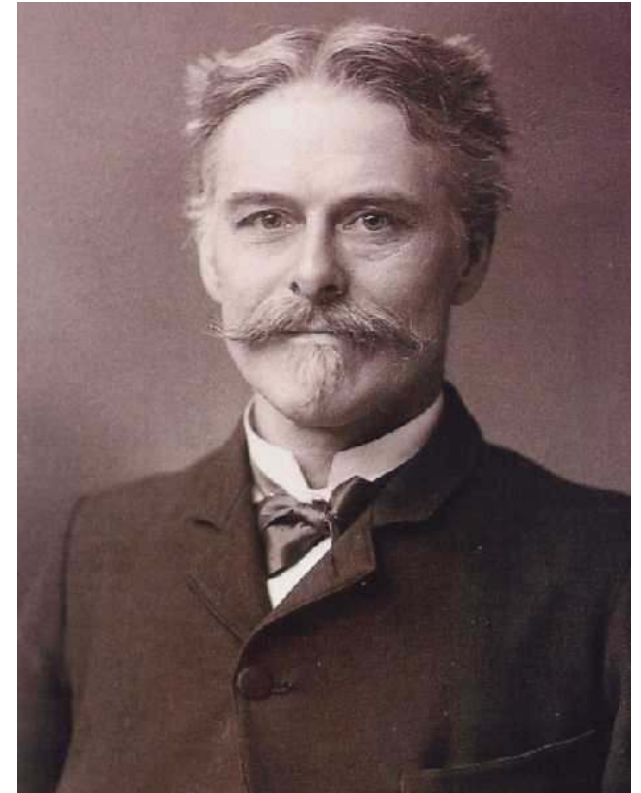


● Amount of variation in the whole clade

● Amount of variation within subclades

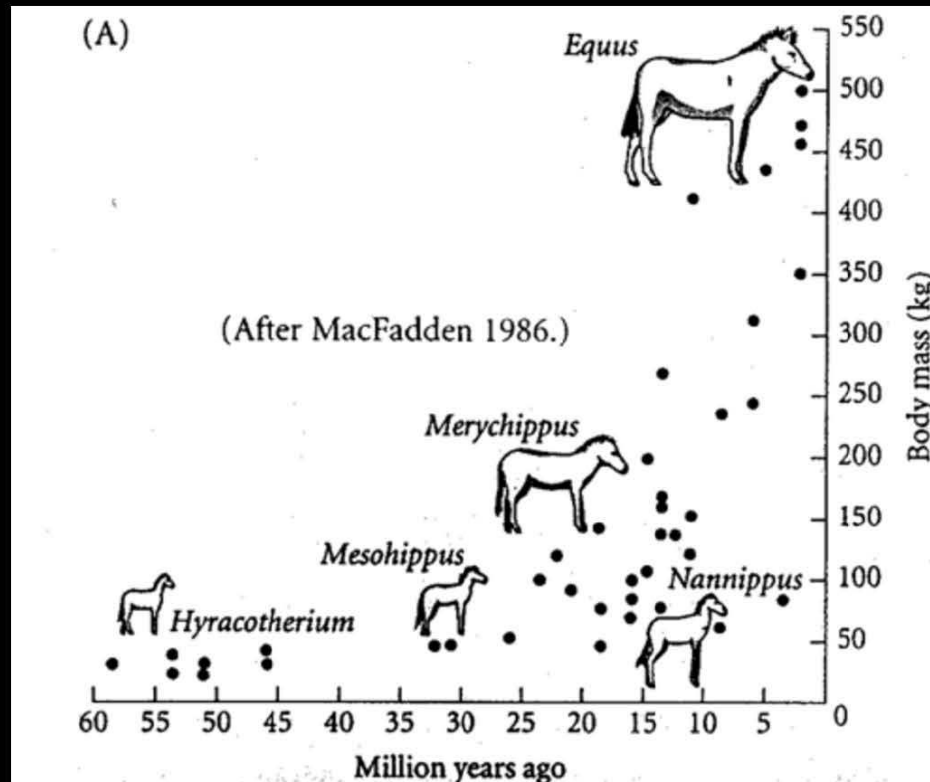
Movimento Browniano com tendência (Trend)

- Incorpora a idéia de mudança tendenciada do fenótipo (atributos) ao longo do tempo
 - Regra de Cope => aumento do tamanho de corpo ao longo do tempo

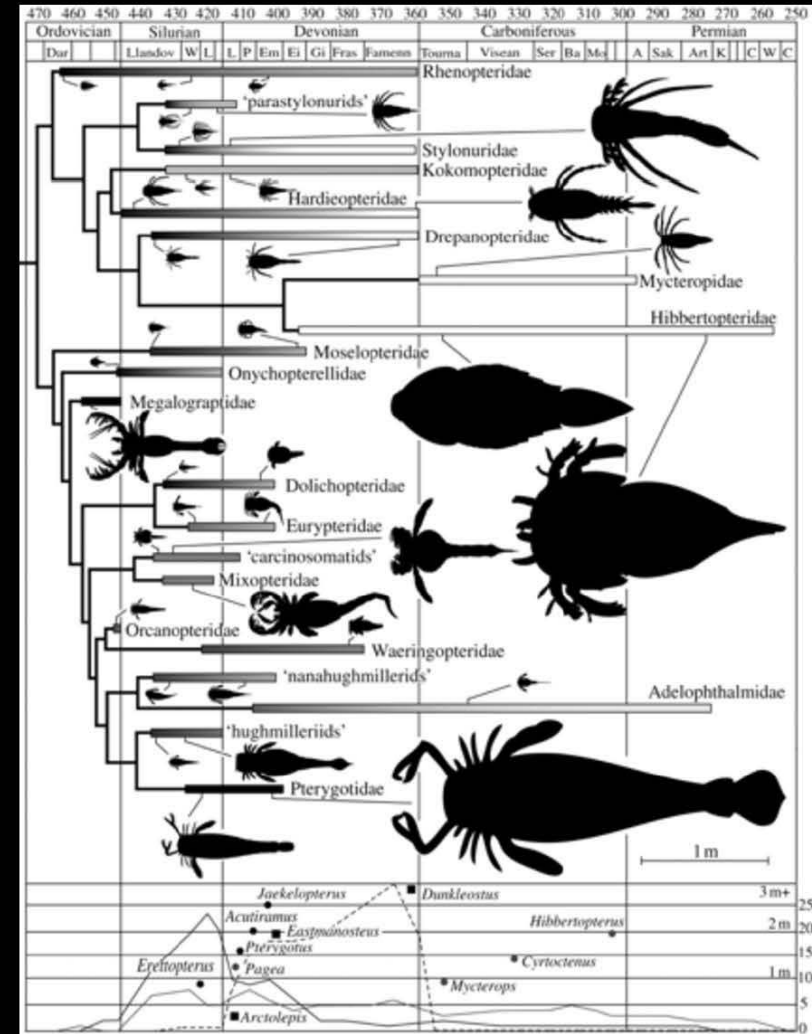


Trend

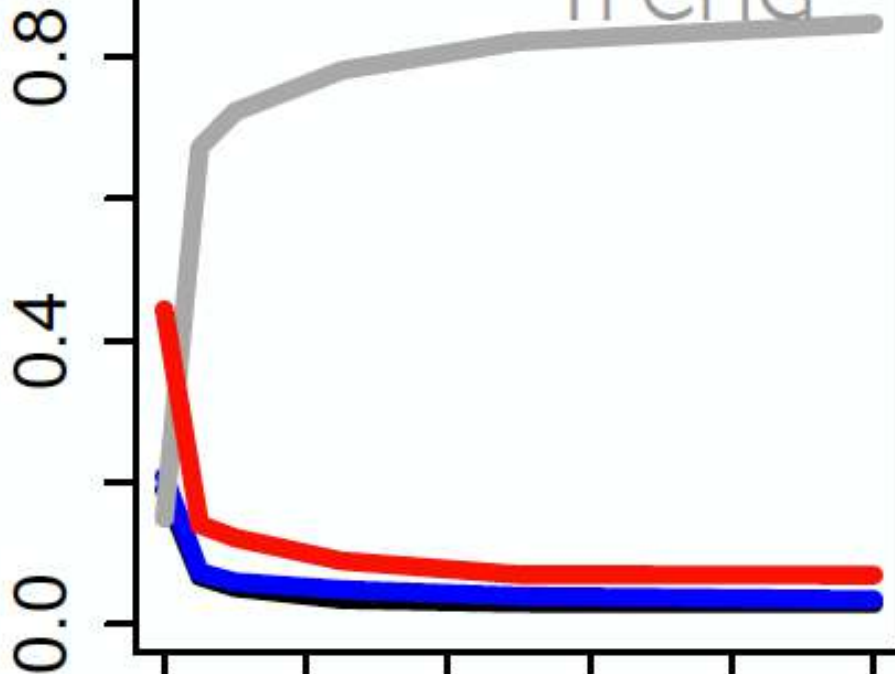
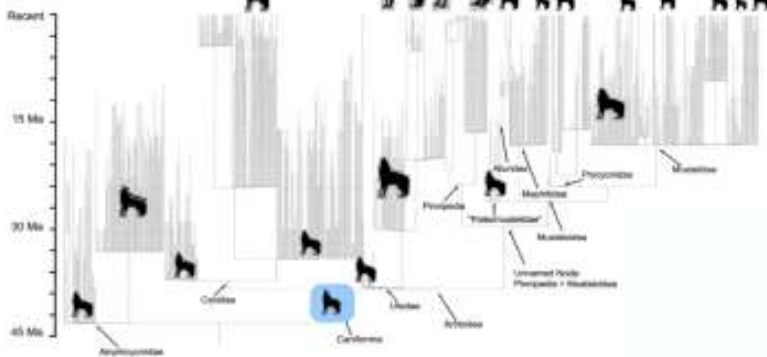
Species within lineages tend to get bigger over time (Cope's rule)



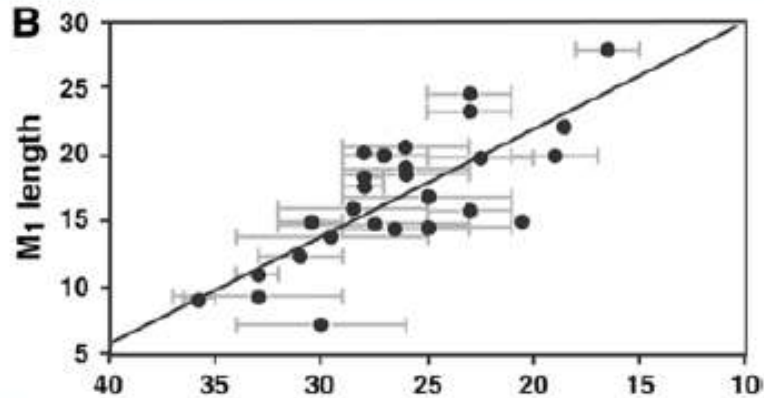
Futuyma 1998



Lamsdell and Braddy 2009



BM with trend



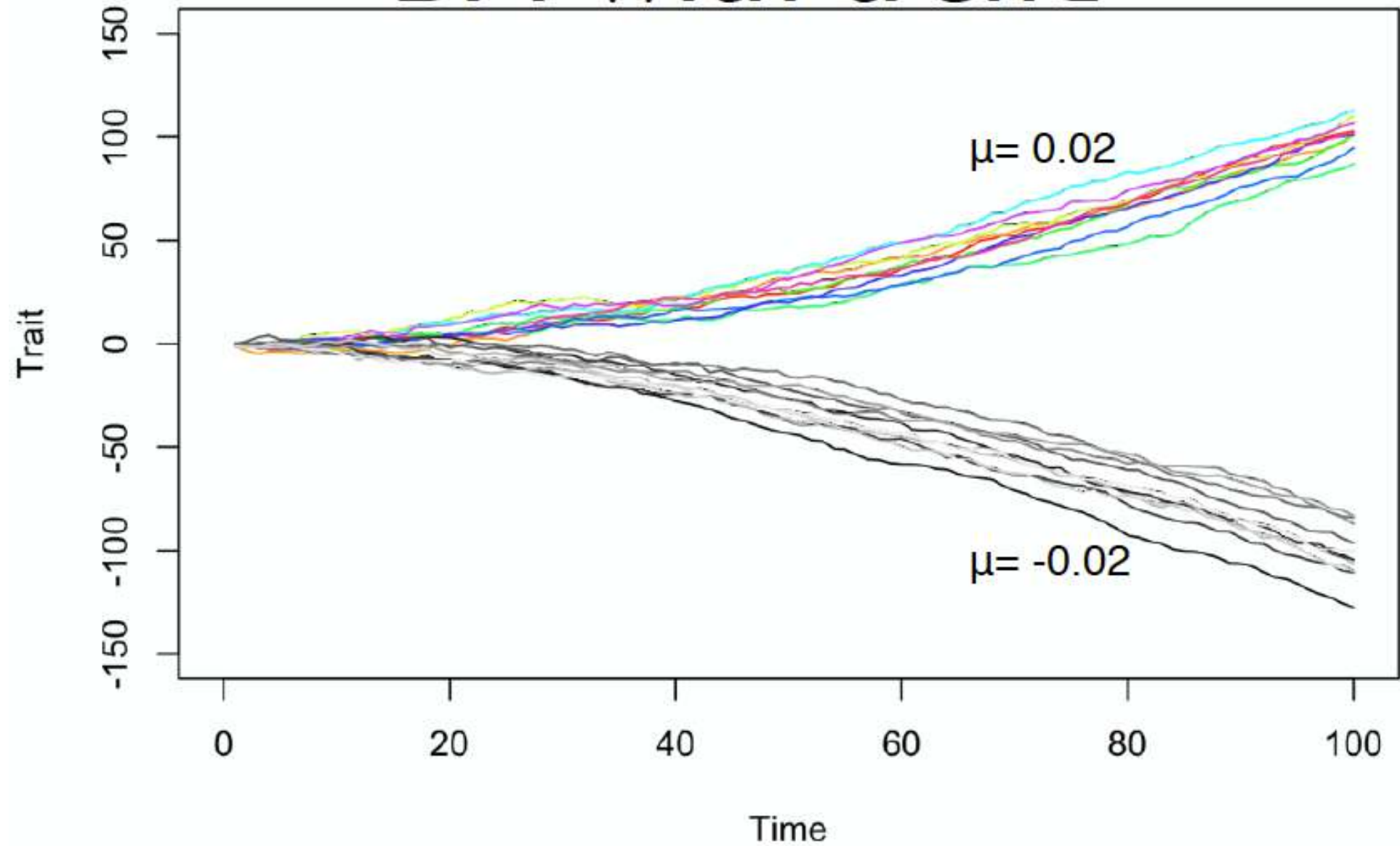
$$dX_{(t)} = \sigma dB_t$$

rate

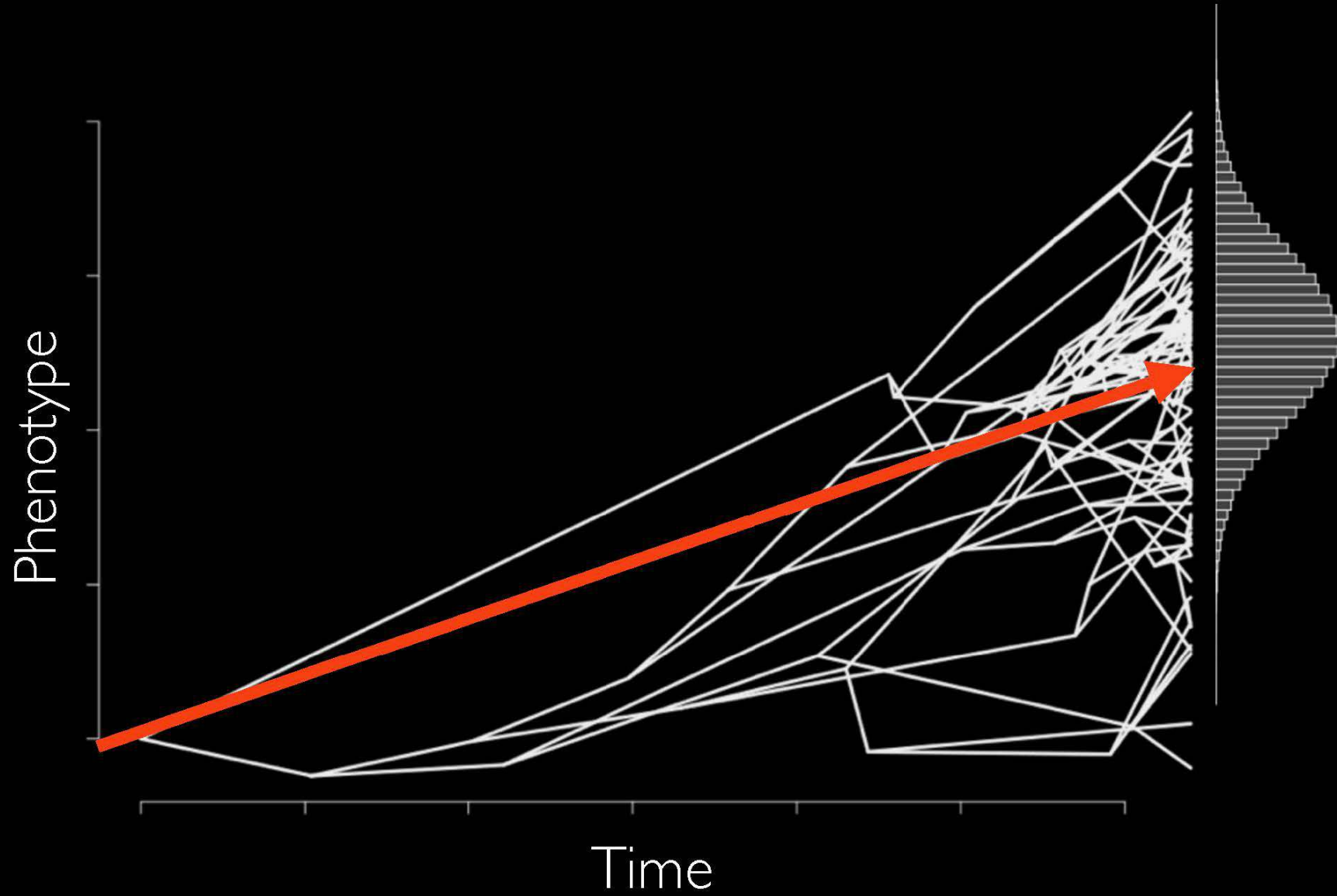
normal
distribution where
mean = $t * \mu$

trait increases
when $\mu > 0$,
decreases when
 $\mu < 0$

BM with trend



trend



under Trend, the variance increases through time as with BM but the expected trait value does not stay the same

trait values for species

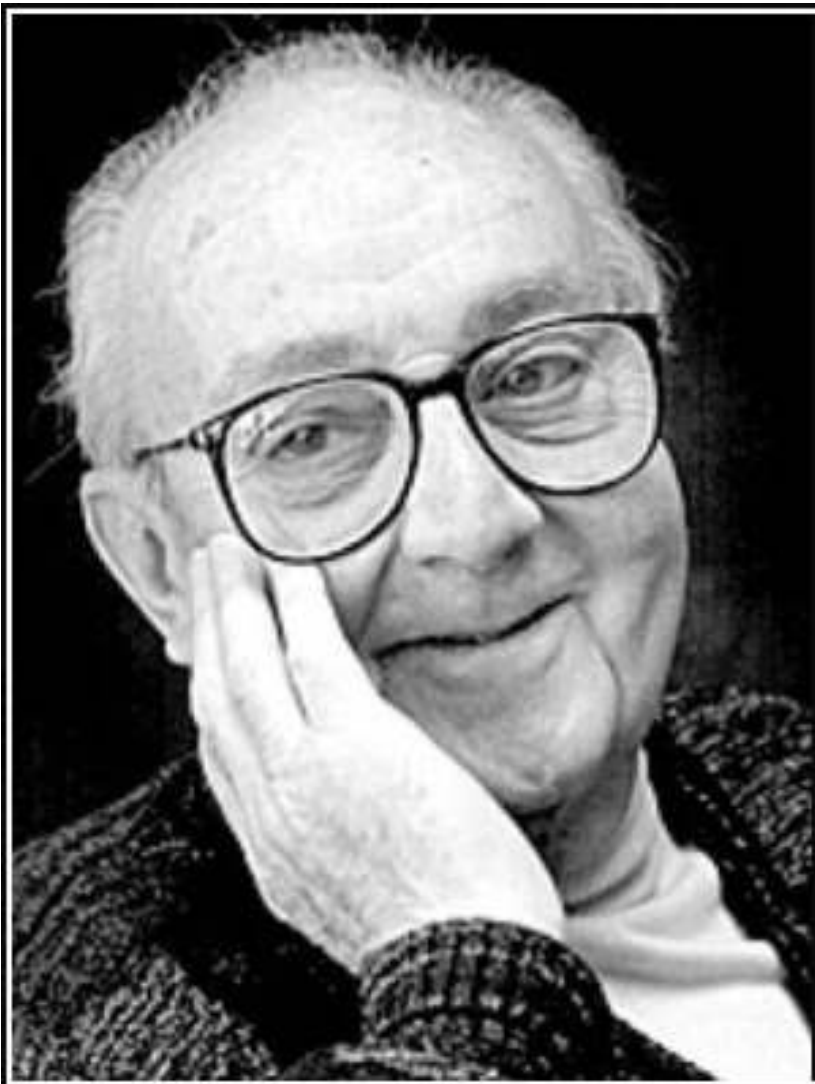
variance-covariance matrix * rate

$$\log(\mathbf{L}) = \log \left[\frac{\exp \left\{ -\frac{1}{2} [\mathbf{X} - \mathbf{E}(\mathbf{X})]' \mathbf{V}^{-1} [\mathbf{X} - \mathbf{E}(\mathbf{X})] \right\}}{\sqrt{(2\pi)^N \times \det(\mathbf{V})}} \right].$$

expected value is a function of:

1. the starting state,
2. the strength of the trend,
3. the direction of the trend

Como comparar o ajuste
de modelos aos dados?



All models are wrong, but some are
useful.

— *George E. P. Box* —

AZ QUOTES



Brian O'Meara

@omearabrian



Seguindo

.@mbutler808: "Don't invest all your hopes and dreams in a single model -- it's a freakin' model!"
#quantTT

Ver tradução

RETWEET

1

CURTIDAS

2



14:33 - 10 de ago de 2016



1

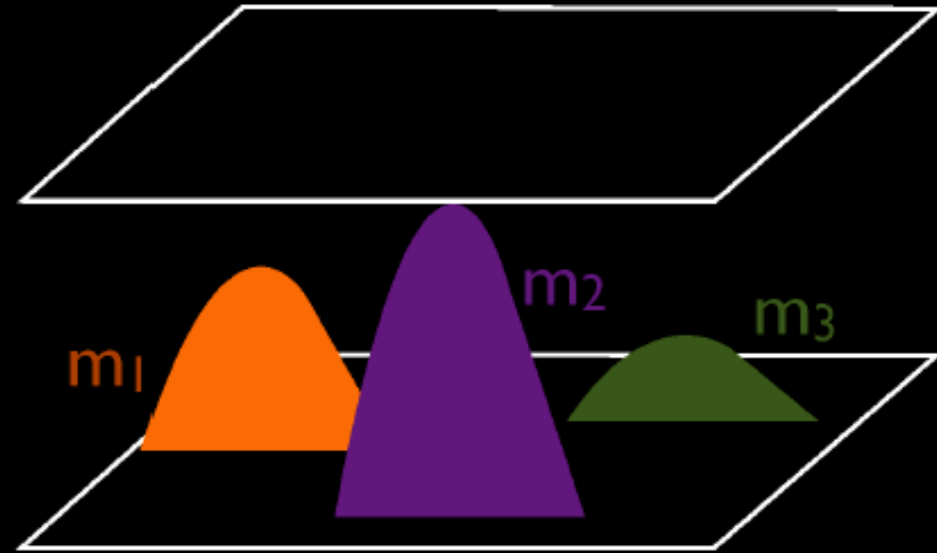


2



Akaike Information Criterion

- Compete models with each other
- All models are wrong
- Choose the model that is the most efficient approximation of the truth



AIC penalizes the likelihood by the number of extra parameters

$$AIC = 2k - 2\ln(L)$$

$$AIC_c = AIC + \frac{2k(k+1)}{n-k-1}$$

$$w_i = \frac{\exp\left(-\frac{1}{2}\Delta_i\right)}{\sum_{r=1}^R \exp\left(-\frac{1}{2}\Delta_r\right)}$$

k	BM	EB	OU
root state	✓	✓	✓
σ^2	✓	✓	✓
scalar	×	✓	✓
total	2	3	3

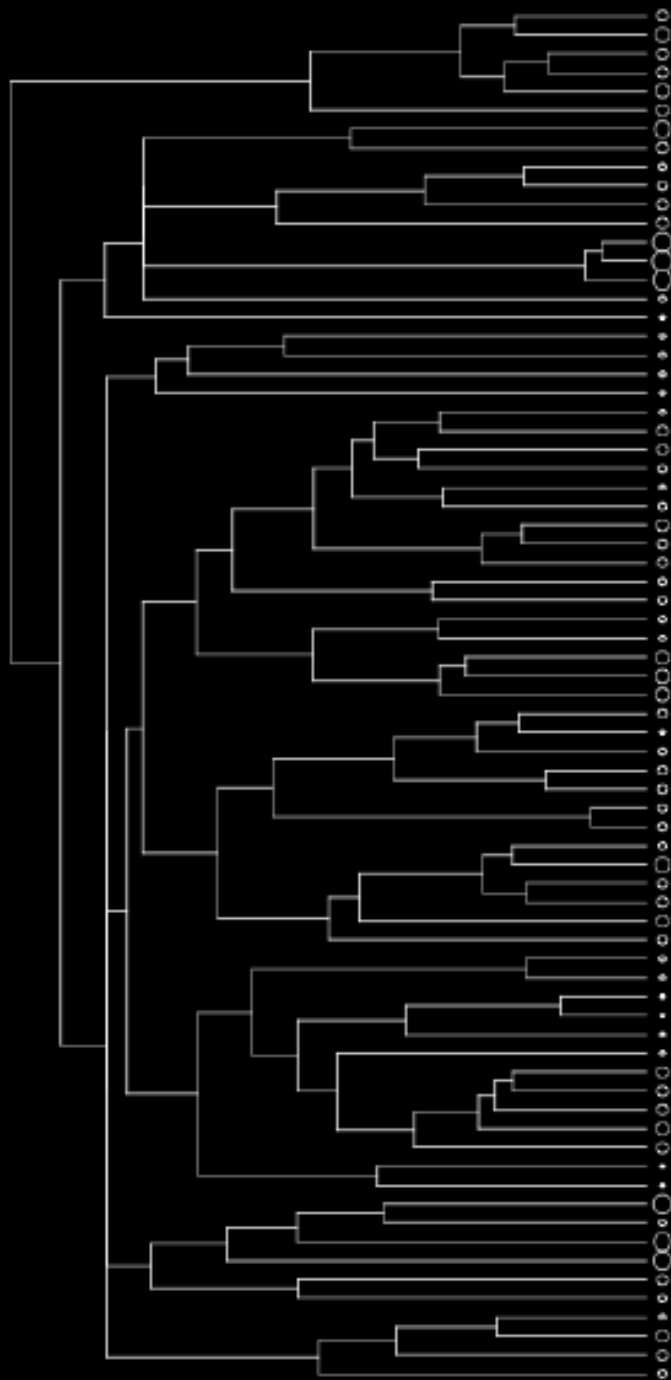
Example: *Anolis* lizards

Lizards on Caribbean
islands

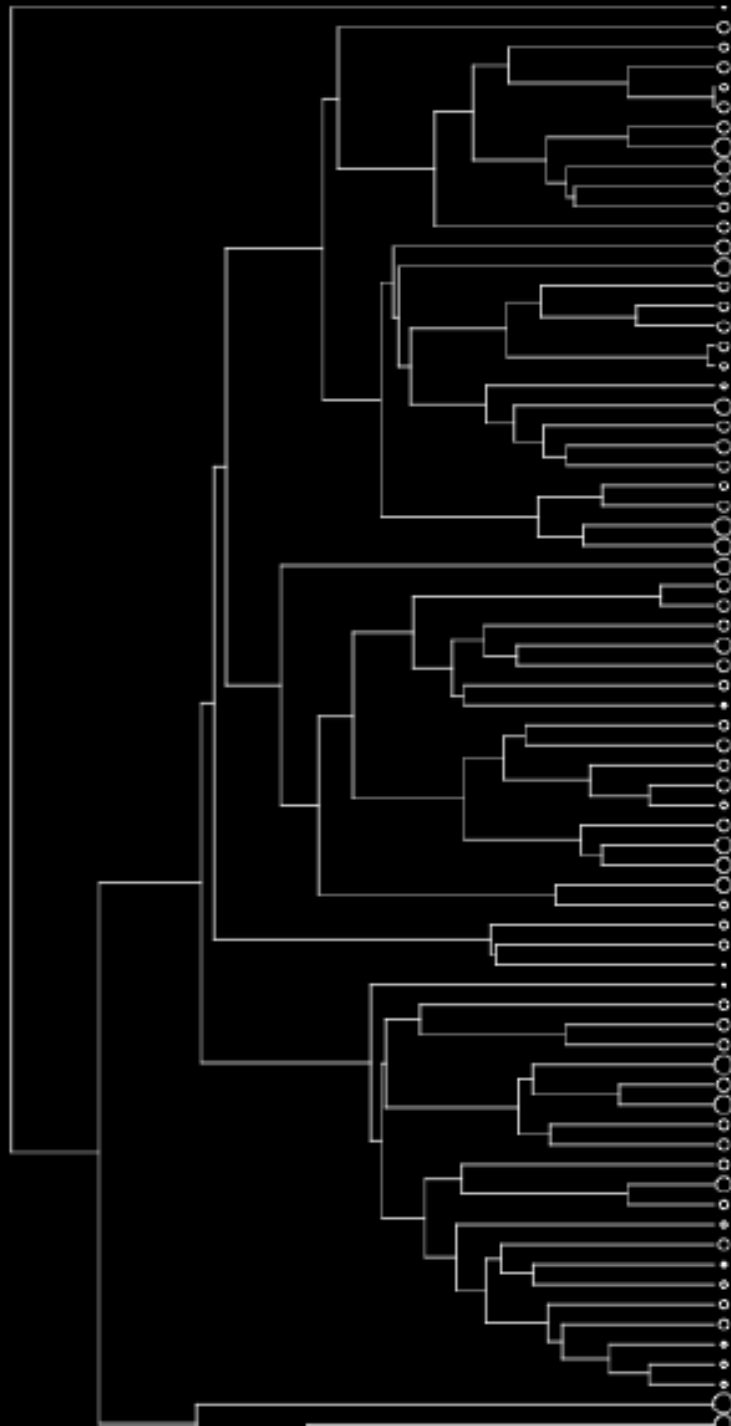
Phylogenetic and body
size data for 73 species
(out of ~140 total)



Anolis baleatus



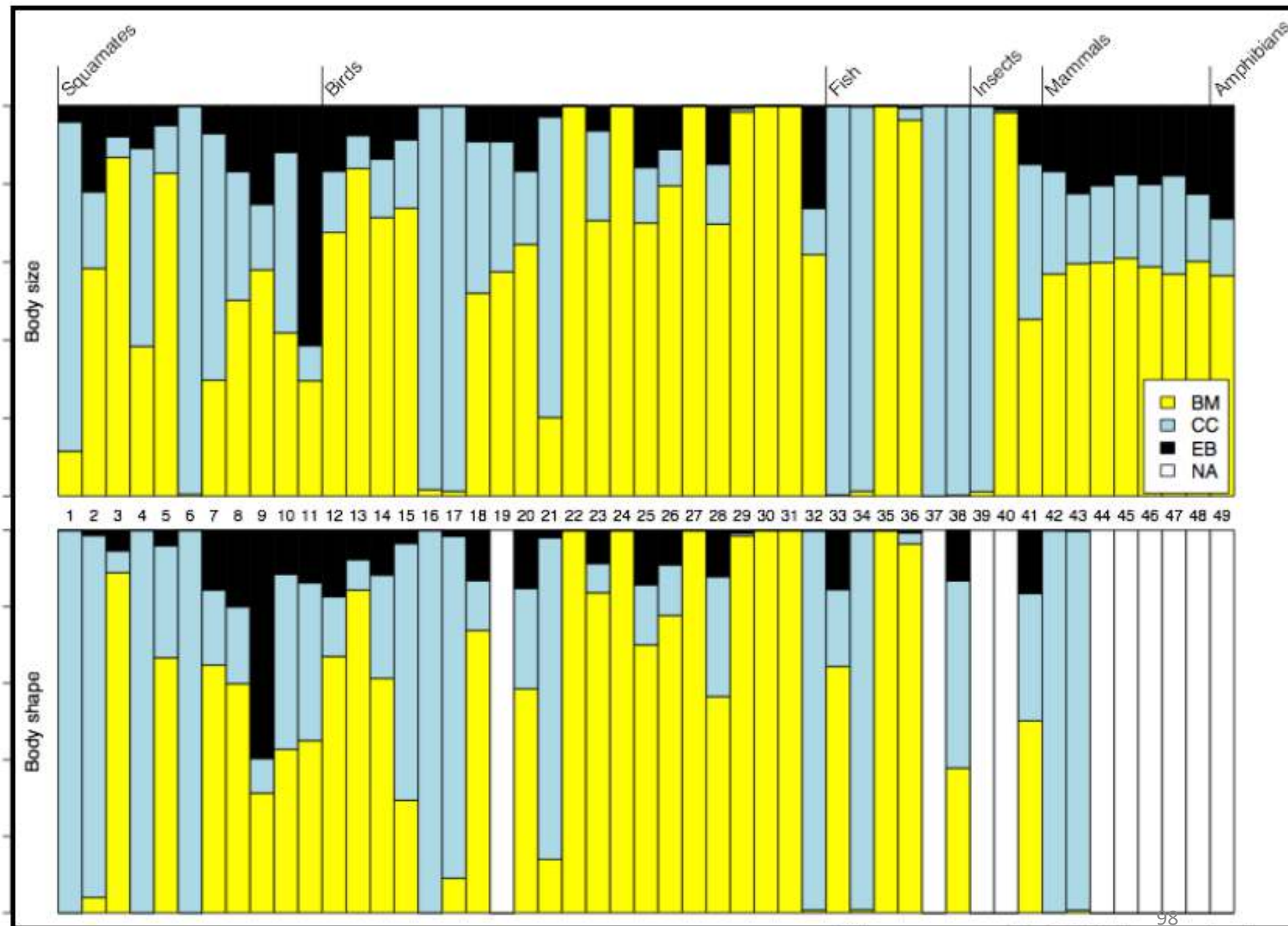
Model	Parameter estimates	lnL	Akaike weight
BM	$\sigma^2 = 0.004$	-18.2	0.58
EB	$\sigma^2 = 0.006$ $r = -0.01$	-18.1	0.2
OU	$\sigma^2 = 0.004$ $\alpha = 0$	-18.2	0.22



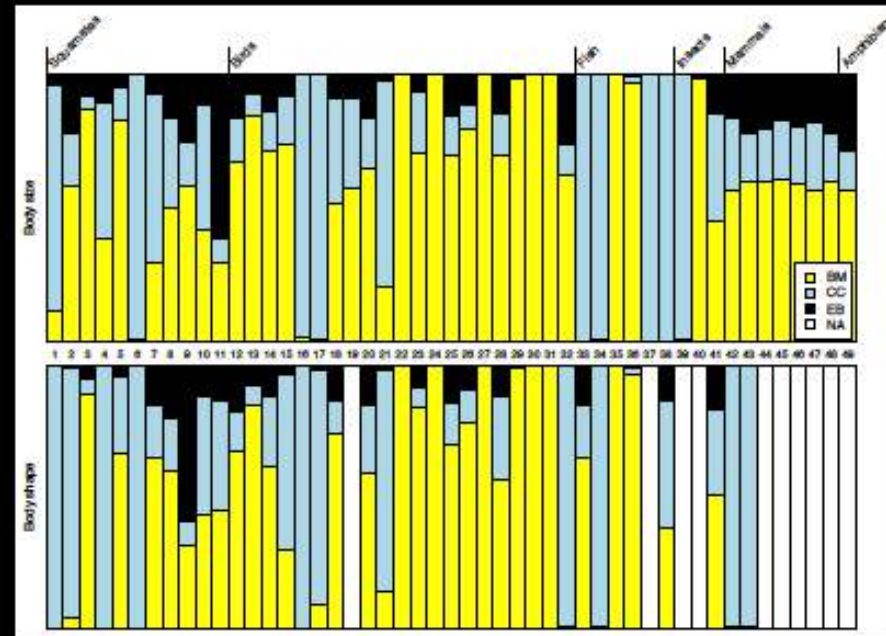
Cichlids in Lake Tanganyika

Model	Parameter estimates	lnL	Akaike weight
BM	$\sigma^2 = 0.02$	-62.3	0
EB	$\sigma^2 = 0.02$ $r = 0$	-62.3	0
OU	$\sigma^2 = \dots$ $\alpha = \dots$	-33.3	1

Pouca suporte para modelos EB para tamanho de corpo em geral, mesmo em exemplos clássicos de radiação adaptativa



- “Adaptive radiation” pattern very rare in this data set
- Constraints dominate over long time periods
- Brownian motion is sometimes a poor fit to real data





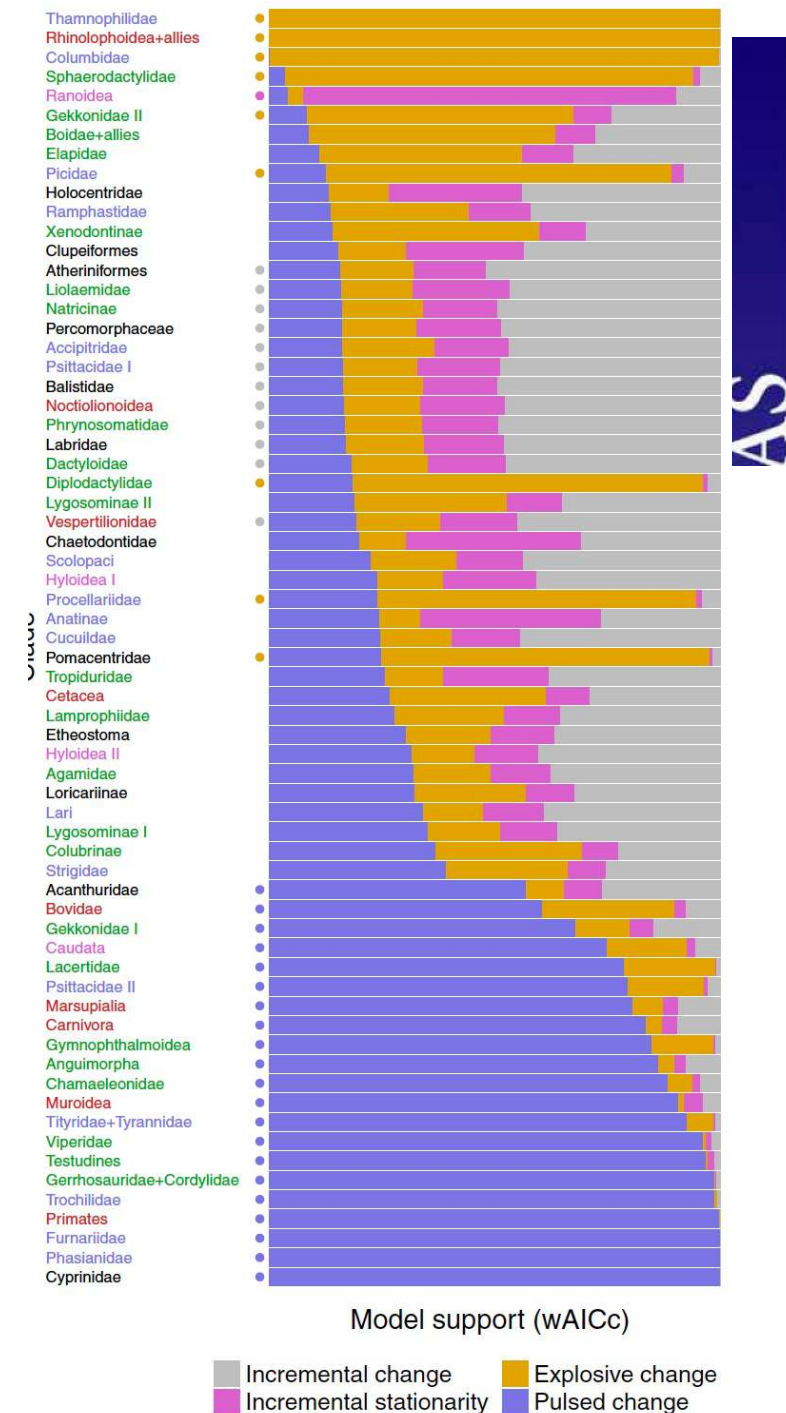
Pulsed evolution shaped modern vertebrate body sizes

Michael J. Landis^a and Joshua G. Schraiber^{b,c,1}

^aDepartment of Ecology and Evolutionary Biology, Yale University, New Haven, CT 06520; ^bDepartment of Biology, Temple University, Philadelphia, PA 19122; and ^cInstitute for Genomics and Evolutionary Medicine, Temple University, Philadelphia, PA 19122

Edited by Neil H. Shubin, The University of Chicago, Chicago, IL, and approved October 6, 2017 (received for review June 18, 2017)

The relative importance of different modes of evolution in shaping phenotypic diversity remains a hotly debated question. Fossil data suggest that stasis may be a common mode of evolution, while modern data suggest some lineages experience very fast rates of evolution. One way to reconcile these observations is to imagine that evolution proceeds in pulses, rather than in increments, on geological timescales. To test this hypothesis, we developed a maximum-likelihood framework for fitting Lévy processes to comparative morphological data. This class of stochastic processes includes both an incremental and a pulsed component. **We found that a plurality of modern vertebrate clades examined are best fitted by pulsed processes over models of incremental change, stationarity, and adaptive radiation.** When we compare our results to theoretical expectations of the rate and speed of regime shifts for models that detail fitness landscape dynamics, we find that our quantitative results are broadly compatible with both microevolutionary models and observations from the fossil record.



The many shapes of diversity: ecological and evolutionary determinants of biodiversity through time

S. Castiglione¹, A. Mondanaro¹, F. Carotenuto¹, F. Passaro¹,
M. Fortelius² and P. Raia¹

Results: For nearly 90% of the fossil clades analysed, the best model includes an early diversification phase, and increased extinction rate over time. These features are consistent regardless of whether the main determinants of species diversity in the models are ecological or evolutionary. Clades terminated by mass extinctions did not have shorter duration than other clades but were characterized by earlier divergence and greater species richness than other clades.

Table 2. Selection of best models per clade by means of likelihood ratio test

	Neutral	Adaptive radiation	Delayed-rise	Red Queen	Key innovation	Diversity-dependence	Totals
Best model	1	8	7	19	18	9	62
Mass	0	6	2	9	8	6	31
No mass	1	2	5	10	10	3	31

Measurement errors should always be incorporated in phylogenetic comparative analysis

Daniele Silvestro^{1,2,3†}, Anna Kostikova^{2,3†}, Glenn Litsios^{2,3}, Peter B. Pearman^{4,5} and Nicolas Salamin^{2,3*}

3. Our analyses show that even small measurement errors (10%) consistently bias model selection towards erroneous rejection of BM in favour of more parameter-rich models (most frequently the OU model). Fortunately, methods that explicitly incorporate measurement errors in phylogenetic analyses considerably improve the accuracy of model selection.
4. Our results call for caution in interpreting the results of model selection in comparative analyses, especially when complex models garner only modest additional support.
5. Importantly, as measurement errors occur in most trait data sets, we suggest that estimation of measurement errors should always be performed during comparative analysis to reduce chances of misidentification of evolutionary processes.

Diferença entre ajuste (AIC) e “explicação” do modelo (R^2)

- Abordagem de teoria de informação (AIC-like) aponta o melhor modelo entre todos os disponíveis
 - Mas mesmo o melhor modelo (aquele com menor AIC) pode explicar pouco dos dados (ter R^2 baixo)
 - Peso de Akaike pode ser uma estatística melhor nesse caso
- O que fazer nesse caso?
 - Abordagens de “model adequacy”
 - Arburtus (Pennell et al. 2015)

E-ARTICLE

Model Adequacy and the Macroevolution of Angiosperm Functional Traits

Arbutus

Matthew W. Pennell,^{1,*} Richard G. FitzJohn,² William K. Cornwell,³ and Luke J. Harmon¹

ORIGINAL ARTICLE

doi:10.1111/j.1558-5646.2011.01574.x

IS YOUR PHYLOGENY INFORMATIVE? MEASURING THE POWER OF COMPARATIVE METHODS

Phylogenetic
Monte
Carlo

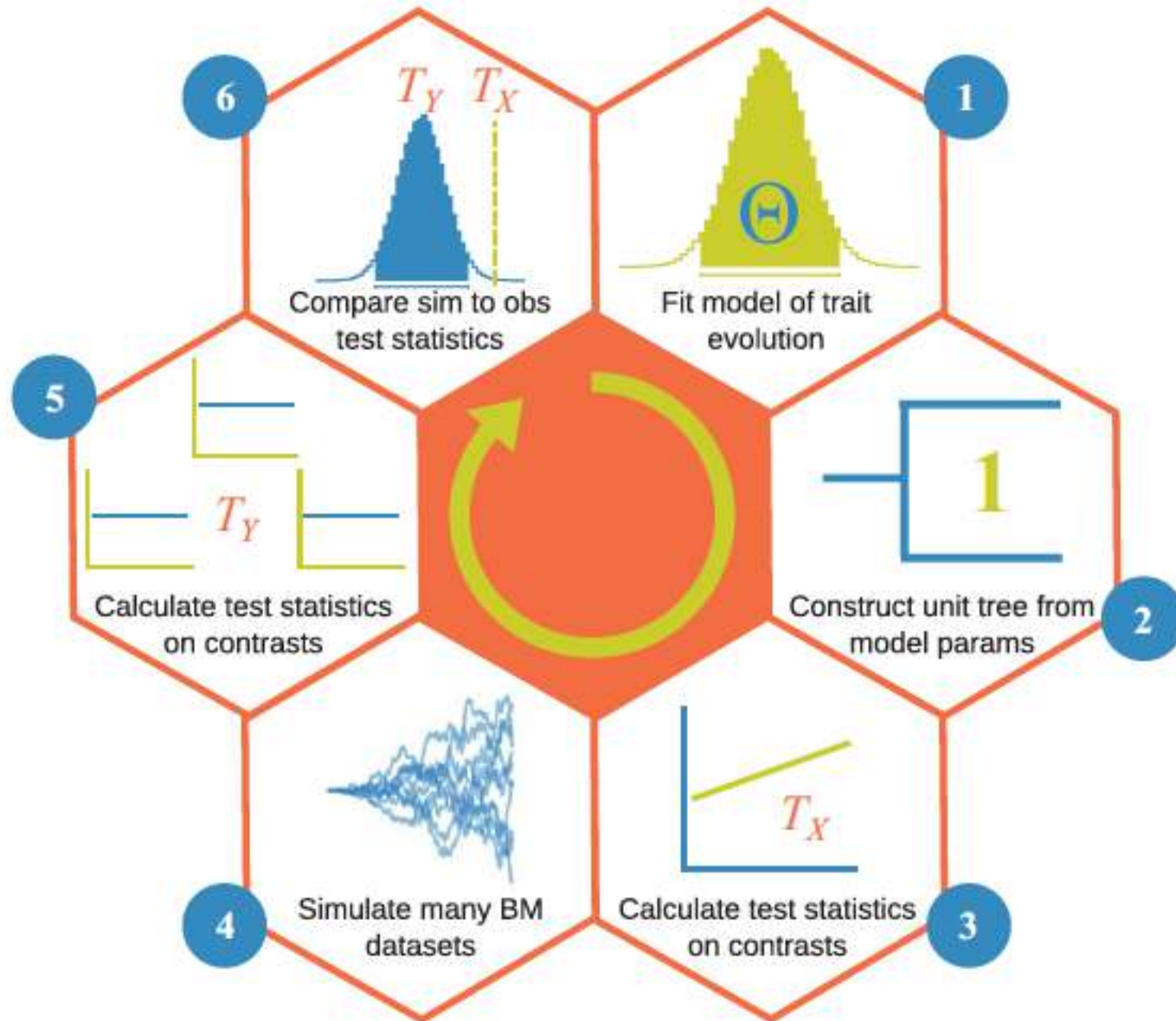
Carl Boettiger,^{1,2} Graham Coop³ and Peter Ralph³

2240

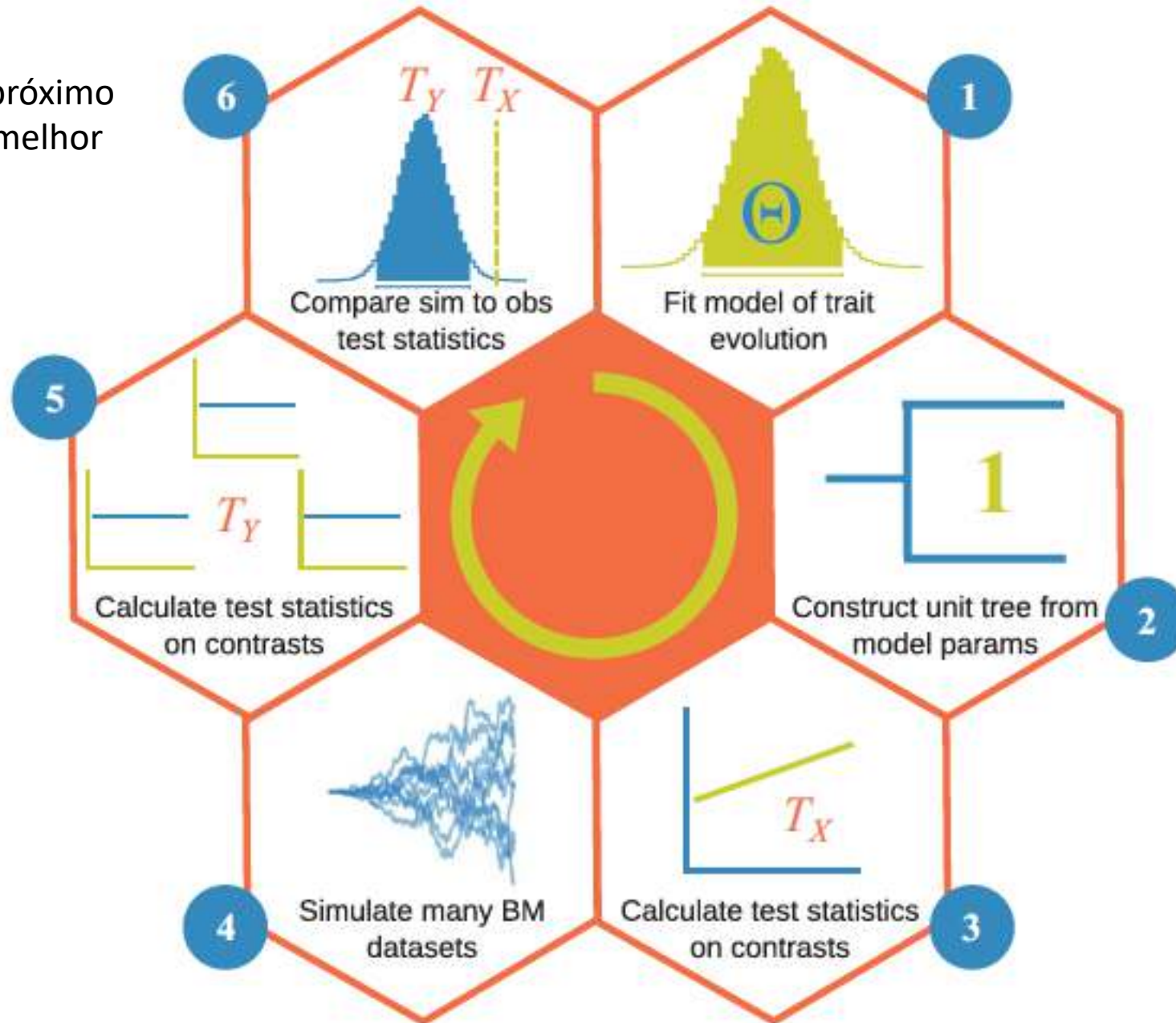
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Evolution 66-7: 2240–2251

Arbutus

- Comparar o ajuste de modelos em relação ao universo de todos os modelos possíveis
- Focado na adequabilidade de modelos para evolução de atributos contínuos.
- Também pode ser usado para PGLS



Quanto mais próximo
do simulado melhor



Modificar a árvore de
maneira a descrever a
história do trait

Table 1: Description of test statistics used to assess model adequacy

Test statistic	Description
M_{SIG}	The mean of the squared contrasts. This is equivalent to the restricted maximum likelihood estimator of the Brownian motion rate parameter σ^2 (Garland et al. 1992; Rohlf 2001). M_{SIG} is a metric of overall rate. Violations detected by M_{SIG} indicate whether the overall rate of trait evolution is over- or underestimated.
C_{VAR}	The coefficient of variation (standard deviation/mean) of the absolute value of the contrasts. If C_{VAR} calculated from the observed contrasts is greater than that calculated from the simulated contrasts, it suggests that we are not properly accounting for rate heterogeneity across the phylogeny. If C_{VAR} from the observed is smaller, it suggests that contrasts are even more than the model assumes. We use the coefficient of variation rather than the variance because the mean and variance of contrasts can be highly correlated.
S_{VAR}	The slope of a linear model fitted to the absolute value of the contrasts against their expected variances (following Garland et al. 1992). Each (standardized) contrast has an expected variance proportional to the sum of the branch lengths connecting the node at which it is computed to its daughter lineages (Felsenstein 1985). Under a model of Brownian motion, we expect no relationship between the contrasts and their variances. We use it to test whether contrasts are larger or smaller than we expect based on their branch lengths. If, for example, more evolution occurred per unit time on short branches than long branches, we would observe a negative slope. If S_{VAR} calculated from the observed data deviates substantially from the expectations, a likely explanation is branch length error in the phylogenetic tree.
S_{ASR}	The slope of a linear model fitted to the absolute value of the contrasts against the ancestral state inferred at the corresponding node. We estimated the ancestral state using the least squares method suggested by Felsenstein (1985) for the calculation of contrasts. (We note that this is not technically an ancestral state reconstruction [see Felsenstein 1985]; it is more properly thought of as a weighted average value for each node.) We used this statistic to evaluate whether there is variation in rates relative to the trait value. For example, do larger organisms evolve proportionally faster than smaller ones?
S_{HGT}	The slope of a linear model fitted to the absolute value of the contrasts against node depth (after Purvis and Rambaut 1995). This is used to capture variation relative to time. It is alternatively known as the “node-height test” and has been used to detect early bursts of trait evolution during adaptive radiations (for uses and modifications of this test see Freckleton and Harvey 2006; Slater and Pennell 2014).
D_{CDF}	The D statistic obtained from a Kolmogorov-Smirnov test from comparing the distribution of contrasts to that of a normal distribution with mean 0 and standard deviation equal to the root of the mean of squared contrasts (the expected distribution of the contrasts under Brownian motion; see Felsenstein 1985; Rohlf 2001). We chose this to capture deviations from normality. For example, if traits evolved via a “jump-diffusion”-type process (Landis et al. 2013) in which there were occasional bursts of rapid phenotypic evolution (Pennell et al. 2013), the tip data would no longer be multivariate normal owing to a few contrasts throughout the tree being much larger than the rest (i.e., the distribution of contrasts would have heavy tails).

A close-up photograph of a baby with light brown hair and blue eyes, wearing a green and white shirt. The baby has a determined, slightly pouting expression. The background is a blurred beach with sand and water.

YES!

**AGORA BORA BOTAR A MÃO NA
MASSA!**

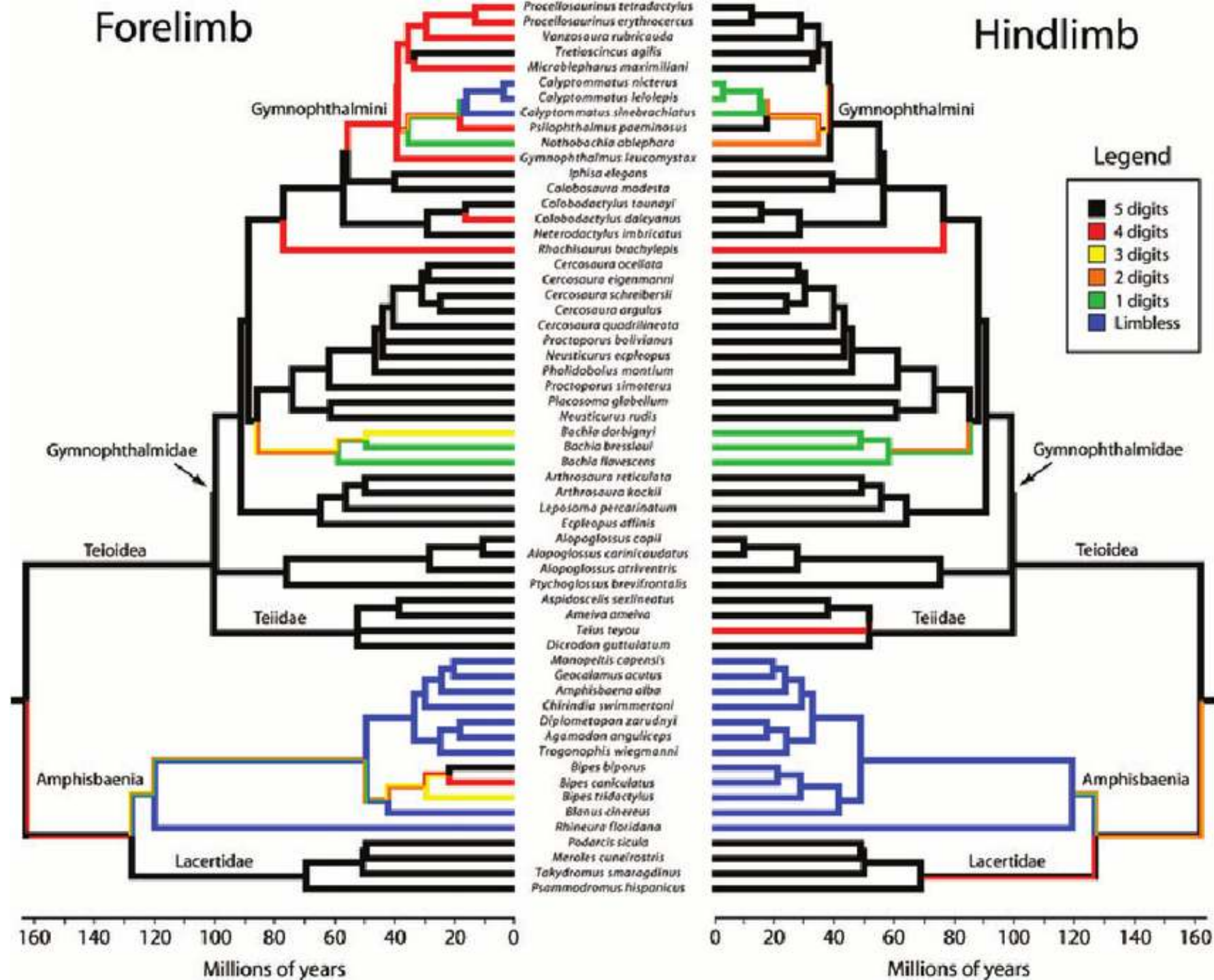
Modelos evolutivos para caracteres categóricos



Barrow Island, W. A. © Ryan Ellis



Nick Hart



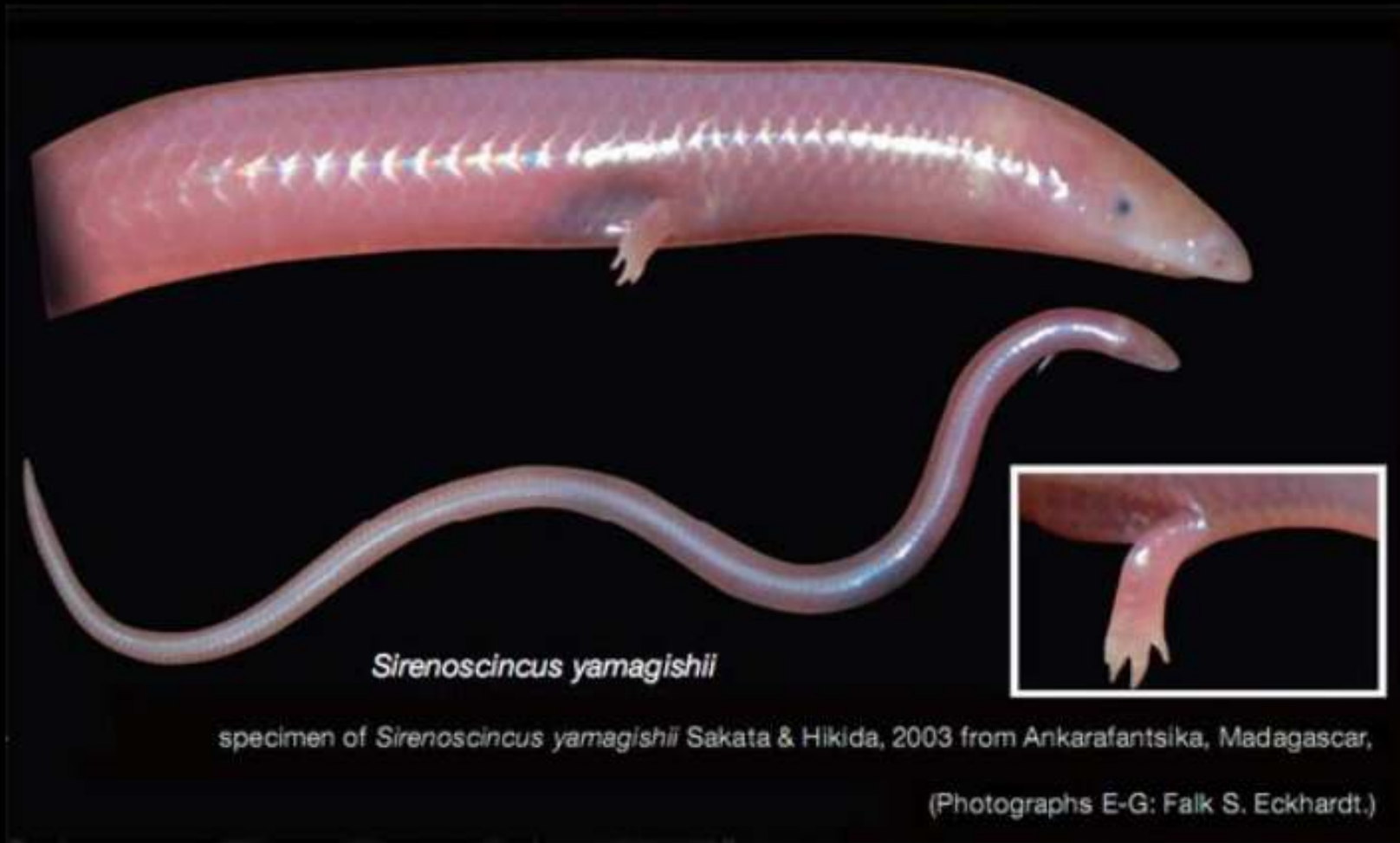
Quantas vezes répteis Squamata perderam
seus membros?



Sem membros
Estado 0

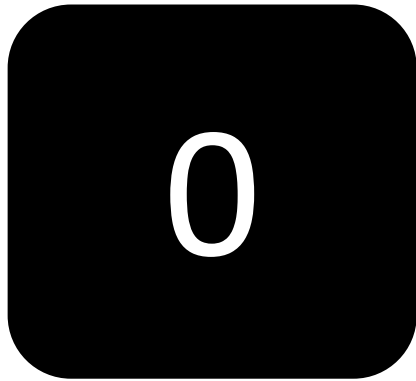


Com membros
Estado 1

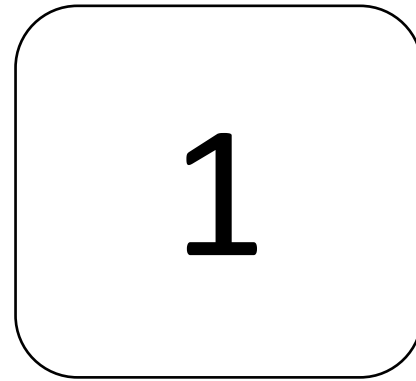


**WHAT IF I TOLD
YOU**

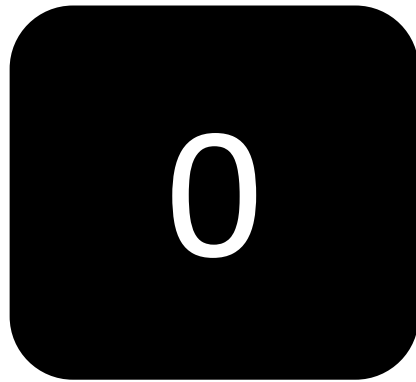
NOTHING MAKES SENSE ANYMORE



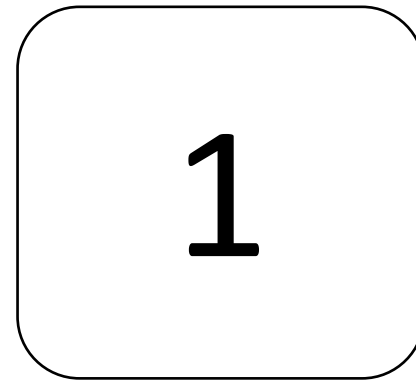
Sem membros



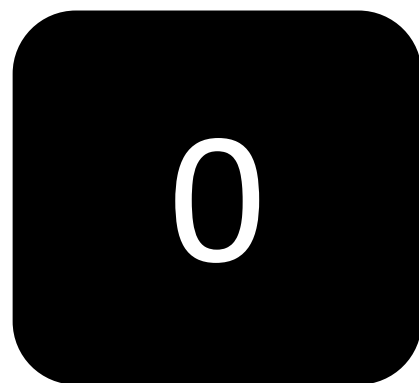
Com membros



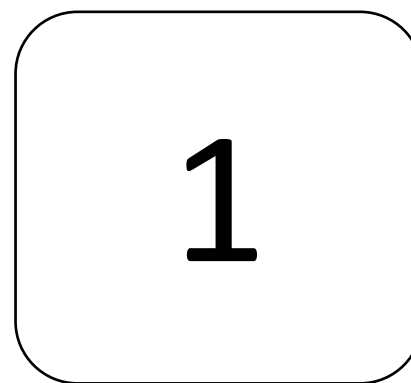
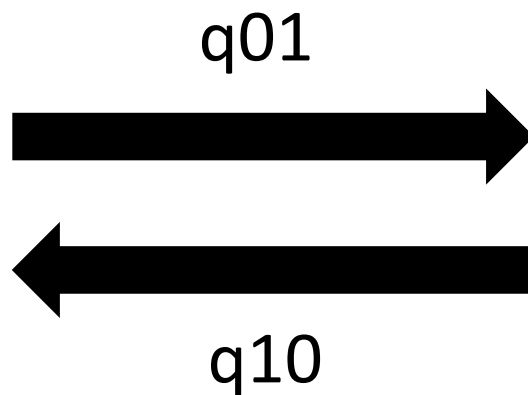
Sem membros



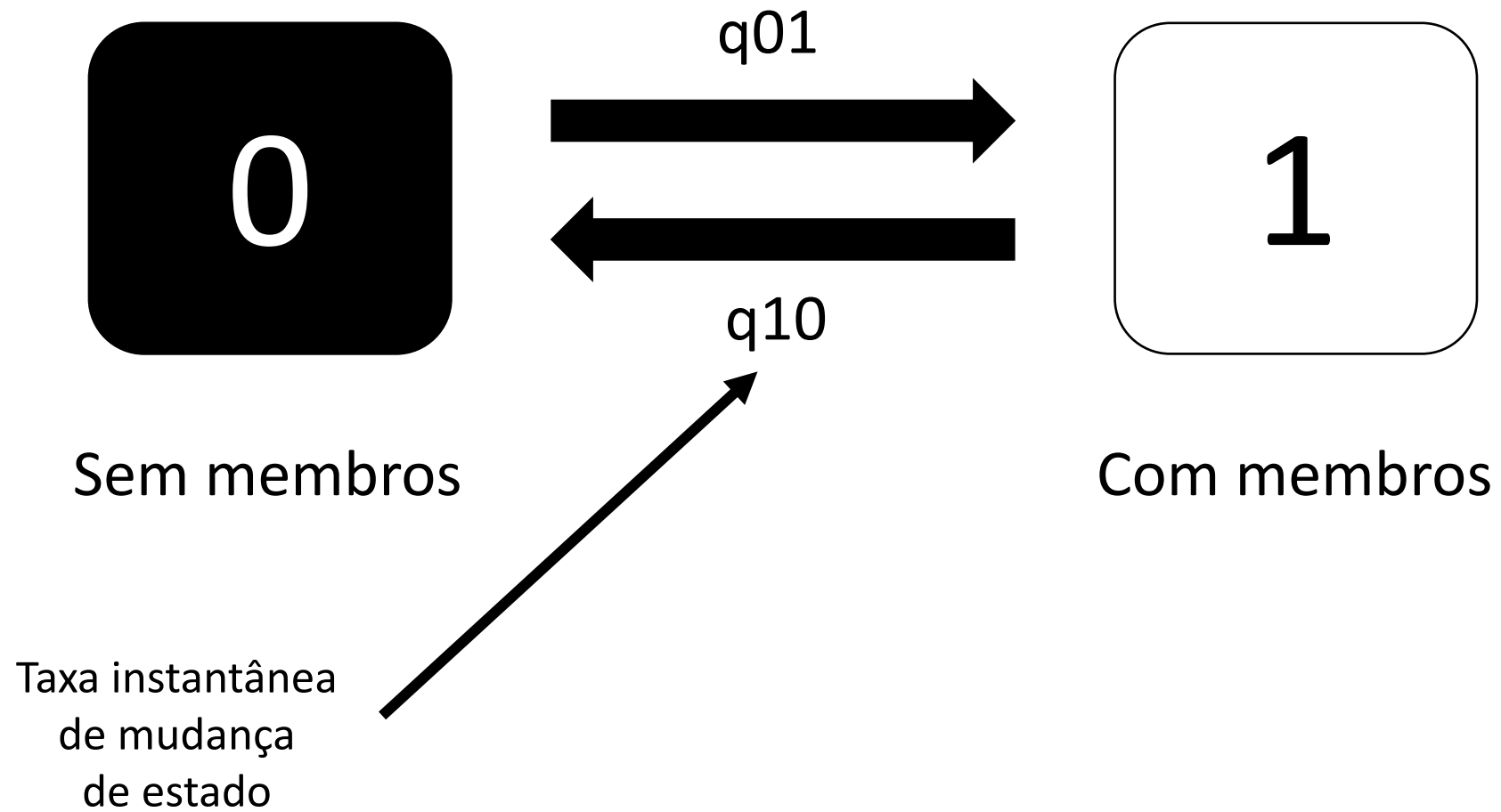
Com membros



Sem membros



Com membros



THIS SEEMS SORT OF



FAMILIAR

Mas porque isso é importante?

a GTR nucleotides

	A	G	C	T
A	-	r_{AG}	r_{AC}	r_{AT}
G	r_{AG}	-	r_{GC}	r_{GT}
C	r_{AC}	r_{GC}	-	r_{CT}
T	r_{AT}	r_{GT}	r_{CT}	-

b Binary correlation

	00	01	11	10
00	-	r_A	0	r_B
01	r_C	-	r_D	0
11	0	r_E	-	r_F
10	r_G	0	r_H	-

d Ordered transitions

	0	1	2	3
0	-	r_{01}	0	0
1	r_{10}	-	r_{12}	0
2	0	r_{21}	-	r_{23}
3	0	0	r_{32}	-

Além de serem necessários em métodos de inferência filogenética "model-based", esses modelos de substituição utilizam o mesmo arcabouço (processos de Markov) que métodos comparativos que iremos estudar, tais como correlação de caracteres binários e multiestado

0

1

0

1

?

t

Matriz de transição instantânea

$$Q = \begin{bmatrix} -q_{01} & q_{01} \\ q_{10} & -q_{10} \end{bmatrix}$$

Matriz de transição instantânea

$$Q = \begin{bmatrix} -q_{01} & q_{01} \\ q_{10} & -q_{10} \end{bmatrix} \quad \text{Soma da linhas} = 0$$

1

?

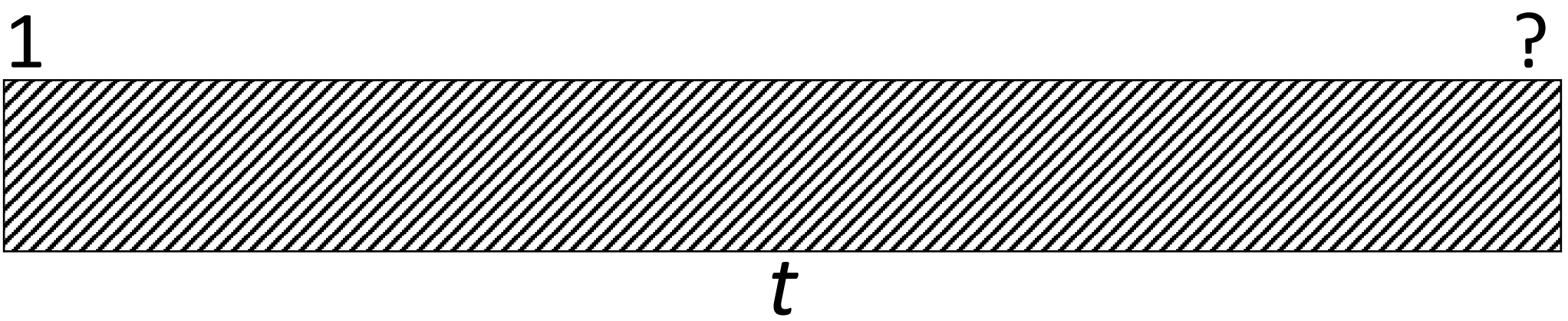
t

Qual a probabilidade de depois
do tempo t o estado 1 mudar para 0?

$$P_{10}(t)$$

Exponenciação de matriz

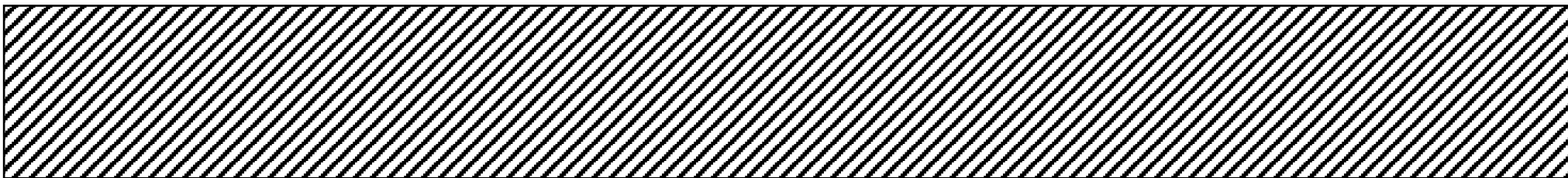
$$P(t) = e^{Qt}$$



$$Q = \begin{bmatrix} -q_{01} & q_{01} \\ q_{10} & -q_{10} \end{bmatrix}$$

1

?

 t

$$q_{01} = 0.1$$

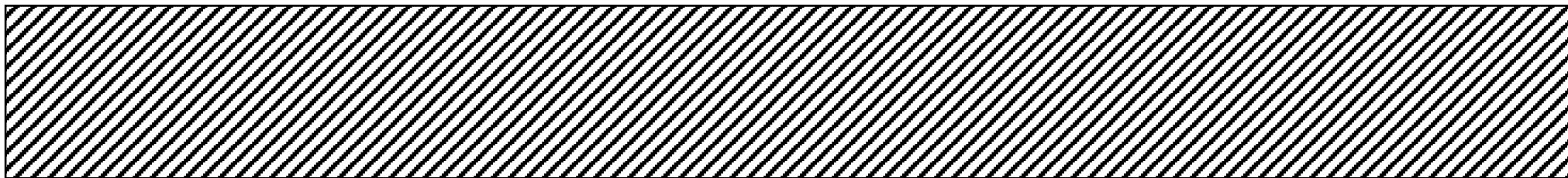
$$q_{10} = 0.1$$

$$t = 5 \text{ mya}$$

$$Q = \begin{bmatrix} -q_{01} & q_{01} \\ q_{10} & -q_{10} \end{bmatrix}$$

1

?

 t

$$q_{01} = 0.1$$

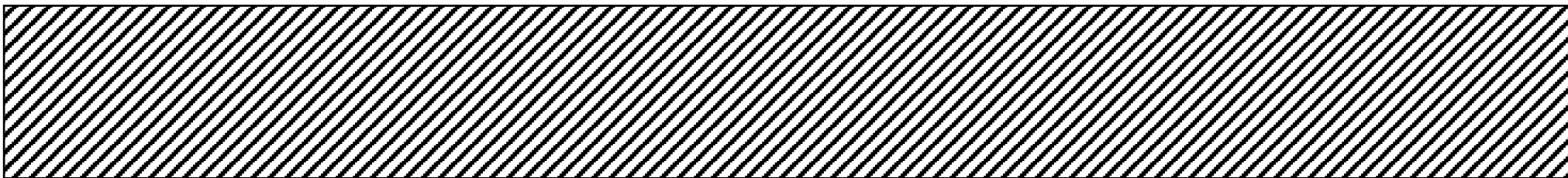
$$q_{10} = 0.1$$

$$t = 5 \text{ mya}$$

$$Q_t = \begin{bmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$$

1

?

 t

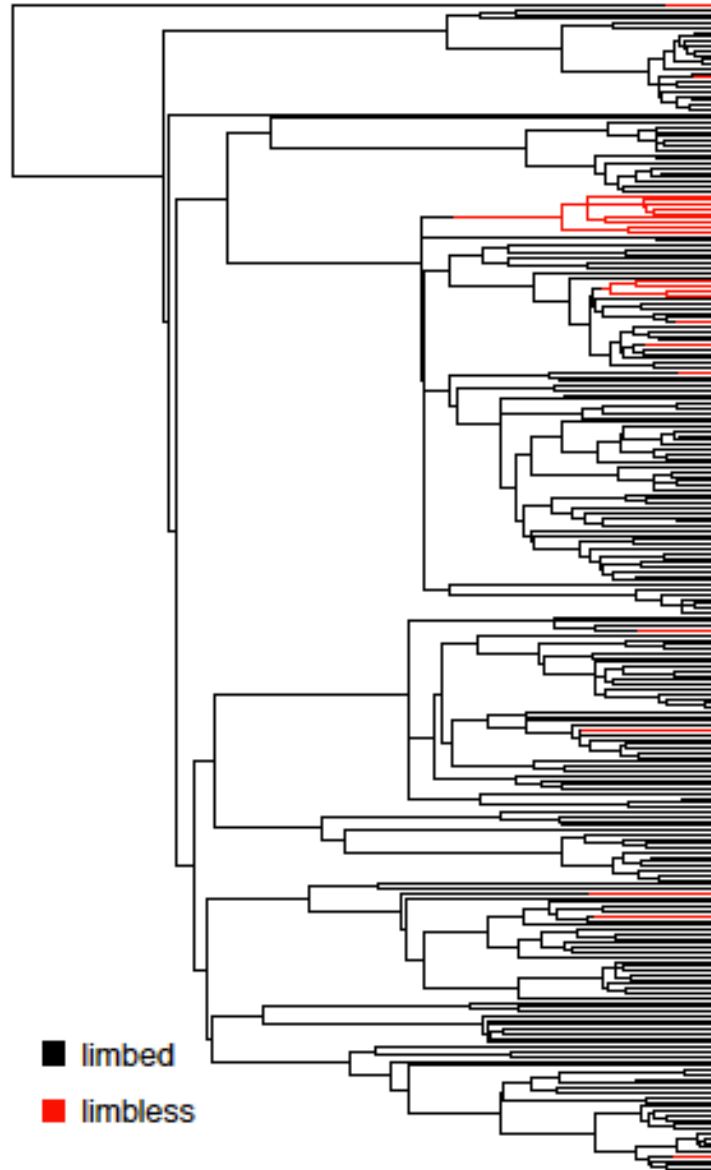
$$q_{01} = 0.1$$

$$q_{10} = 0.1$$

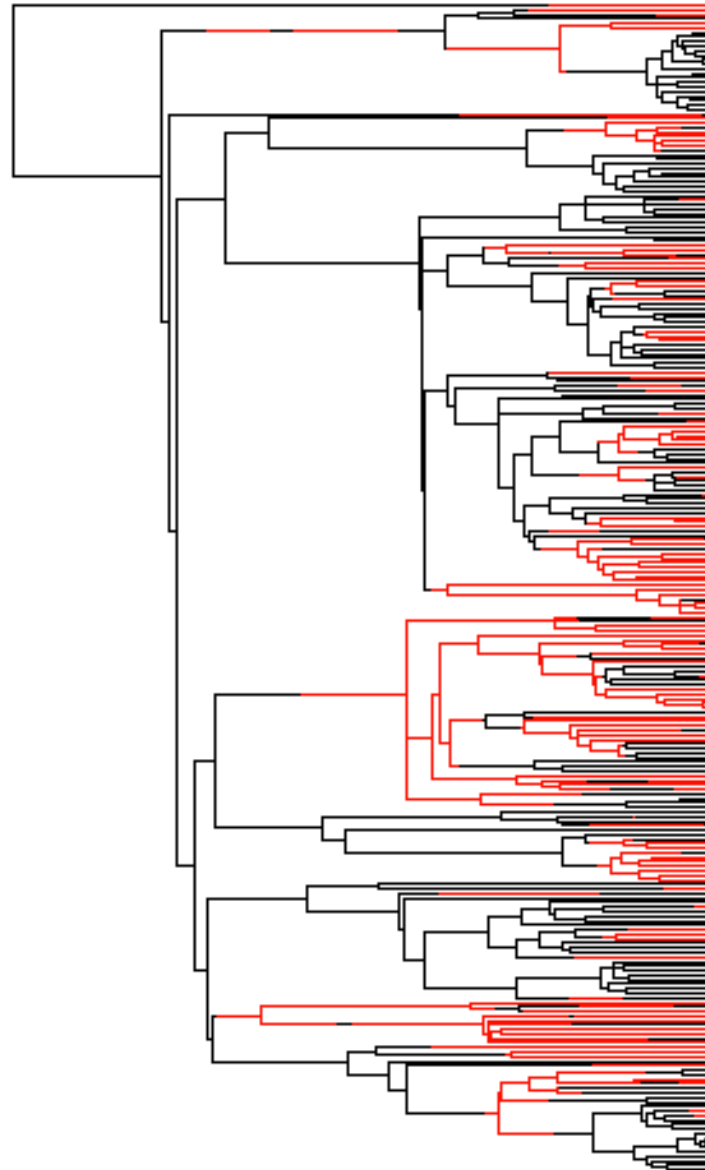
$$t = 5 \text{ mya}$$

$$e^{Qt} = \begin{bmatrix} 0.68 & 0.32 \\ 0.32 & 0.68 \end{bmatrix}$$

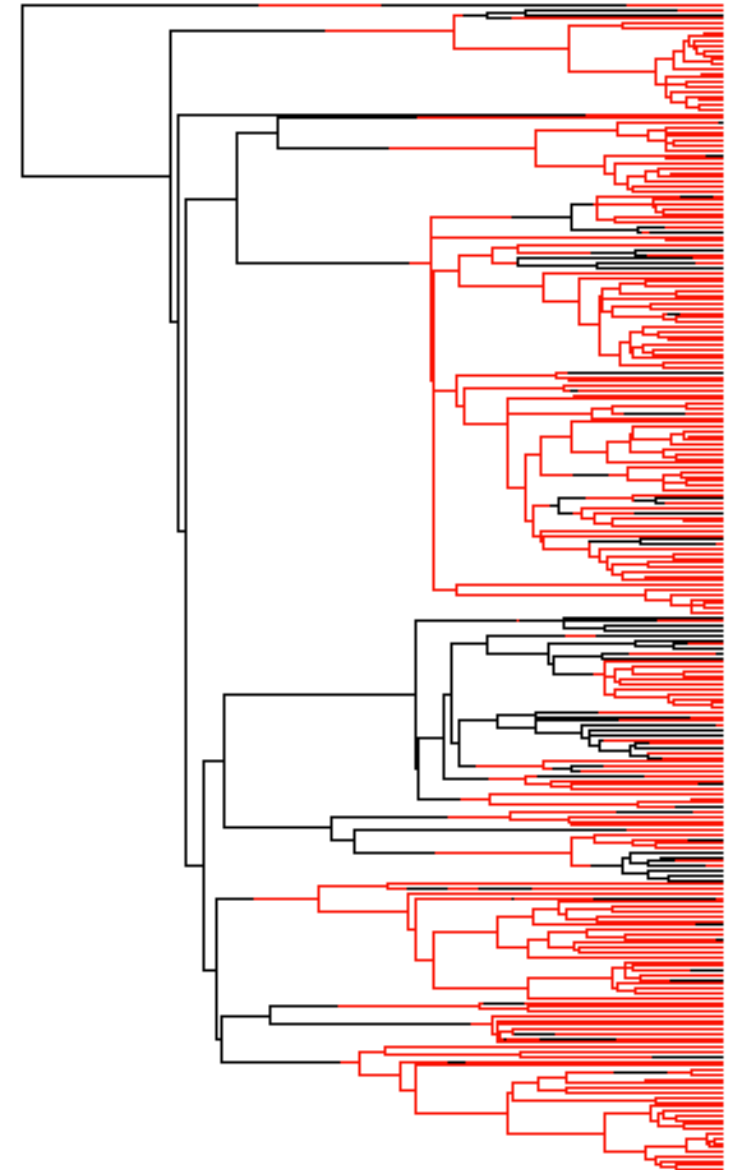
Equal rates (slow)

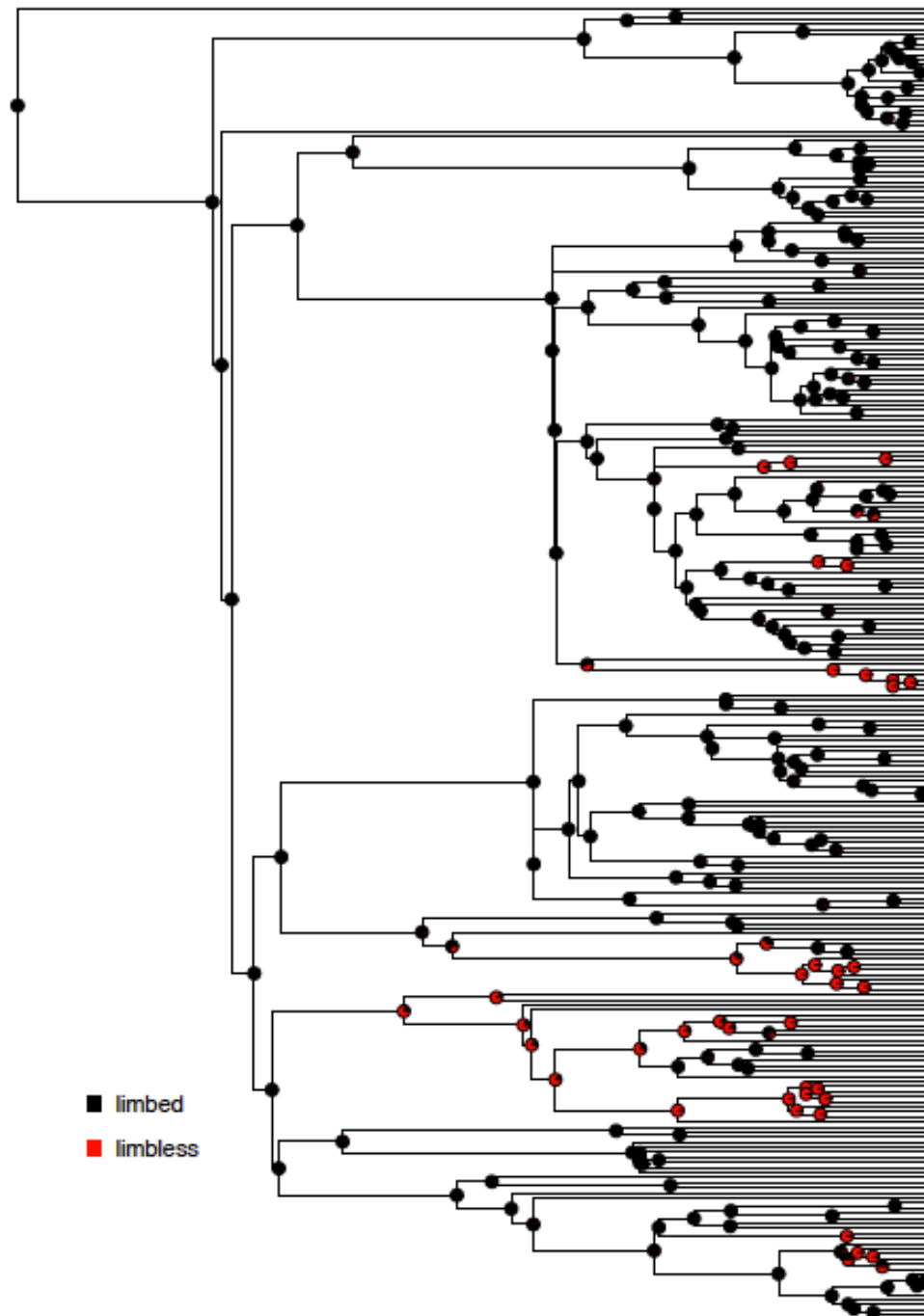


Equal rates (fast)



Unequal rates

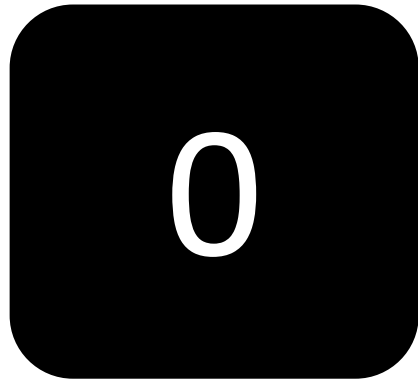




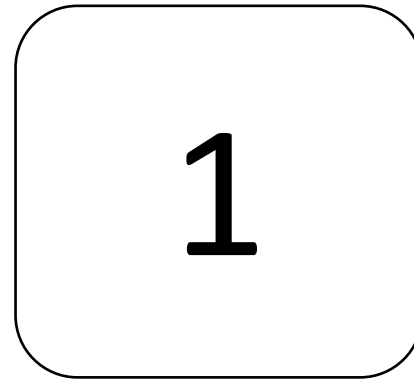
Podemos também
usar esses modelos
para estimar
estados ancestrais

Qual seria o resultado de uma análise com dados reais?

Equal rates

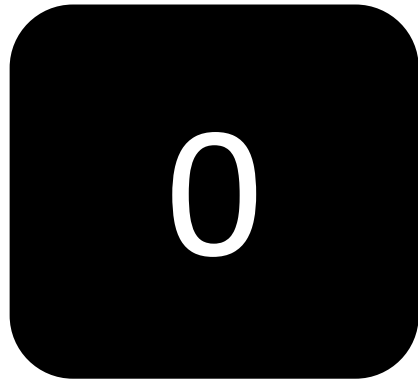


Sem membros

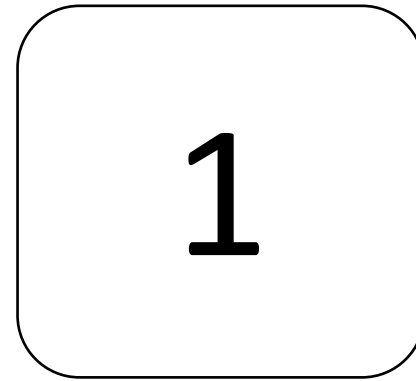


Com membros

Unequal rates



Sem membros



Com membros

Equal Rates Model

$$\ln L = -80.5$$

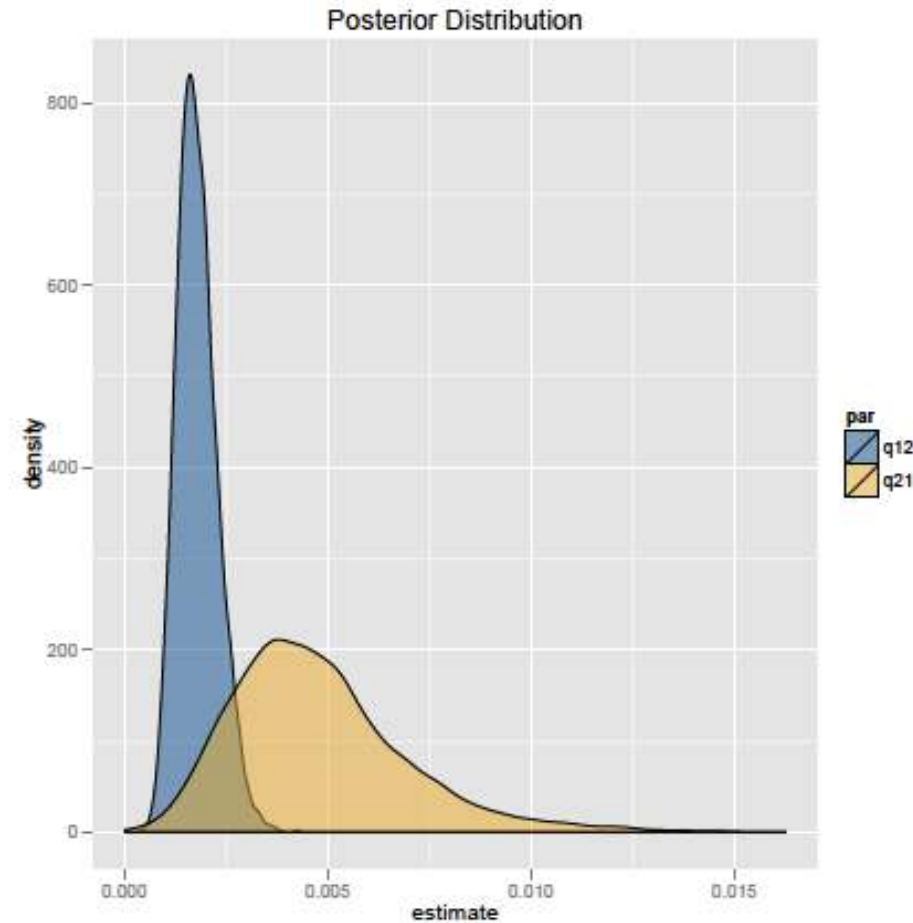
$$Q_{ER} = \begin{bmatrix} -0.0019 & 0.0019 \\ 0.0019 & -0.0019 \end{bmatrix}$$

Unequal Rates Model

$$\ln L = -79.4$$

$$Q_{ARD} = \begin{bmatrix} -0.0016 & 0.0016 \\ 0.0038 & -0.0038 \end{bmatrix}$$

We can also do this with Bayesian analyses



`geiger::fitDiscreteMCMC`

`geiger::fitContinuousMCMC`

Caracteres multiestado



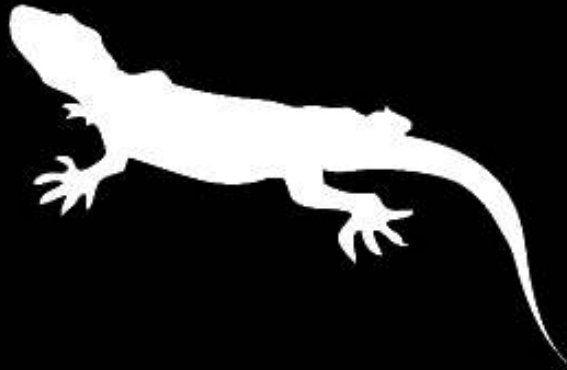
No limbs

State 0



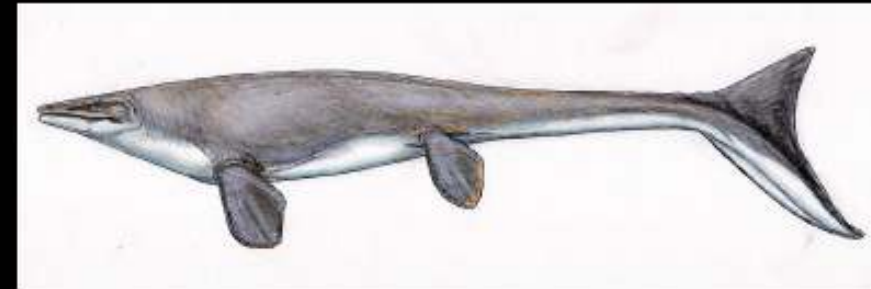
Front limbs only

State 1



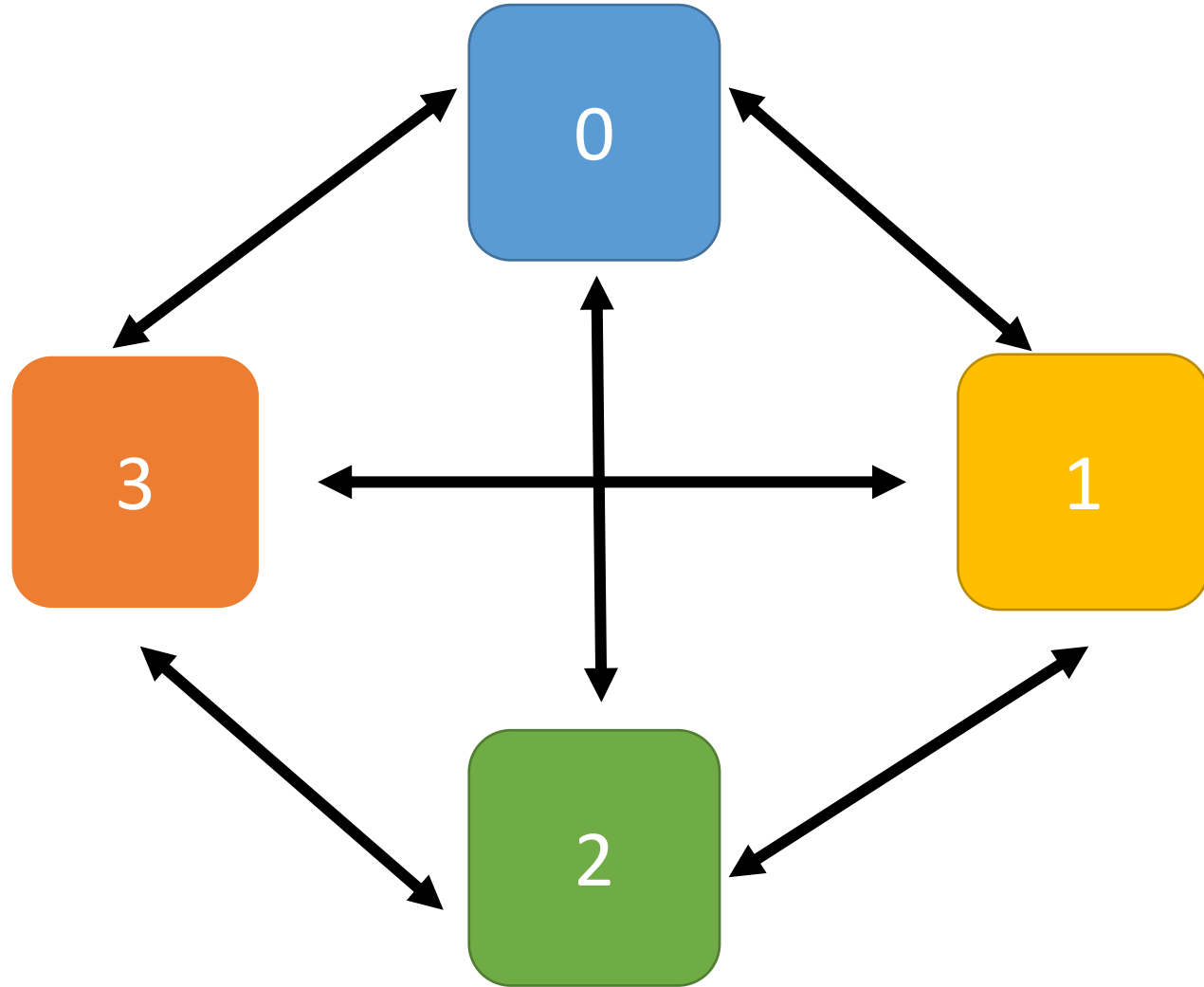
Limbs

State 2



"Fins"

State 3



Matriz de transição instantânea

$$Q = \begin{bmatrix} - & q_{01} & q_{02} & q_{03} \\ q_{10} & - & q_{12} & q_{13} \\ q_{20} & q_{21} & - & q_{23} \\ q_{30} & q_{31} & q_{32} & - \end{bmatrix}$$

Matriz de transição instantânea

$$Q = \begin{bmatrix} - & q_{01} & q_{02} & q_{03} \\ q_{10} & - & q_{12} & q_{13} \\ q_{20} & q_{21} & - & q_{23} \\ q_{30} & q_{31} & q_{32} & - \end{bmatrix}$$

Equal rates (ER)
1 parâmetro

Matriz de transição instantânea

$$Q = \begin{bmatrix} - & q_{01} & q_{02} & q_{03} \\ q_{10} & - & q_{12} & q_{13} \\ q_{20} & q_{21} & - & q_{23} \\ q_{30} & q_{31} & q_{32} & - \end{bmatrix}$$

Symmetric (SYM)

6 parâmetros

Matriz de transição instantânea

$$Q = \begin{bmatrix} - & q_{01} & q_{02} & q_{03} \\ q_{10} & - & q_{12} & q_{13} \\ q_{20} & q_{21} & - & q_{23} \\ q_{30} & q_{31} & q_{32} & - \end{bmatrix}$$

All Rates Differ (ARD)
12 parâmetros

A close-up photograph of a baby with light brown hair and blue eyes, wearing a green and white shirt. The baby has a serious, determined expression. The background is a blurred beach scene with sand and water.

YES!

**AGORA BORA BOTAR A MÃO NA
MASSA!**

Mínimos Quadrados Generalizados (GLS)

Ecological Applications, 16(1), 2006, pp. 20–32
© 2006 by the Ecological Society of America

STATISTICS FOR CORRELATED DATA: PHYLOGENIES, SPACE, AND TIME

ANTHONY R. IVES^{1,3} AND JUN ZHU²

THE PHYLOGENETIC REGRESSION

BY A. GRAFEN†

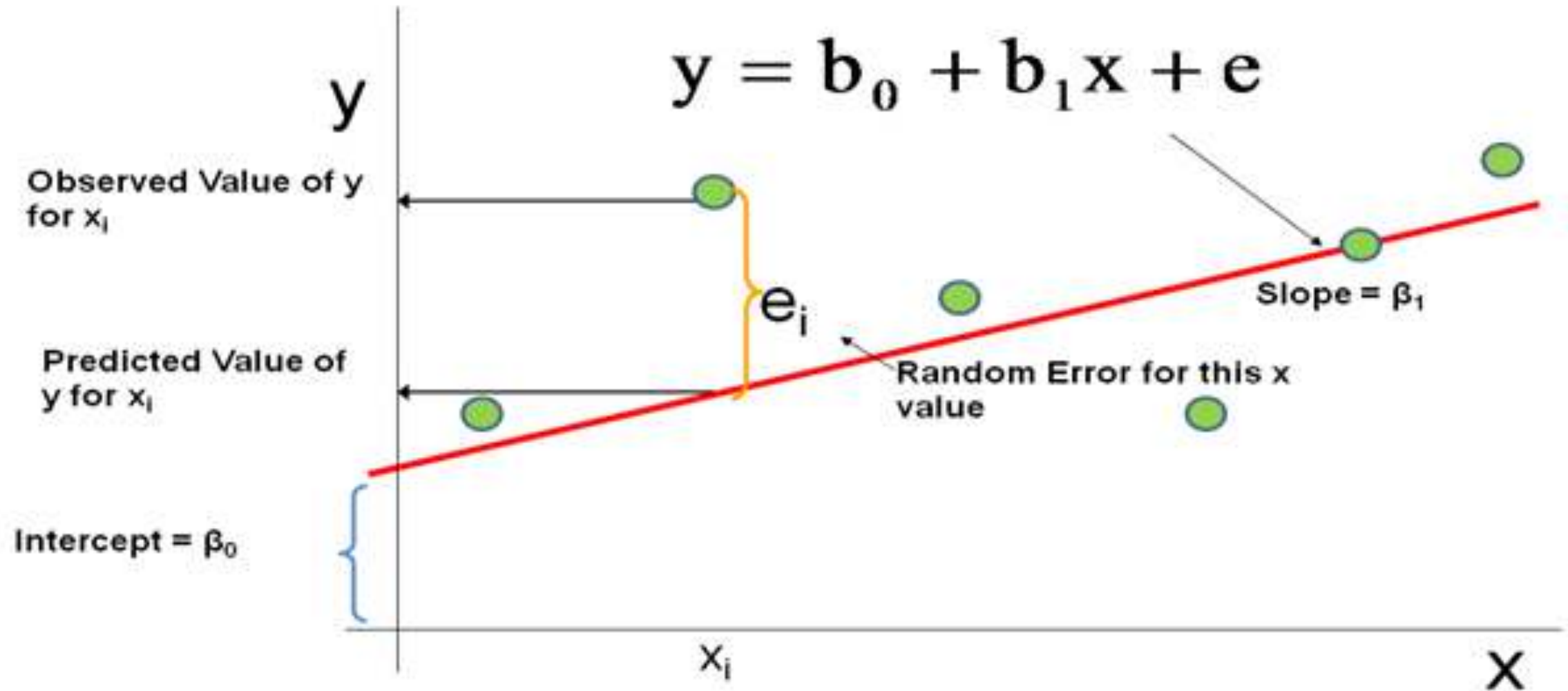
- Normalmente empregado para testar a associação entre dois atributos
- Muito flexível
 - Preditor e resposta contínuos
 - Preditores discretos binários ou ordinais (multiestado)
- Para resposta binária use uma regressão logística filogenética
 - Disponível no pacote `phylolm::phylglm`
 - Ou uma abordagem Bayesiana no pacote `MCMCglmm`
- Pode incorporar diferentes modelos evolutivos
 - Manipulação da estrutura de correlação (matriz de variância-covariância)
 - Customizado ou já disponível

Antes, relembRANDo o modelo linear (OLS)

The diagram shows the OLS regression equation $y_i = b_0 + b_1 x + e$ with five labels and arrows pointing to its components:

- Estimated (or predicted) y value**: points to y_i
- Estimate of the regression intercept**: points to b_0
- Estimate of the regression slope**: points to b_1
- Independent variable**: points to x
- Error term**: points to e

$$y_i = b_0 + b_1 x + e$$



Pressupostos do OLS (Teorema Gauss-Markov)

Matemática

$$y = \mathbf{X}\beta + \varepsilon$$

$\mathbf{X}$ é uma matriz $n \times k$

$$E(\varepsilon \varepsilon' | \mathbf{X}) = \sigma^2 \mathbf{I}$$

$$\varepsilon | \mathbf{X} \sim N(0, \sigma^2 \mathbf{I})$$

Tradução pra língua humana

Relação linear entre y e $\mathbf{X}$

Colunas de $\mathbf{X}$ são independentes

Homocedasticidade e ausência de autocorrelação

Resíduos têm distribuição normal com média zero e variância homogênea

Pressuposto de independência e homocedasticidade

Matemática

$$\text{var}[\varepsilon_i|X] = \sigma^2 \text{ para qualquer } i$$

$$\text{cov}[\varepsilon_i, \varepsilon_j|X] = 0 \text{ para qualquer } i$$

Tradução pra língua humana

Variância no erro é a mesma (σ^2) para qualquer elemento da matriz

Sabendo o valor da variável numa unidade amostral não nos diz nada do valor dela em outra u.a.

$$E(\epsilon\epsilon'|X) = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix}$$

$$E(\epsilon\epsilon'|X) = \sigma^2 \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = \sigma^2 I$$

Chapter 5

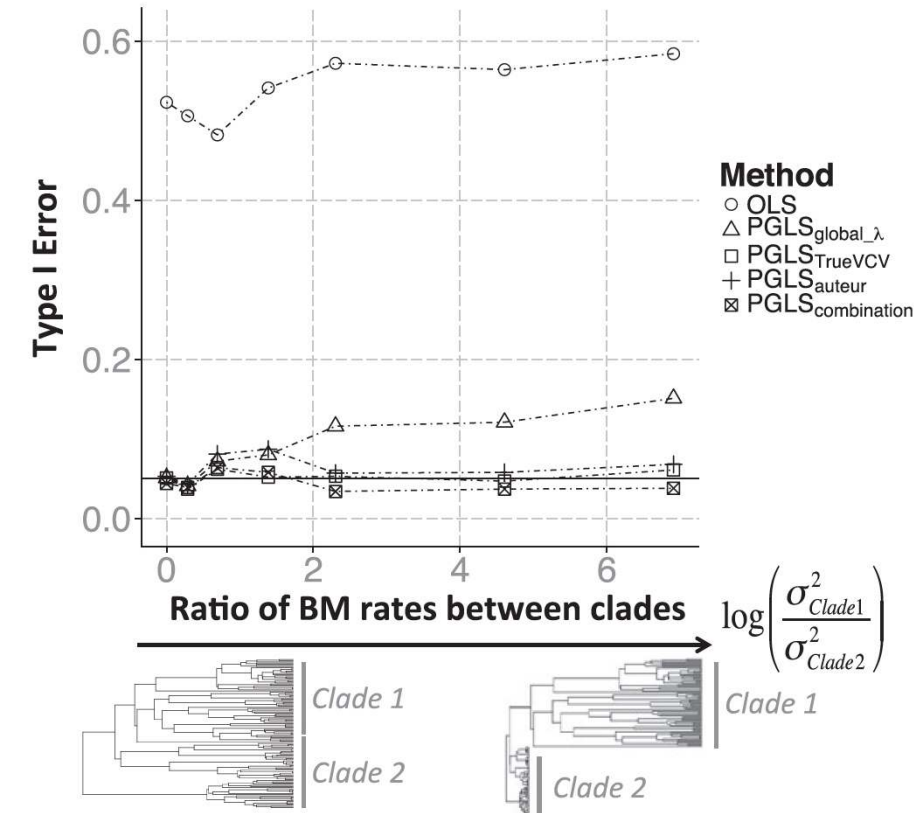
A Primer on Phylogenetic Generalised Least Squares

Matthew R. E. Symonds and Simon P. Blomberg

- Assume linearidade entre as variáveis resposta e preditora(s)
- Estimativa vai depender do modelo evolutivo incorporado

Improving phylogenetic regression under complex evolutionary models

FLORENT MAZEL,^{1,2,6} T. JONATHAN DAVIES,^{3,4} DAMIEN GEORGES,^{1,2} SÉBASTIEN LAVERGNE,^{1,2}
 WILFRIED THUILLER,^{1,2} AND PEDRO R. PERES-NETO⁵



regression. We found that PGLS has good power but unacceptable type I error rates. This finding is important since this method has been increasingly used in comparative analyses over the last decade. To address this issue, we propose a simple solution based on transforming the underlying variance–covariance matrix to adjust for model heterogeneity within PGLS. We suggest that heterogeneous rates of evolution might be particularly prevalent in large phylogenetic trees, while most current approaches assume a homogenous rate of evolution. Our analysis demonstrates that overlooking rate heterogeneity can result in inflated type I errors, thus misleading comparative analyses. We show that it is possible to correct for this bias even when the underlying model of evolution is not known a priori.

Chapter 5

A Primer on Phylogenetic Generalised Least Squares

Matthew R. E. Symonds and Simon P. Blomberg

- Assume linearidade entre as variáveis resposta e preditora(s)
- Estimativa vai depender do modelo evolutivo incorporado
- Erro comum num PGLS
 - Testar sinal filogenético nas variáveis resposta e/ou preditora
- O que é correto: testar sinal filogenético nos resíduos

Ajustando um PGLS

- O que é melhor: ajustar um modelo em que simultaneamente se estima o "sinal" filogenético dos resíduos (usando ML), p.ex. usando o lambda do Pagel e se estima os parâmetros do modelo

Methods in Ecology and Evolution



British Ecological Society

Methods in Ecology & Evolution 2010, 1, 319–329

doi: 10.1111/j.2041-210X.2010.00044.x

Phylogenetic signal and linear regression on species data

Liam J. Revell*

Independent Contrasts and PGLS Regression Estimators Are Equivalent

SIMON P. BLOMBERG^{1,*}, JAMES G. LEFEVRE², JESSIE A. WELLS¹, AND MARY WATERHOUSE³

- Os estimadores dos Contrastes Filogenéticos Independentes (PIC) são equivalentes aos do PGLS quando se assume Movimento Browniano

Diagnosticando um PGLS

